

From Reservoir Replay to Prospective Imagined-Rehearsal Gating in Sequential RLScope

A chronological experimental chapter on retention, model semantics, critic failure, controlled actor recovery, and adaptation safety

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Continual Reinforcement Learning Research Notes

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Abstract

This chapter records twelve linked studies in a goal-conditioned DreamerV3 agent operating in RLScope. Studies I–VII establish the base mechanism. A goal-stratified episode reservoir yields transient multi-goal retention but not stable continual learning; the original critic is severely optimistic and supplies 94.4%–97.3% of the six-step standard-IR target; and critic-free reward-only IR repairs both synthetic actor corruption and natural forgetting. Counterfactual supervision raises held-out completion-goal accuracy from 33.3% to 97.4%, while a separately bounded value head removes gross scale bias. A five-percent bounded-value correction then improves sampled reward-only success from 73.8% to 87.6%, but fails a deterministic weakest-goal safety rule.

Studies VIII–XII ask when that effective but potentially harmful adaptation should run. Fixed entropy, Monte Carlo Posterior Batching (MCPB) Jensen-Shannon (JS), and two-stage consequence gates reduce updates and imagined transitions, but none preserves always-on reward-only performance. On a frozen competent chop task and weak bury task, JS activates more often on the weak task yet reduces bury success from 65.3% to 32.0% sampled and 48.0% deterministic; the consequence gate activates in the wrong direction. An offline 90-episode audit explains why: ten-state JS ranks later underperformance at pooled AUROC 0.501, whereas imagined failure reaches 0.669 and rises to 0.772 by 25 states. JS retains modest information about whether failed episodes are repairable, but its thresholds do not transfer across goals. A paired virtual-update gate then tentatively updates on one imagined batch and commits only when an independent, common-random-number batch predicts improvement. This prospective test succeeds on chop and partly on kill, but catastrophically fails on bury; first virtual improvement ranks true IR need at only AUROC 0.470–0.528.

Study XII autopsies that bury failure over 2,010 sampled episodes. Always-on reward-only IR raises success from 34.0% to 60.7%, and a 127-update cap reaches 61.3%; caps of 16–80 updates instead fall to 11.3%–20.7%. Randomly distributing at most 46 updates reaches 54.0%, while the virtual-positive selector reaches exactly zero despite committing 48.6 updates per episode. Inverting its decision recovers only 17.3%. An exact $\beta = .005$ value mixture is not a safe regularizer: always-on mixed IR also reaches zero, although random scheduling again reaches 54.0%. Thus the catastrophe combines a partial-repair region with harmful selection and extreme seed sensitivity. A dedicated world-model audit does not pass its no-intervention check (only 14.3% paired action agreement), so reward-model exploitation remains plausible but unproven. The combined result is a constructive negative: reward-only IR is a strong recovery operator, but the present world model cannot yet safely judge the real behavioral value of its own actor updates. Persistent Stage 5C adaptation remains deferred in favour of RNG-isolated model auditing, stronger performance estimation, uncertainty-aware acceptance, and hard update safety.

Contents

1	Research programme and chronological evidence chain	5
2	Study I: sequential reservoir replay and checkpoint IR	7
2.1	Task sequence	7
2.2	Agent and training configuration	7

2.3	Replay conditions	8
2.4	Evaluation-time imagined rehearsal	9
2.5	Evaluation protocol and integrity	9
3	Study I results	10
3.1	Execution and systems outcome	10
3.2	Frozen-policy checkpoint matrix	11
3.3	Forgetting	12
3.4	Effect of iterative rehearsal	13
4	Study I interpretation	15
4.1	What reservoir replay did establish	15
4.2	Why balanced replay may still collapse	15
4.3	Why ungated IR can hurt	15
5	Study I limitations and threats to validity	16
6	Decision point after Study I	16
7	Interim conclusion after Study I	17
8	Study II: locating the failure inside the learned IR target	17
8.1	Study IIa method: immutable same-state head audit	17
8.2	Study IIa results: event recognition without goal semantics	18
8.3	Study IIb method: exact six-step imagination-path audit	19
8.4	Study IIb results: IR optimizes the critic, not imagined reward	20
8.5	Study II interpretation and resulting prediction	21
9	Study III: controlled actor corruption and recovery	21
9.1	Question and experimental controls	21
9.2	Recovery objectives	22
9.3	Evaluation scale and integrity	23
9.4	Study III results	24
9.5	Study III interpretation	27
10	Study IV: critic-free recovery from natural forgetting	28
10.1	Question and prospective design	28
10.2	Execution and integrity	28
10.3	Absolute recovery results	28
10.4	Paired effects and task-specific exceptions	30
10.5	Study IV interpretation and claim boundary	31
11	Study V: critic provenance and calibration	31
11.1	Fixed state bank and estimands	32
11.2	The critic is already wrong on real posterior states	32
11.3	Horizon decomposition: no late runaway	33
11.4	Actor selection amplifies the critic error	33

11.5 Study V interpretation and claim boundary	34
12 Study VI: counterfactual goal semantics and bounded-value repair	35
12.1 Disjoint repair and held-out state banks	35
12.2 Paired actor-adaptation assay	35
12.3 Counterfactual labels repair held-out semantics	36
12.4 The bounded head repairs scale and ranking	37
12.5 Behavioral result: reward-only remains strongest	38
12.6 Study VI interpretation and claim boundary	41
13 Study VII: conservative bounded-value dose and stability	41
13.1 Causal interpolation and fixed controls	41
13.2 Pairing, budget, and decision rule	42
13.3 Small value doses strongly improve sampled behavior	42
13.4 Deterministic control rejects automatic advancement	43
13.5 Study VII decision and claim boundary	46
14 Study VIII: interpretable gating and compute accounting	46
14.1 Question, signals, and frozen selection	46
14.2 Fixed gates save work but lose the recovery signal	47
14.3 Study VIII decision	48
15 Study IX: weak-versus-competent IR necessity discrimination	48
15.1 Calibration and confirmation design	48
15.2 The JS direction is plausible, but behavioral discrimination fails	49
15.3 Study IX decision	50
16 Study X: offline performance-gate audit	50
16.1 Prospective features and retrospective labels	50
16.2 Imagined failure predicts performance; JS does not calibrate globally	50
16.3 Study X decision	51
17 Study XI: paired virtual-update acceptance	51
17.1 Causal gate and lifecycle	51
17.2 Mixed task behavior rejects the virtual score as a safety test	52
17.3 Compute and completion speed	53
17.4 Study XI decision and claim boundary	53
18 Study XII: bury-bones virtual-gate autopsy	53
18.1 Question and controlled design	53
18.2 Behavioral result: update placement dominates nominal dose	54
18.3 Virtual selection is worse than random selection	55
18.4 A tiny value term is a qualitative intervention	56
18.5 Action attractors and the failed no-intervention audit	57
18.6 Study XII decision and claim boundary	58

19 Cross-study synthesis	58
20 Revised experimental ladder	60
21 Global limitations	61
22 Chapter conclusion	62
A Complete frozen-policy evaluation table	64
B Reservoir IR effects and paired intervals	64
C Reproducibility manifest	65
D Study II–XI reproducibility manifest	65
E Complete Study III success table	67
F Complete Study IV success tables	67
G Complete Study V diagnostic tables	68
H Complete Study VI success tables	68
I Complete Study VII success tables	69
J Complete Study VIII–XII result tables	70

1 Research programme and chronological evidence chain

The long-term research direction is continual reinforcement learning over an RLscape task landscape: train on goals $A \rightarrow B \rightarrow C \rightarrow \dots$, repeatedly evaluate all previous goals, and measure both forgetting and transfer speed relative to a non-IR baseline. The immediate question was narrower. The preceding FIFO experiment suggested that later phases remove most early-task data, potentially weakening the goal-conditioned reward model and value function that IR relies on. We therefore asked:

1. Does goal-stratified reservoir replay preserve earlier goal competence better than global FIFO replay?
2. Does a retained multi-goal world model and critic make actor-only IR useful at evaluation time?
3. Can the system train continuously for 1.5M steps with research-grade checkpoint and evaluation integrity?
4. If critic-free rehearsal repairs synthetic actor corruption, does it also recover behavior lost naturally during sequential training?
5. Does critic error begin on real posterior states, emerge with imagination depth, or arise because the actor exploits the learned model?
6. Can explicit counterfactual event labels and a separately bounded success-value head repair the learned teaching signals, and does that repair improve real-environment IR?
7. Can a small bounded-value correction improve reward-only IR under fully paired adaptation randomness without damaging the weakest goal or deterministic policy?
8. Can actor entropy, MCPB JS disagreement, or imagined consequence evidence decide when episode-local IR should run while reducing unnecessary compute?
9. Do those fixed gates distinguish an IR-responsive weak task from an already competent task, and which prospective signals actually predict later performance?
10. Can a tentative update be accepted or rolled back using only paired held-out imagination, without observing the real outcome?
11. Does the catastrophic bury failure arise from insufficient update dose, from the virtual selector choosing harmful updates, or from exploitation of a miscalibrated learned reward signal?

The first experiment contributes a fully persisted three-goal sequential run, six immutable training checkpoints, 24 paired frozen/IR evaluation units, two action-selection modes, and a direct comparison to the previous FIFO run. It is an exploratory, single-seed systems experiment rather than a population-level statistical claim. The later experiments deliberately reuse the same pinned checkpoints. This converts a costly multi-day training run into a sequence of increasingly controlled mechanism tests.

The chapter is organized as a research diary without sacrificing paper-style separation between method, result, and interpretation:

1. **Study I — continual training and replay.** Establish whether reservoir replay preserves usable cross-goal knowledge and whether ordinary actor-only IR helps at checkpoints.
2. **Study II — subsystem and imagination audits.** Hold the replay-state bank fixed and locate whether failure lies in factual event recognition, counterfactual goal semantics, continuation, critic calibration, or the exact IR objective.
3. **Study III — controlled actor corruption and recovery.** Hold the world model and critics fixed, damage only the actor, and causally ablate horizon, entropy, and critic bootstrap.
4. **Study IV — natural forgetting recovery.** Apply the successful critic-free objective to unmodified, naturally forgotten checkpoint actors and measure recovery and harm against paired frozen baselines.
5. **Study V — critic provenance and calibration.** Evaluate identical real posterior states, recorded-action imagination, and actor-selected imagination to identify where critic optimism originates and how it changes with horizon.
6. **Study VI — counterfactual and bounded head repair.** Freeze the representation and actor, repair reward/continuation semantics and value scale from preserved experience, then repeat a paired real-environment IR assay.

7. **Study VII — conservative value dose and stability.** Interpolate exactly between reward-only and bounded-value teaching, repeat fully paired adaptation seeds, and test dose, horizon, correction clipping, and actor-update caps against a prespecified worst-goal rule.
8. **Study VIII — interpretable gating.** Audit entropy, MCPB JS, and consequence signals, freeze thresholds, then compare gated and always-on variants with explicit compute accounting.
9. **Study IX — necessity discrimination.** Test the frozen gates on one calibrated weak task and one competent task from the same immutable checkpoint.
10. **Study X — offline performance-gate audit.** Use prospective state prefixes and retrospective paired outcomes to separate predicted underperformance from predicted IR utility.
11. **Study XI — paired virtual-update gate.** Tentatively adapt on one imagined batch, validate on an independent paired batch, and commit or exactly roll back without real feedback.
12. **Study XII — bury-bones virtual-gate autopsy.** Reproduce the sampled bury catastrophe, separate dose from update selection, invert the selector, add exact $\beta = .005$ mixed controls, and audit imagined-reward/action attractors.

To keep implementation artifacts and thesis terminology aligned, the canonical repository stages map to the paper studies as follows. Historical run names beginning with **m** denote engineering milestones and are retained only for provenance.

Table 1: Canonical experiment identifiers used by the repository registry.

Canonical ID	Experiment	Chapter location
Stage 0	Sequential reservoir baseline and checkpoint IR	Study I
Stage 1A	Same-state goal-conditioned head audit	Study IIa
Stage 1B	Exact imagination-path audit	Study IIb
Stage 2	Controlled actor corruption and recovery	Study III
Stage 3A	Critic provenance and calibration audit	Study V
Stage 3B	Natural forgetting recovery	Study IV
Stage 4A	Counterfactual goal-head repair	Study VI
Stage 4B	Conservative bounded-value dose and stability	Study VII
Stage 5A	Interpretable gating for imagined rehearsal	Study VIII
Stage 5B	IR necessity discrimination	Study IX
Stage 5B.1	Offline performance-gate audit	Study X
Stage 5B.2	Paired virtual-update gate	Study XI
Stage 5B.2a	Bury-bones virtual-gate autopsy	Study XII

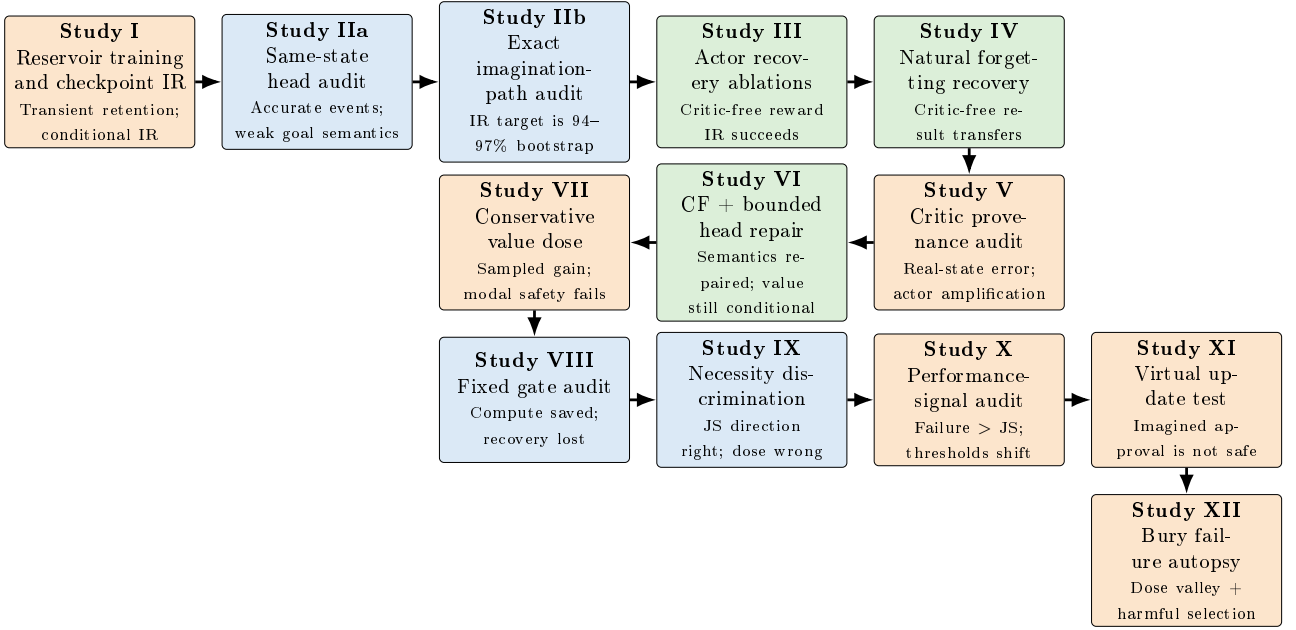


Figure 1: Chronological experimental logic. Each study was designed after observing the previous result; Studies II–XII reuse immutable checkpoints and therefore isolate mechanisms without another continual-training run.

2 Study I: sequential reservoir replay and checkpoint IR

2.1 Task sequence

The agent trained on three goals in a fixed curriculum:

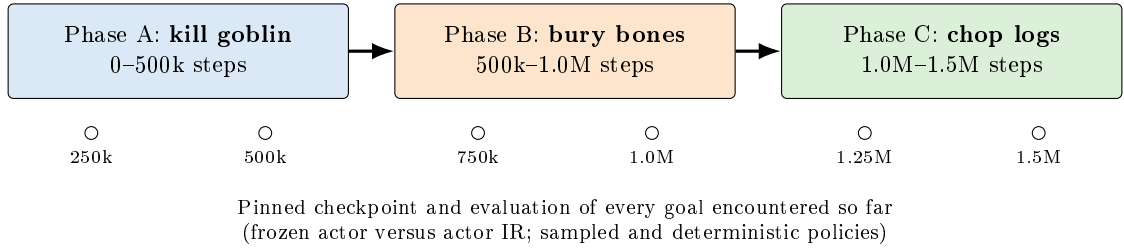


Figure 2: Sequential curriculum and evaluation ladder. There is no return phase in this experiment. The checkpoints permit later ablations without retraining the full curriculum.

Each episode was capped at 200 actions and used sparse binary task-completion reward. The action interface was a flat 28×18 click grid, giving 504 discrete click locations. This was intentionally held fixed between replay conditions so the comparison isolates replay, not environment design.

2.2 Agent and training configuration

The implementation uses DreamerV3 [3] with the repository’s “50m” preset. The observed trainable parameter count was approximately 77.3M; the preset name is therefore a configuration label, not the measured parameter total. The recurrent state-space model (RSSM) uses a deterministic state of width 4,096 and a 32×32 categorical stochastic state. Training used batches of 8 sequences, sequence length 32, imagination horizon 15, training ratio 32, and bfloat16 computation on a 24GB GPU.

Let o_t be the image, a_t the click action, and $s_t = (h_t, z_t)$ the Dreamer latent state. The goal-agnostic dynamics

are

$$h_t = f_\phi(h_{t-1}, z_{t-1}, a_{t-1}), \quad (1)$$

$$z_t \sim q_\phi(z_t \mid h_t, e_\phi(o_t)) \quad \text{during representation learning}, \quad (2)$$

$$z_t \sim p_\phi(z_t \mid h_t) \quad \text{during imagination}. \quad (3)$$

Five RLScope goals are represented by a five-dimensional one-hot vector g_t , although only three goal IDs occur in this experiment. The behavior feature is

$$x_t = [h_t; z_t; g_t]. \quad (4)$$

The reward, continuation, actor, value, and slow-value heads consume x_t :

$$\hat{r}_t \sim p_\phi(r_t \mid h_t, z_t, g_t), \quad \hat{c}_t \sim p_\phi(c_t \mid h_t, z_t, g_t), \quad (5)$$

$$a_t \sim \pi_\theta(a_t \mid h_t, z_t, g_t), \quad v_\psi = v_\psi(h_t, z_t, g_t). \quad (6)$$

The RSSM transition and visual representation remain goal-agnostic. Thus the shared latent should model what is happening on screen, while the heads interpret that state under the requested goal.

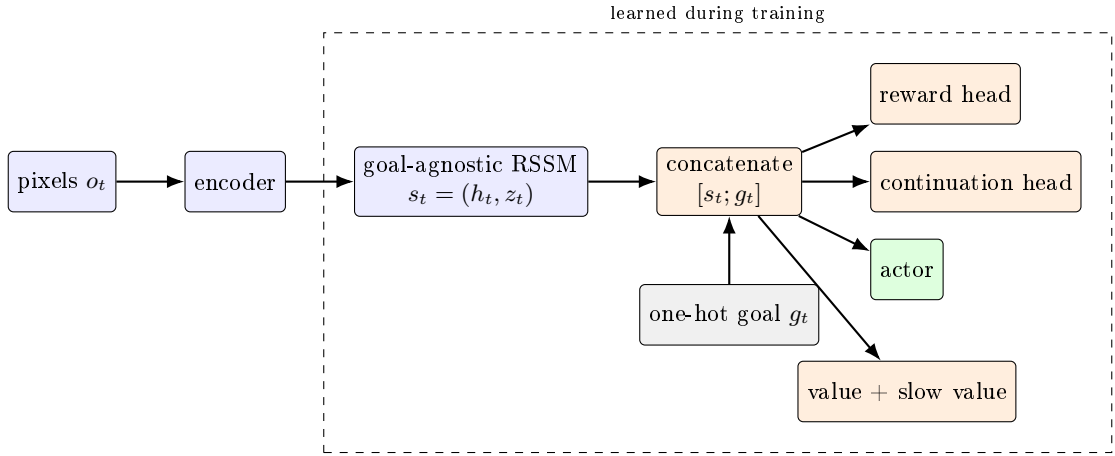


Figure 3: Goal-conditioning architecture. During evaluation-time IR, only the actor parameters (green) are updated; the world model, reward/continuation heads, and critic/value functions are frozen.

2.3 Replay conditions

FIFO baseline. The comparison run used the original global replay with capacity 200,000 sequence starts. Once full, newly collected experience evicts the oldest experience regardless of goal. Uniform training batches therefore increasingly reflect the current phase.

Goal-stratified episode reservoir. The new replay stores at most $M_g = 1,000$ complete episodes independently for each represented goal, for a total capacity of 3,000 episodes. Within goal g , the n_g th observed episode is processed by Algorithm R [4]:

$$\Pr(\text{admit episode } n_g) = \min \left(1, \frac{M_g}{n_g} \right). \quad (7)$$

If admitted after the group is full, it uniformly replaces one of the M_g resident episodes. Consequently every historical episode for a goal has equal probability M_g/n_g of residing in that goal’s reservoir. Training samples a represented goal uniformly, an episode uniformly within that goal, and then a random sequence start. Short fragments are concatenated with explicit `is_first` boundaries so the recurrent state resets correctly.

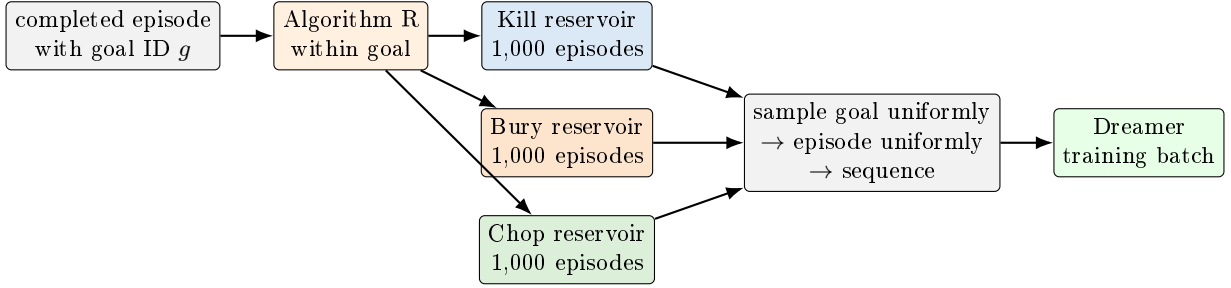


Figure 4: Goal-stratified episode reservoir. Statistical retention is uniform over the episode stream within each goal; optimization sampling is balanced across represented goals.

This balance changes the effective emphasis on new data. The global training ratio remains 32, but the expected share of updates drawn from the current goal is 1, 1/2, and 1/3 in phases A, B, and C. In this limited sense, current-goal exposure is approximately 32, 16, and 10.7 replayed samples per environment step. This is a plausible source of slower acquisition or instability and is important when interpreting the comparison.

2.4 Evaluation-time imagined rehearsal

IR performs temporary test-time optimization of the actor from the agent’s current latent state. For each environment action, Monte Carlo Posterior Batching (MCPB) draws $N = 128$ posterior latent samples, imagines $H = 6$ steps under the frozen world model, and takes one actor gradient update with learning rate 4×10^{-5} . MCPB is posterior batching: it does not denote particles and does not perform particle filtering, weighting, or resampling. The objective is the Dreamer actor objective: increase imagined, critic-estimated return while retaining the configured entropy bonus $\eta = 3 \times 10^{-4}$,

$$\theta' \leftarrow \theta - \alpha \nabla_{\theta} \left[-\mathbb{E}_{\hat{\tau} \sim p_{\phi}, \pi_{\theta}} \left(\hat{A}_{\psi}(\hat{\tau}) + \eta \mathcal{H}[\pi_{\theta}(\cdot \mid \hat{s}, g)] \right) \right]. \quad (8)$$

The precise implementation uses the repository’s normalized imagined-advantage actor loss. The world model, reward/continuation models, critic, and slow critic are frozen. Actor updates accumulate within an episode, are discarded at the episode boundary, and never alter the saved training checkpoint. There is no confidence, entropy, or value-based gate: IR runs at every action from the first action ($k = 0$).

Conceptually, IR asks whether the retained model and critic contain enough goal-specific knowledge to repair a locally damaged actor without collecting gradient updates from the real reward stream. It is not hindsight relabeling [1]: replay goals and rewards are not relabeled.

2.5 Evaluation protocol and integrity

At 250k-step intervals, every goal encountered by that point was evaluated. For each checkpoint–goal cell we ran:

- 100 frozen and 100 IR episodes with sampled actions;
- 50 frozen and 50 IR episodes with deterministic actions.

Frozen and IR conditions used matching seed schedules and goal requests. Across 12 checkpoint–goal cells this yielded 3,600 episodes in 48 complete conditions (24 frozen/IR pairs). Requested episode counts, goal IDs, reset counts, return codes, and checkpoint parameter digests passed the evaluator’s integrity checks. Because frozen and IR conditions ran in separate RLScope process lifetimes, their initial image hashes were not identical despite matching seed/goal identities. The analysis therefore treats them as seed-paired aggregate comparisons, not guaranteed pixel-identical counterfactual trajectories.

The FIFO evaluation predates a fix to the strict frame-hash pairing checker: its queue recorded failure even though individual condition summaries completed. We use the complete first attempts at checkpoints through 1.5M steps descriptively. This caveat does not affect the reservoir run’s own completion status.

3 Study I results

3.1 Execution and systems outcome

The reservoir experiment completed all six training milestones and all 24 paired evaluation units without a training restart. Training took approximately 62.3 hours, evaluation 19.7 hours, and the full workflow approximately 82 hours. This satisfies the operational objective: a multi-day continual experiment can train, checkpoint, resume evaluation, and terminate cleanly.

At 1.5M steps the reservoir contained exactly 1,000 episodes per goal, 18,574 historical episodes had been considered for admission, and the retained episodes comprised 265,802 transitions. The latest sampling interval drew 655, 652, and 693 batches from kill, bury, and chop respectively, demonstrating near-uniform goal selection. Replay occupied 28.5, GiB of host memory and the full process 34.3, GiB, leaving 22.9, GiB available on the 62, GiB machine. Acting and training throughput were approximately 6.66 and 213 steps/s. The FIFO endpoint used 23.2, GiB for replay and 26.5, GiB for the process, with 6.92 acting steps/s and 222 training steps/s. The methods are therefore not memory-matched.

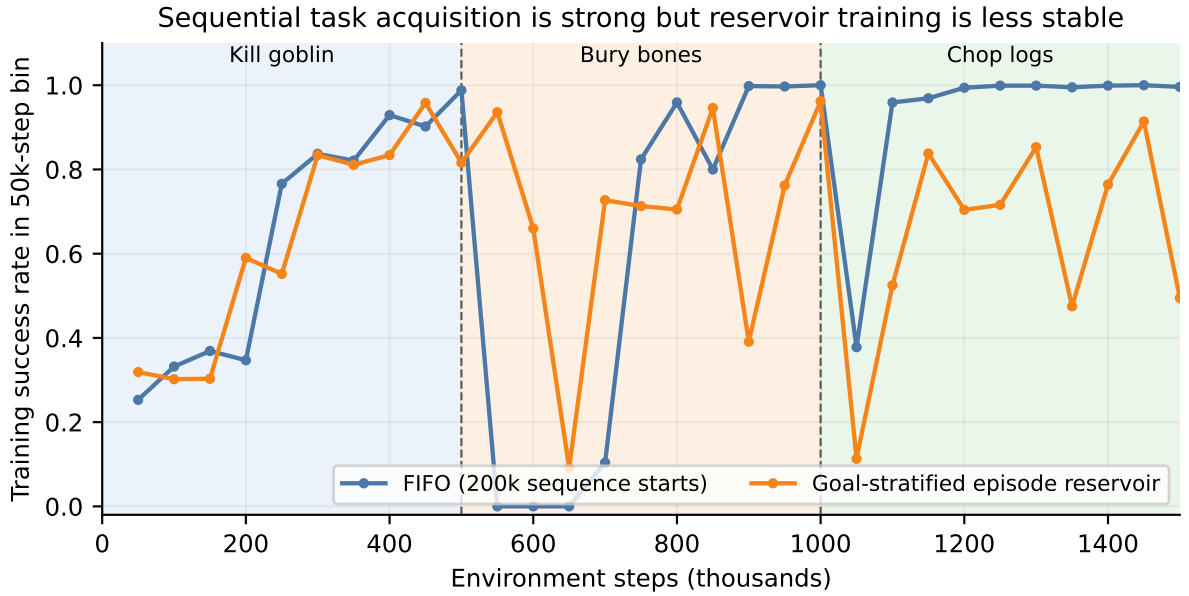


Figure 5: Training success in successive 50k-step bins. FIFO learns each active task cleanly after an initial delay. Reservoir replay reaches high success on every phase but exhibits repeated large collapses, including late in the final phase. These are behavioral training metrics, not held-out evaluation.

Reservoir training reached rolling 100-episode peaks of 99%, 100%, and 100% on kill, bury, and chop, proving that the architecture and action interface can solve every chosen goal. Final rolling success was only 85%, 54%, and 16%. In the last 10k steps of each phase, success was 84.2%, 37.5%, and 11.3%. The main failure is therefore retention/stability, not basic reachability.

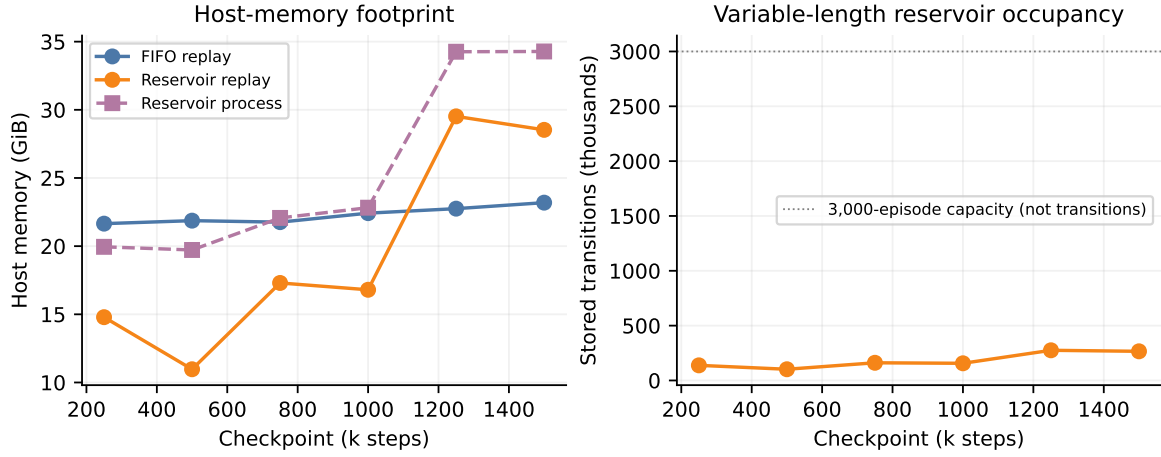


Figure 6: Resource use at pinned milestones. Episode capacity does not fix transition count because episode lengths vary. The 3,000-episode reservoir ended larger than the 200k-start FIFO buffer, making the comparison intentionally algorithmic rather than memory-matched.

3.2 Frozen-policy checkpoint matrix

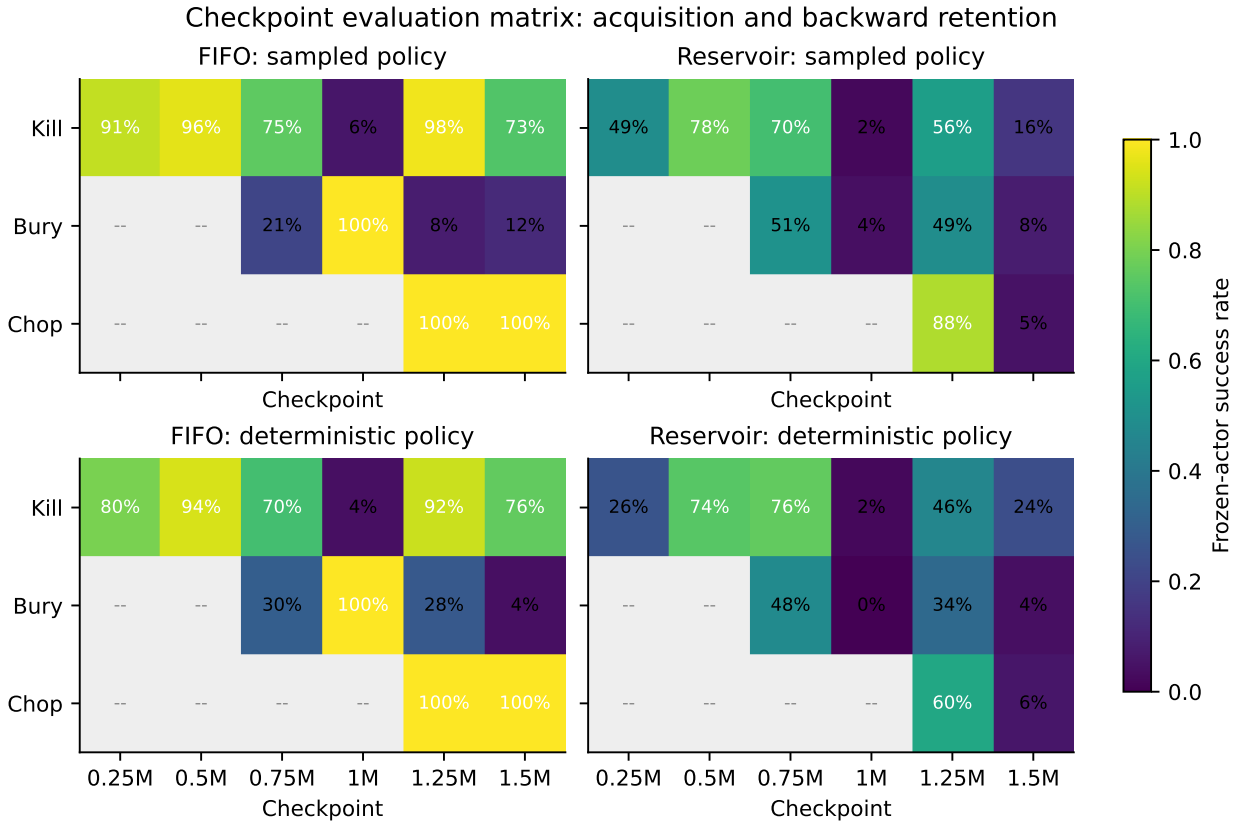


Figure 7: Frozen-actor success at all evaluable checkpoint-goal cells. Gray cells correspond to goals not yet introduced. The reservoir temporarily improves balanced retention but undergoes synchronized collapses at 1.0M and 1.5M steps.

Across all available cells, FIFO frozen success averaged 65.0% sampled and 64.8% deterministic. Reservoir frozen success averaged 39.7% and 33.3%, a substantial overall deficit. On acquisition checkpoints—the end of the phase in which a goal was introduced—reservoir trailed FIFO by 38.8 points sampled and 48.3 points deterministic. Goal-balanced replay is therefore not a free retention improvement: it materially reduces or destabilizes current-task learning under the present update schedule.

The result is not uniformly negative. At 750k steps, after only 250k bury-bones training, reservoir sampled success was 70% on kill and 51% on bury; FIFO achieved 75% and 21%. The reservoir’s worst task was therefore 30 points better. At 1.25M, after 250k chop training, reservoir simultaneously achieved 56%, 49%, and 88%; FIFO achieved 98%, 8%, and 100%. The reservoir’s worst task was 49% rather than 8%. This is the clearest positive evidence: stored old-goal episodes, goal-balanced sampling, and goal-conditioned heads can jointly recover multiple skills while a new goal is being learned.

That balance was transient. At 1.0M, reservoir kill and bury collapsed to 2% and 4%; at 1.5M all three goals were 16% or worse. FIFO showed selective forgetting instead: its final sampled scores were 73%, 12%, and 100%. FIFO’s retained mean was much higher, but its weakest old goal remained poor.

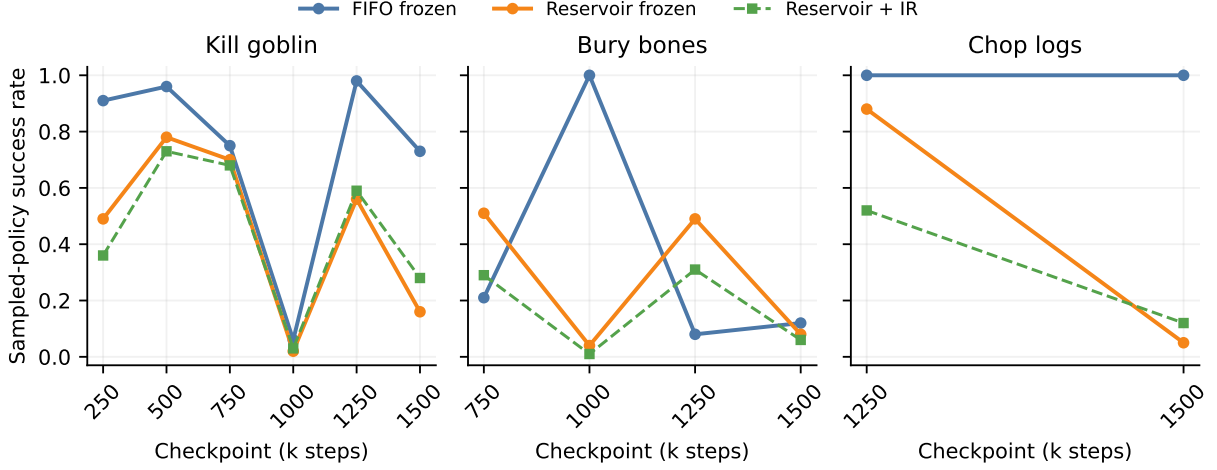


Figure 8: Sampled-policy trajectories for each goal. The dashed green line adds actor IR to the reservoir checkpoint. Reservoir skill recovery between 1.0M and 1.25M is followed by broad collapse between 1.25M and 1.5M.

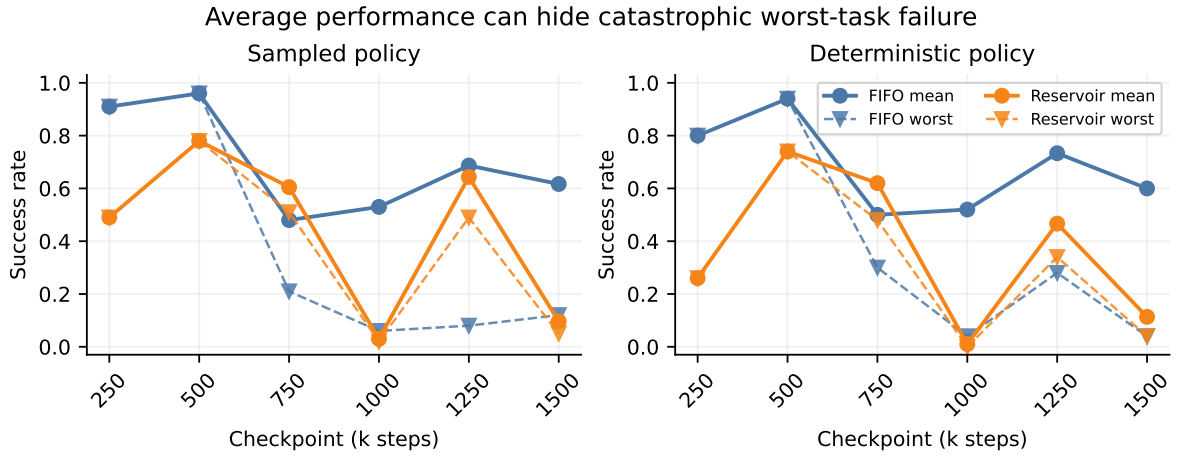


Figure 9: Mean and worst encountered-goal performance. Reservoir replay has useful mid-phase worst-task advantages, while both methods can hide severe forgetting behind a reasonable mean.

3.3 Forgetting

Forgetting is reported as the maximum checkpoint success for a goal minus its final-checkpoint success. In the sampled reservoir evaluations it was 62 points for kill, 43 for bury, and 83 for chop. FIFO forgetting was 25, 88, and 0 points. Deterministic reservoir forgetting was 52, 44, and 54 points, versus 18, 96, and 0 for FIFO. Reservoir failure was broad; FIFO failure was concentrated in bury bones.

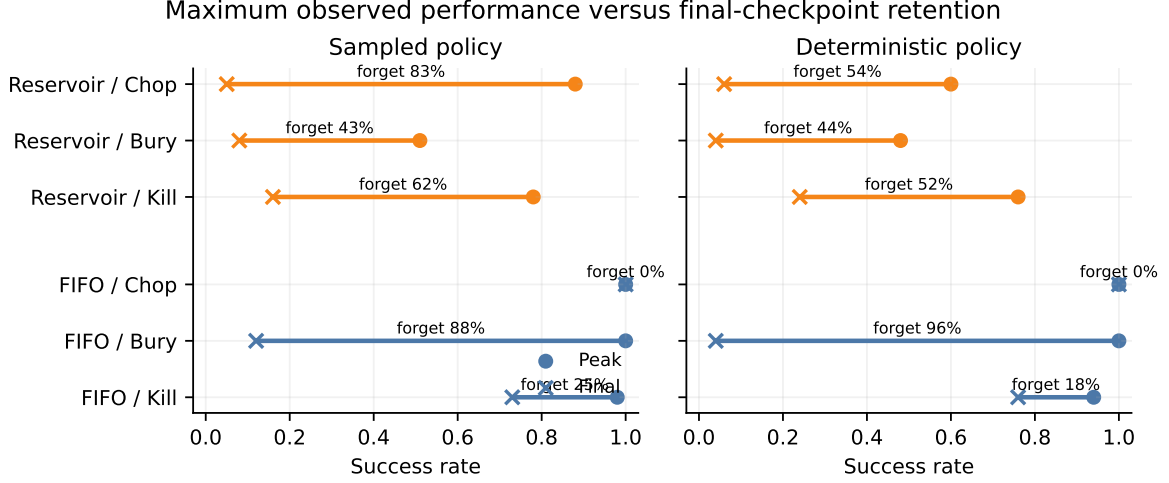


Figure 10: Peak-to-final forgetting. Circles mark the best observed checkpoint and crosses mark the final checkpoint. A useful continual learner should keep both markers high and close together.

3.4 Effect of iterative rehearsal

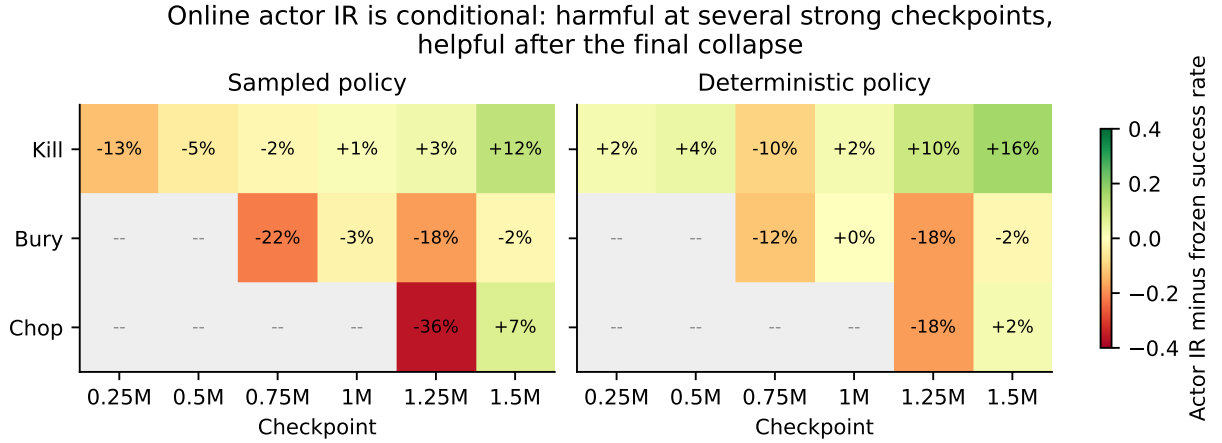


Figure 11: IR effect in the reservoir run. Positive values mean actor-only IR improved success over the frozen checkpoint. Strong harm is concentrated at several high-performing checkpoint-goal pairs; final-checkpoint effects are small but consistently positive for kill and chop.

Across 24 mode-specific comparisons, IR improved 10, tied 1, and harmed 13. Averaged over all reservoir checkpoint-goal cells, sampled success fell from 39.7% to 33.2% (−6.5 points), while deterministic success fell from 33.3% to 31.3% (−2.0 points). The strongest harms were sampled chop at 1.25M (−36 points), sampled bury at 750k (−22), and both sampled and deterministic bury at 1.25M (−18).

At the final checkpoint—the most directly relevant corrupted-policy case—the pattern reversed. Mean sampled success rose from 9.7% to 15.3%, and deterministic success from 11.3% to 16.7%. Sampled kill improved from 16% to 28% (paired bootstrap 95% interval for the 12-point change: [2, 23] points); chop improved from 5% to 12%; bury fell from 8% to 6%. Deterministic kill improved from 24% to 40%, although its 50-episode interval was wide ([−4, 36] points).

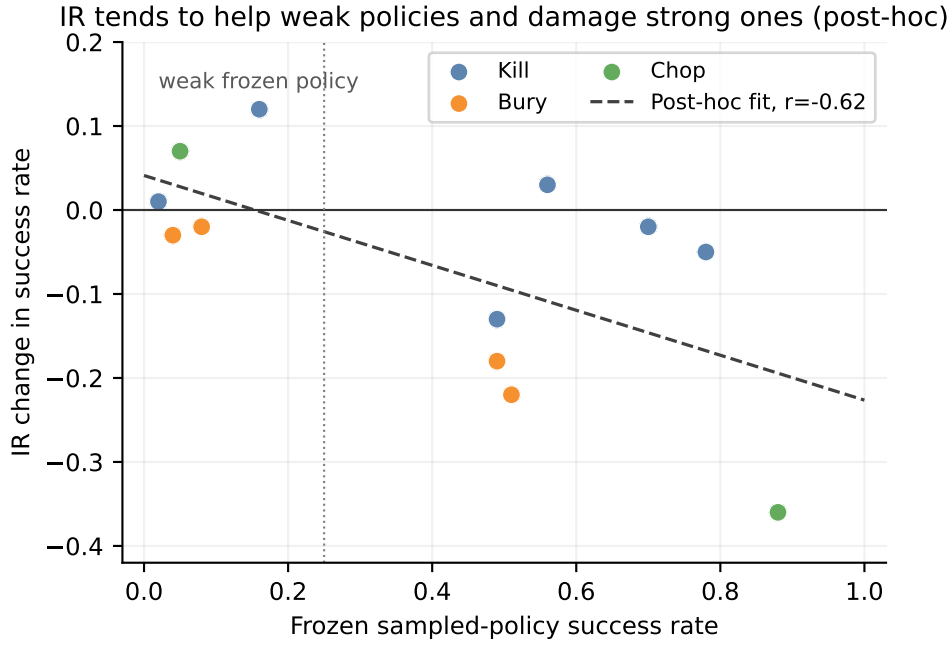


Figure 12: Post-hoc relationship between frozen policy strength and sampled-policy IR effect. The correlation is $r = -0.50$. For cells with frozen success at most 25% ($n = 10$), mean IR effect was +3.3 points and six improved; above 25% ($n = 14$), mean effect was -9.6 points. This threshold was selected after seeing the data and is a hypothesis for a future gate, not a confirmatory result.

The negative association supports the original intuition behind gated IR: adapt when the actor appears damaged, and avoid perturbing a competent policy. It does not validate entropy as the gate variable, because this experiment did not record or intervene on a prespecified entropy threshold. Entropy also remained in the IR objective and may have dominated short-horizon updates in some states.

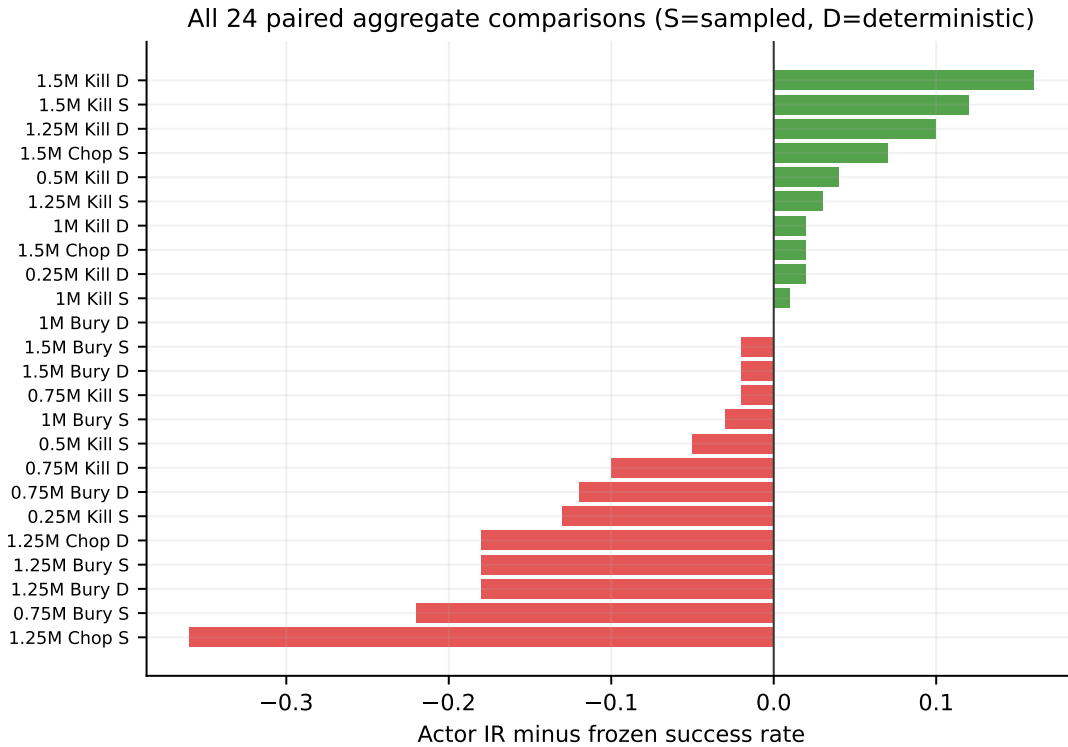


Figure 13: Every reservoir IR comparison, sorted by effect. S and D denote sampled and deterministic evaluation. The distribution is visibly asymmetric: a few large harms dominate several modest gains.

Table 2: Aggregate success rates. Values are unweighted means over available checkpoint–goal cells.

Replay	Policy mode	Frozen	Actor IR	IR change
FIFO	Sampled	0.650	0.620	−0.030
FIFO	Deterministic	0.648	0.623	−0.025
Reservoir	Sampled	0.397	0.332	−0.065
Reservoir	Deterministic	0.333	0.313	−0.020
Reservoir, final only	Sampled	0.097	0.153	+0.056
Reservoir, final only	Deterministic	0.113	0.167	+0.053

4 Study I interpretation

4.1 What reservoir replay did establish

The experiment rules out the claim that old-task data are wholly inaccessible under the new workflow. All goal strata were full, goal sampling was balanced, and earlier policies recovered sharply during chop training: from 1.0M to 1.25M, sampled kill rose from 2% to 56%, bury from 4% to 49%, while chop reached 88%. That combination would be very unlikely if replay groups, reset boundaries, goal IDs, or head conditioning were completely broken.

The experiment does *not* prove that the reward model and critic are well calibrated for every goal. Behavioral success is an end-to-end diagnostic. A model can contain enough signal for recovery while still having biased reward predictions, value misranking, or compounding imagination error. Direct head-calibration probes remain necessary.

4.2 Why balanced replay may still collapse

Several mechanisms are consistent with the observed synchronized swings:

1. **Optimization interference.** Shared encoder, RSSM, and large actor receive gradients induced by multiple visually similar but behaviorally distinct goals. Balanced data availability does not guarantee compatible gradients.
2. **Reduced current-task dose.** Once three strata exist, only about one third of replay batches target the active goal, while the nominal training ratio remains fixed. The reservoir condition is not update-dose matched to FIFO.
3. **Episode-uniform versus transition-uniform sampling.** Complete-episode selection changes the weight of short successful and long failed episodes relative to FIFO sequence-start sampling.
4. **Head/world-model drift.** The RSSM is goal-agnostic and shared. Even if every goal is represented in replay, representation changes can move the inputs seen by reward, value, and actor heads.
5. **Single-run stochasticity.** With one seed, a late policy collapse cannot yet be separated from an unlucky optimization trajectory.

The result therefore argues for measuring subsystem health, not merely increasing replay capacity.

4.3 Why ungated IR can hurt

IR trusts imagined returns. When the actor is already competent, one update at every real action creates many opportunities to move away from a successful action mode because of value error, model exploitation, posterior sampling noise, or the entropy term. A six-step horizon reduces compounding error but does not eliminate it. Conversely, a badly damaged actor has more room for improvement, which matches the final-checkpoint recovery and the post-hoc negative correlation in Figure 12.

The next IR study should therefore begin with a controlled actor-corruption experiment: choose a checkpoint with high task success and intact goal-conditioned heads, perturb only the actor to known severity levels, and measure recovery under frozen versus adapted actors. That design isolates whether IR can repair policy damage when its model and critic are known to be competent.

5 Study I limitations and threats to validity

- **One seed.** All uncertainty intervals quantify episode-level evaluation variability, not training-seed variability. They must not be interpreted as generalization across runs.
- **Replay conditions are not capacity matched.** FIFO capacity is counted in sequence starts; reservoir capacity is counted in complete episodes and consumed more host memory.
- **Optimization exposure differs.** Goal-balanced sampling necessarily decreases the current task’s fraction as goals accumulate. A fair algorithmic ablation may need either ratio compensation or a tunable current/old mixture.
- **Evaluation pairing is imperfect.** Seed and goal schedules match, but separate process lifetimes produced different initial frame hashes. Episode-order paired bootstrap intervals [2] are informative, not exact paired-environment causal estimates.
- **FIFO summaries are legacy artifacts.** The old evaluation supervisor marked the queue failed under a subsequently fixed frame-hash rule. Completed first-attempt summaries are retained for descriptive comparison.
- **No direct model calibration.** Reward-head accuracy, value calibration, imagined-return bias, and goal separability were not measured. Behavioral success alone cannot locate the failing subsystem.
- **Post-hoc gate analysis.** The 25% frozen-performance split and its correlation were discovered after observing outcomes. A future run must prespecify and validate any gate.
- **Environment scope.** The three chosen tasks use the same 504-way click grid and are solvable with left clicks. Results need not transfer to the two harder RLScape goals or to continuous robotic control.

6 Decision point after Study I

The checkpoint archive makes the following sequence possible without immediately repeating the multi-day run:

1. **Head audit at strong and collapsed checkpoints.** For a fixed image/latent set, measure goal-conditioned reward prediction, continuation, value ranking, and imagined-versus-real return. Include wrong-goal counterfactuals to quantify whether g changes the heads functionally.
2. **Actor corruption and recovery.** Start from the strongest checkpoint for each goal; apply controlled actor noise or partial reinitialization; compare frozen, ungated IR, IR without entropy, and IR with smaller learning rates/horizons.
3. **Prespecified IR gate.** Gate on a measurable signal available at test time—for example ensemble disagreement, low predicted value relative to a checkpoint baseline, or sustained high entropy. Entropy alone should be treated as one candidate, not an assumption.
4. **Replay-mixture ablation.** Keep Algorithm R retention but compare uniform goal sampling with a current-goal-biased mixture, such as 50% current and 50% uniformly old. Match total update dose where possible.
5. **Stability replication.** Repeat the best replay/IR configuration over multiple seeds, then extend from three to all five RLScape goals.
6. **Return phase.** Finally revisit A after $A \rightarrow B \rightarrow C$ to quantify relearning speed, forward/backward transfer, and whether IR reduces real-environment samples.

Hindsight relabeling remains a possible later extension for sparse rewards, but it is not the first diagnosis: all three tasks already reached near-perfect training windows, so the immediate bottleneck is maintaining and exploiting learned competence.

7 Interim conclusion after Study I

The reservoir implementation worked, the long-running workflow was operationally robust, and every task was individually learnable inside the sequential run. Yet goal-stratified reservoir replay did not solve continual learning by itself. It produced compelling but temporary balanced retention, then ended in a synchronized behavioral collapse substantially worse than FIFO’s final mean. Actor-only IR was also not uniformly beneficial: fixed every-action updates often damaged strong actors, while providing modest recovery at the final weak checkpoint.

The most useful outcome is therefore diagnostic. Old experience and goal-conditioned knowledge can support recovery, but neither balanced replay nor ungated test-time adaptation guarantees stability. The next experiments should exploit the saved checkpoints to separate actor damage from reward/value/world-model failure and to design an IR gate before spending another multi-day training budget.

8 Study II: locating the failure inside the learned IR target

Study I left an ambiguity that could not be resolved from behavioral success alone. A weak actor may fail because the policy parameters are damaged while the world model, reward head, continuation head, and critic remain useful. Alternatively, the actor may be reacting rationally to an inaccurate learned objective. This distinction is decisive for IR: test-time actor optimization is only useful when the frozen components supply a trustworthy teaching signal.

Study II therefore compares the **strong 1.25M checkpoint** with the **collapsed 1.5M checkpoint**. At 1.25M, sampled-policy success was 56%/49%/88% on kill/bury/chop. At 1.5M, every task was at most 16%, with a mean of 9.7%. The analysis has two linked parts. Study IIa evaluates the learned heads on real replay trajectories. Study IIb executes the exact actor-IR imagination path, without updating parameters, to measure where its return and gradient signals originate.

8.1 Study IIa method: immutable same-state head audit

The state bank was drawn from the pinned 1.25M replay archive, not recollected separately under each checkpoint. For every trained goal it contains 64 successful and 64 failed episodes, yielding 384 episodes and 48,299 states. Both checkpoints therefore receive the same image sequences, actions, reset boundaries, rewards, continuations, and factual goal labels. This design removes changes in environment sampling from the strong–weak comparison.

Each sequence is encoded under each checkpoint to obtain its posterior latent trajectory. At every state, the reward, continuation, online value, and slow-value heads are queried under all five one-hot goal IDs: the three trained goals and the unused *light fire* and *catch fish* slots. “Factual” means the goal used when that replay episode was collected. “Counterfactual” means that the latent state and every other input are held fixed while only the one-hot goal is changed.

For reward and continuation, the most important counterfactual query is a completion state for goal i evaluated under another goal $j \neq i$. The desired targets are

$$r(s^{\text{complete}(i)}, g_i) = 1, \quad c(s^{\text{complete}(i)}, g_i) = 0, \quad (9)$$

$$r(s^{\text{complete}(i)}, g_j) = 0, \quad c(s^{\text{complete}(i)}, g_j) = 1 \quad (j \neq i). \quad (10)$$

The second line was never supplied by ordinary on-policy replay: every observed completion appears only with its active goal. The audit measures whether one-hot conditioning nevertheless induced that semantic distinction.

Value calibration compares predicted values with recorded discounted returns along the replay trajectory. This is a diagnostic rather than an exact reconstruction of Dreamer’s critic loss: replay actions are off-policy relative to each audited checkpoint, whereas Dreamer trains lambda returns under imagined policy rollouts. The comparison remains valuable because both checkpoints are evaluated on identical states, empirical returns remain in $[0, 1]$, and the strong-to-weak direction can be compared directly. Parameter digests before and after both audits are identical.

8.2 Study IIa results: event recognition without goal semantics

Factual reward prediction remains excellent. Reward AUROC is 0.99993 at 1.25M and 0.99991 at 1.5M. Mean reward prediction on genuine completion states is 0.961 and 0.957, while ordinary-state false positives remain very small. Thus replay retention preserved a highly accurate detector of visible completion-like events through the behavioral collapse.

The reward head scarcely uses the requested goal. On the same completion states, the factual-minus-counterfactual reward margin is only 0.0033 at 1.25M and 0.0018 at 1.5M. Choosing the goal with the largest reward prediction identifies the correct goal only 39.6% and 38.0% of the time despite five goal slots. The head has learned “a task completed” much more strongly than “the requested task completed.”

Continuation has the same semantic failure. The continuation head accurately predicts stopping on factual completion states and continuing on ordinary states. Yet mean stop probability on wrong-goal completion queries is already 93.9%/94.3% at the strong/weak checkpoint, compared with only 95.6%/95.9% for the matching goal. Reward and continuation are therefore separate heads but share the same missing counterfactual supervision.

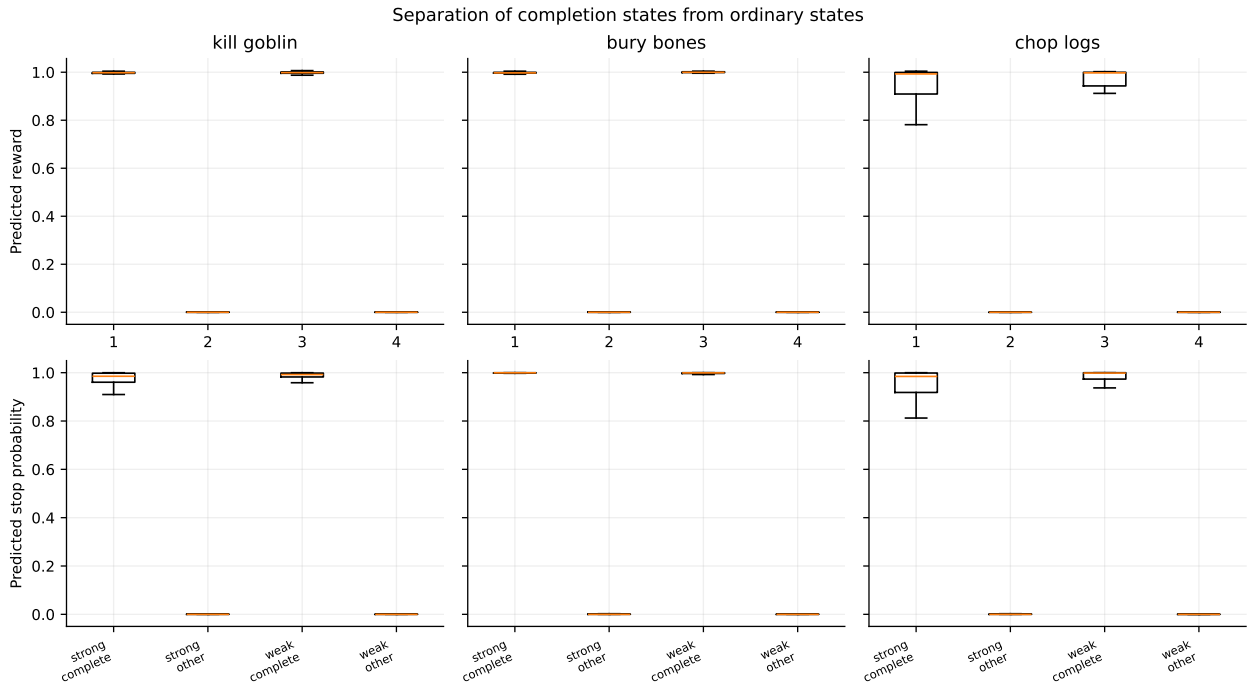


Figure 14: Study IIa factual event recognition. Reward (top) and stopping (bottom) almost perfectly separate completion states from ordinary states at both checkpoints. This figure is deliberately paired with the counterfactual analysis: factual accuracy alone can hide the fact that the same completion state receives almost the same output under every requested goal.

The critic is severely overoptimistic and worsens after collapse. At 1.25M, mean factual critic predictions are 1.62, 1.73, and 0.79 for kill, bury, and chop, whereas empirical discounted-return means are 0.20, 0.10, and 0.21. At 1.5M, values rise further to 1.99, 2.00, and 1.36 even though real policy success collapses. Value-return Pearson correlations deteriorate from 0.182/0.107/0.307 to 0.048/0.064/0.061. At episode starts, value barely separates successful from failed kill trajectories, reverses their ordering for bury, and weakens for chop.

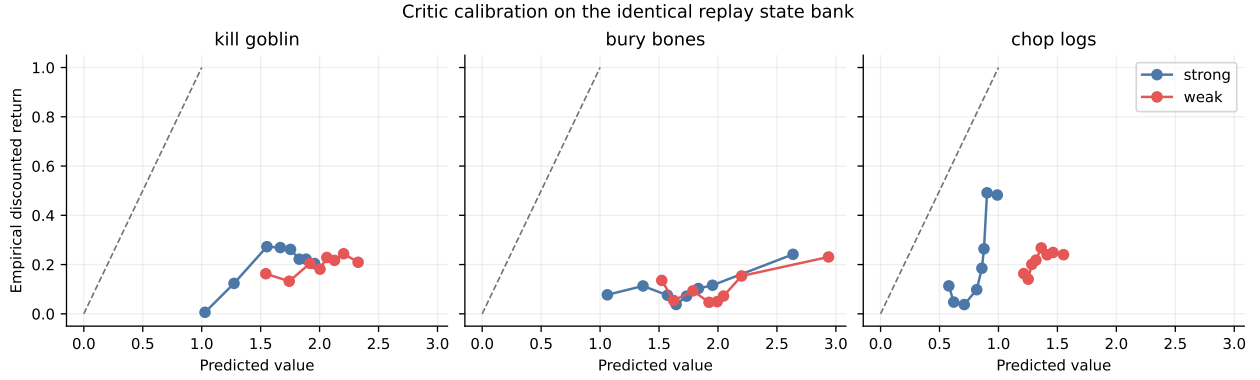


Figure 15: Study IIa critic calibration on the identical replay-state bank. The dashed line is ideal calibration. Predicted values are far above recorded discounted returns, and the weak checkpoint (red) is generally more optimistic despite much worse real behavior.

The slow critic does not provide an independent safety signal. Its mean absolute disagreement with the online critic is only 0.006–0.017; both share the same optimism and drift. Posterior categorical entropy changes only from 0.564 to 0.543 and deterministic-state RMS from 0.256 to 0.262. These scalar summaries do not rule out task-relevant representation drift, but they rule out a gross posterior entropy or magnitude collapse as the immediate explanation.

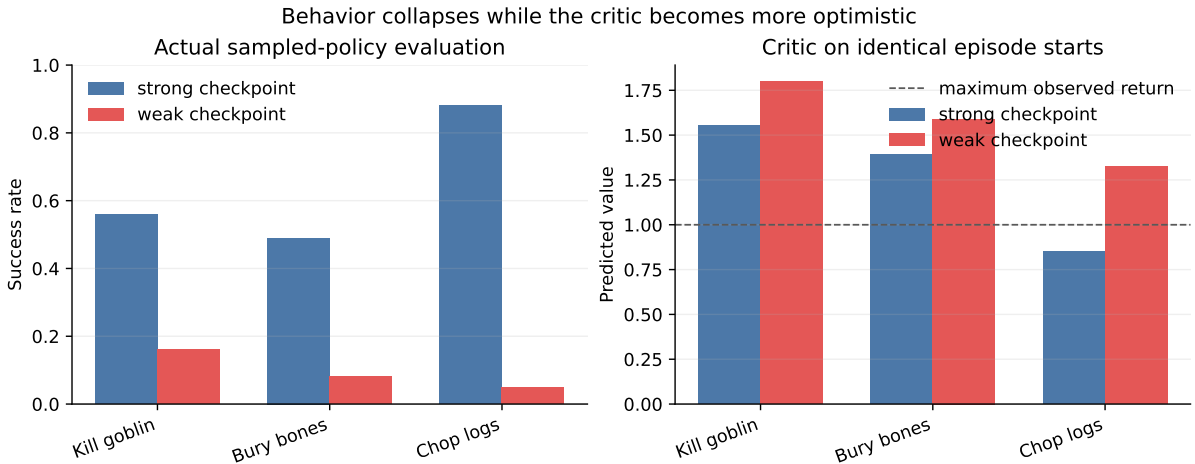


Figure 16: Behavior and critic confidence move in opposite directions between the strong and weak checkpoints. This inverse relationship motivated an audit of the exact imagined return used by actor IR rather than treating critic output as a valid gate.

8.3 Study IIb method: exact six-step imagination-path audit

Study IIb reuses the same 384-episode selection. Four deterministic, nonterminal anchors are requested from each episode; short trajectories yield 1,433 valid anchors. At every anchor, the agent reconstructs the posterior latent state and sweeps all five goal IDs. For each anchor–goal pair, it draws 128 paired posterior samples and executes the same $H = 6$ policy imagination used by Study I actor IR. In total, each checkpoint produces 7,165 anchor–goal rollouts and 5,502,720 imagined action transitions.

No optimizer update is applied. Instead, the audit records, at each imagined time step:

- predicted reward and the incidence of reward-like events;
- continuation and predicted stopping events;
- online and slow critic values;
- lambda return, split into reward-only and critic-bootstrap contributions;

- actor reinforce/advantage term, entropy term, and policy entropy;
- actual replay goal, probed goal, anchor location, and success/failure stratum.

Randomness is paired across goal probes so differences are attributable to the one-hot goal rather than different posterior samples. Parameter digests again match before and after the audit.

For clarity, write the imagined truncated return abstractly as

$$\hat{G}_t^{\lambda,H} = \underbrace{\sum_{k=0}^{H-1} w_{t,k} \hat{r}_{t+k}}_{\text{reward contribution}} + \underbrace{b_{t,H} v_{\bar{\psi}}(\hat{s}_{t+H}, g)}_{\text{critic bootstrap}}, \quad (11)$$

where the actual implementation uses Dreamer’s lambda-return weighting and continuation discounts. The audit reconstructs both contributions rather than inferring them from the total.

8.4 Study IIb results: IR optimizes the critic, not imagined reward

Critic bootstrap dominates the IR target. At the strong checkpoint, mean factual lambda return is 1.450: only 0.072 is attributable to imagined rewards and 1.378 to bootstrap, a 94.4% bootstrap fraction. At the weak checkpoint, mean return rises to 1.868 despite behavioral collapse: reward contribution falls to 0.039 while bootstrap rises to 1.829, or 97.3%. Ordinary anchors are more than 99% bootstrap driven. Even immediately before recorded completion, bootstrap still contributes 69.1% at the strong checkpoint and 82.4% at the weak checkpoint.

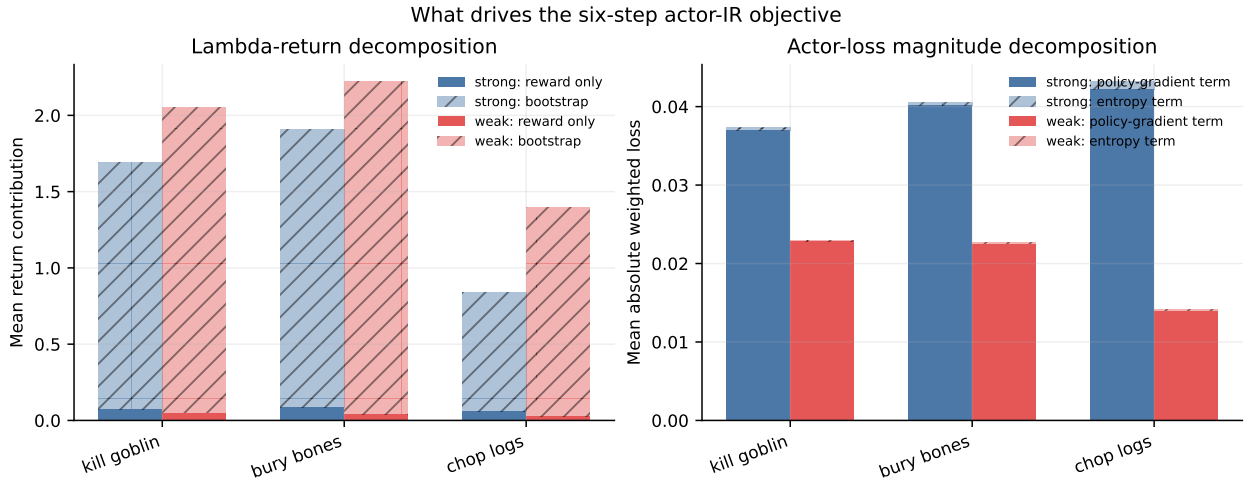


Figure 17: Study IIb decomposition of the exact six-step actor-IR objective. Left: lambda return is almost entirely critic bootstrap, and bootstrap grows at the weak checkpoint while imagined reward shrinks. Right: entropy is small relative to the policy-gradient term in measured scalar loss magnitude.

The weak model imagines fewer successes but larger returns. Factual imagined reward-event incidence falls from 6.76% to 3.52% of MCPB posterior samples, and reward-only return falls from 0.072 to 0.039. Nevertheless, bootstrap increases by 0.451 and total return increases by 0.419. Factual starting value rises from 1.315 to 1.668. The optimism observed on replay states is therefore present on the exact IR path and overwhelms the direction of the reward evidence.

Goal identity scarcely changes the teaching signal. Strong-checkpoint factual and counterfactual imagined returns are 1.450 and 1.447. At the weak checkpoint they are 1.868 and 1.865. On pre-completion anchors, factual versus wrong-goal reward-event rates are 34.0% versus 33.5% at 1.25M and 21.2% versus 20.8% at 1.5M. Requested-goal top-1 identification from imagined return degrades from 52.8% to 41.9%, and the factual-minus-wrong-goal mean margin is only 0.0029/0.0032.

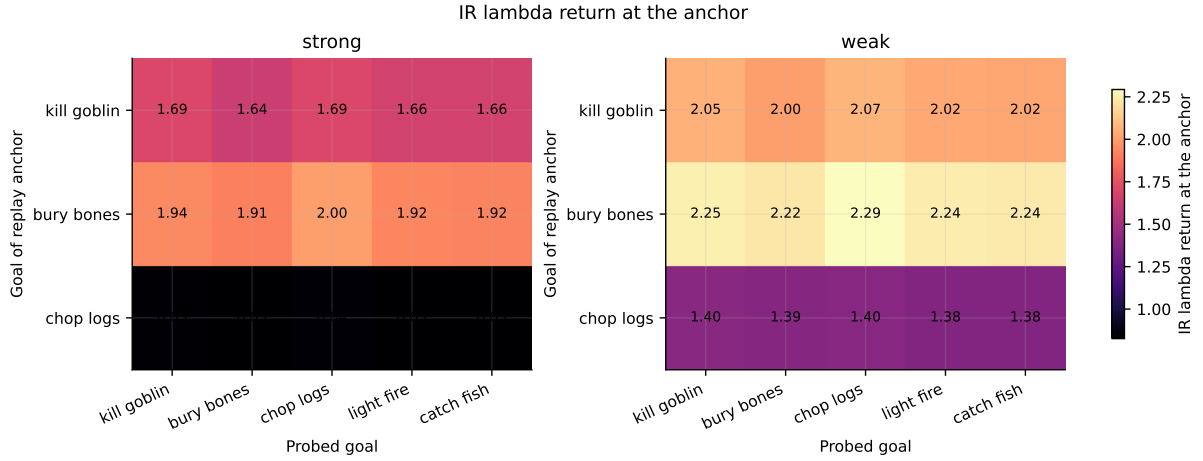


Figure 18: Mean IR lambda return by replay anchor goal (rows) and requested one-hot goal (columns). Row identity dominates column identity: the source trajectory matters far more than the goal requested for rehearsal. The weak checkpoint is uniformly more optimistic.

Reward and stopping events remain meaningful visual detectors: on successful trajectories, factual imagined reward-event incidence is 12.5% (strong) and 6.7% (weak), versus 1.8% and 0.7% on failed trajectories. Events concentrate near factual pre-completion states. The defect is not absence of all model signal; it is the missing same-state, wrong-goal distinction.

Entropy is not the dominant scalar term. Mean absolute factual policy-gradient loss is 0.0398 at 1.25M and 0.0196 at 1.5M. The corresponding entropy terms are 0.000552 and 0.000180, approximately 1.4% and 0.9% of those aggregate magnitudes; mean per-rollout ratios are 2.2% and 1.6%. These are scalar loss comparisons rather than gradient-norm comparisons, so an entropy ablation remains justified. They do not, however, support entropy as the primary explanation.

The actor itself becomes much more deterministic after collapse: factual policy entropy falls from 1.915 to 0.628. Simultaneously, the mean absolute policy-gradient term approximately halves. In other words, the damaged policy has less exploratory diversity while the learned objective intended to repair it becomes less informative and more bootstrap dominated.

8.5 Study II interpretation and resulting prediction

The two audits form a consistent mechanism. Reservoir replay preserved factual examples, so reward and continuation learned excellent visual event detection. It did not create the counterfactual labels in Equation 10; those heads can minimize factual loss while almost ignoring g . The critic then bootstraps on its own predictions through imagined states and develops values that are both too large and increasingly detached from realized success. Standard actor IR mostly optimizes this critic bootstrap.

This does not by itself prove the ultimate origin of critic optimism. Plausible contributors include self-reinforcing bootstrap targets, distribution shift into policy-imagined states, exploitation of model errors, weak goal selectivity, and continual drift in shared representations. What Study II establishes more narrowly is that *the critic is not a trustworthy teacher for the next actor-recovery experiment*. It therefore made a falsifiable prediction: if actor adaptation is mechanically viable, removing critic bootstrap should outperform standard IR; removing entropy alone should not.

9 Study III: controlled actor corruption and recovery

9.1 Question and experimental controls

Study III asks whether actor-only imagined adaptation can recover a deliberately damaged policy when every non-actor component is held fixed. It is a mechanism experiment, not another continual-training run. The source is the pinned 1.25M checkpoint, chosen because all three goals were simultaneously competent and Study II had already characterized its heads and imagination path.

The experiment applies an independent zero mask to actor parameters only. Encoder, RSSM, decoder, reward, continuation, online value, slow value, normalizers, and their optimizer states remain byte-identical to the source. Six mask probabilities, $p \in \{0.1, 0.2, 0.3, 0.4, 0.5, 0.6\}$, are calibrated on 20 sampled episodes per goal using paired reset seeds. The selection target is 35% aggregate retention of clean success, accepted only within 10%–65%, with no goal retaining more than 85% and a penalty for uneven or beneficial corruption.

The selected 40% zero mask affects all 11 actor parameter arrays (3,409,912 parameters) and has actor-relative L_2 distance 0.637 from the source, close to the $\sqrt{0.4}$ scale expected for independent masking. In calibration it retains 35.1% of aggregate clean success: per-goal retention is 50.0%, 20.0%, and 35.3% for kill, bury, and chop. The independent main seed schedule produces a less severe but still substantial sampled decline, from 56.3% clean mean to 28.3% corrupted mean.

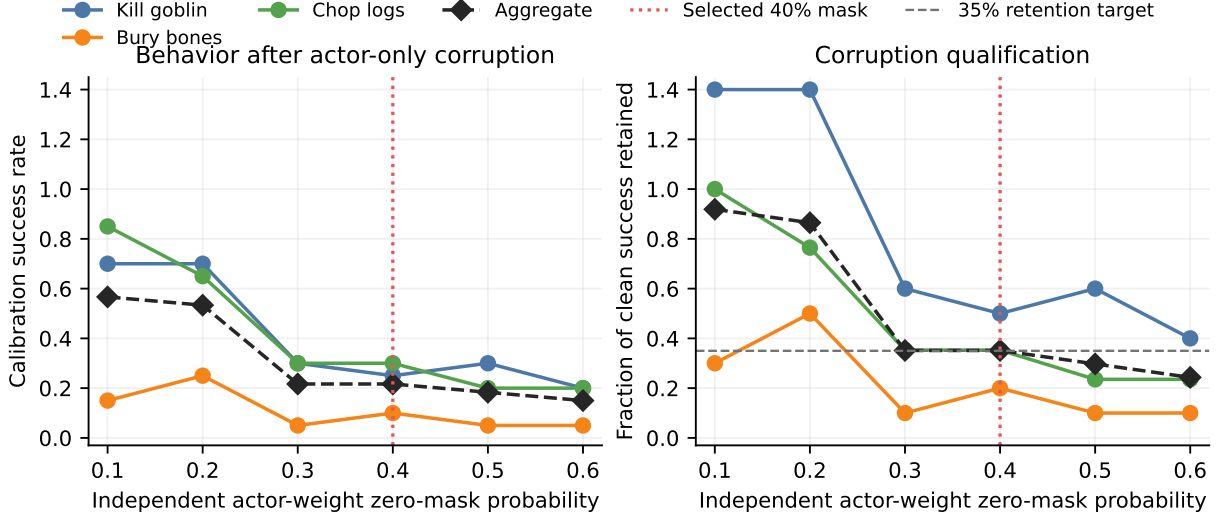


Figure 19: Study III corruption calibration. Six actor-only mask strengths were evaluated before the main assay. The 40% mask is the qualified candidate closest to the prespecified 35% aggregate-retention target while avoiding near-intact individual goals.

Every evaluation episode begins from the same persistent checkpoint parameters. Temporary actor parameters adapt from the first real posterior state and then once per environment action; they accumulate within the episode and are discarded at reset. Each update uses 128 posterior samples, learning rate 4×10^{-5} , and one optimizer step. The world model and both critics are frozen in every arm. Horizons are $H \in \{3, 6, 15, 20\}$.

9.2 Recovery objectives

Fifteen conditions separate three possible causes of failure.

No-adaptation controls. The clean and corrupted actors establish the attainable source behavior and the size of the induced deficit under the same main evaluation schedule.

Standard IR. This is the learned Dreamer actor objective used in Study I: policy gradients from normalized imagined lambda returns, including value bootstrap, plus entropy coefficient 3×10^{-4} . It is tested at all four horizons.

Entropy-off standard IR. This arm keeps the standard imagined return and critic bootstrap but sets the entropy coefficient to zero. It directly tests whether entropy, rather than the return target, caused harmful updates.

Reward-only IR. This arm uses predicted reward and continuation along the imagined rollout but sets terminal value bootstrap and entropy to zero. Abstractly,

$$J_{\text{RO}}(\theta; s, g) = \mathbb{E}_{p_\phi, \pi_\theta} \left[\sum_{k=0}^{H-1} \left(\prod_{j < k} \hat{c}_{t+j} \right) \gamma^k \hat{r}_{t+k} \right], \quad (12)$$

with the implementation retaining Dreamer’s differentiable policy-gradient machinery and giving posterior samples whose imagined trajectories contain zero reward no policy-gradient signal. This is a diagnostic critic-free objective, not an oracle: rewards and continuations are still learned model predictions.

Clean-reference distillation. The frozen clean actor supplies its full categorical action distribution on the current real posterior state. The temporary corrupted actor minimizes cross-entropy to that teacher. This horizon-independent positive control tests whether the corruption can be reversed under the same episode-local actor optimizer without relying on imagined returns.

9.3 Evaluation scale and integrity

Every condition is evaluated on all three goals with 100 sampled and 50 deterministic episodes. The main matrix therefore contains

$$15 \text{ conditions} \times 3 \text{ goals} \times (100 + 50) = 6,750 \text{ real-environment episodes.} \quad (13)$$

Calibration contributes 420 further episodes. In runner terms, all 21 calibration units and all 90 main units completed, giving 111 complete units with no failed attempt or retry. Goal, policy mode, and reset-seed identities match across conditions; the pairing audit reports no mismatch. Requested episode counts, reset counts, return codes, and persistent parameter-integrity digests all pass.

9.4 Study III results

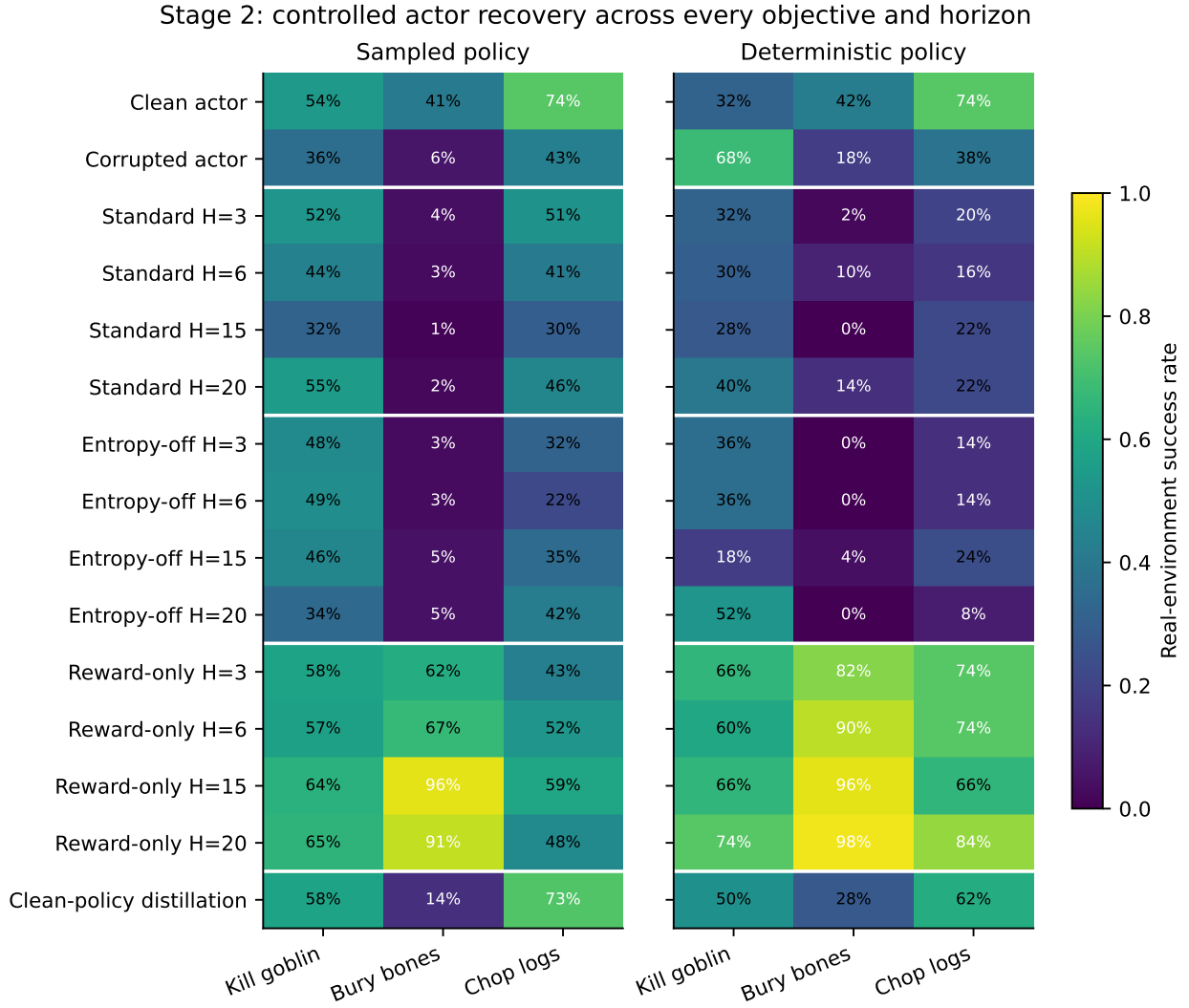


Figure 20: Complete Study III result matrix. Each cell is real-environment task success, not predicted reward. Standard and entropy-off IR remain weak, while reward-only IR recovers the actor and often exceeds the clean source behavior.

Table 3: Aggregate Study III success across all three goals. Sampled totals use 300 episodes per row; deterministic totals use 150. “Best” selects the horizon separately within each objective family and policy mode.

Condition	Sampled mean	Deterministic mean	Interpretation
Clean actor	.563	.493	Source checkpoint
Corrupted actor	.283	.413	Actor-only deficit
Standard IR, best	.357 ($H = 3$)	.253 ($H = 20$)	Weak sampled; destructive deterministic
Entropy-off IR, best	.287 ($H = 15$)	.200 ($H = 20$)	Removing entropy does not repair IR
Reward-only IR, $H = 3$	.543	.740	Near/full aggregate recovery
Reward-only IR, $H = 6$	.587	.747	Exceeds clean mean
Reward-only IR, $H = 15$	.730	.760	Best sampled result
Reward-only IR, $H = 20$	.680	.853	Best deterministic result
Clean-policy distillation	.483	.467	Partial positive-control recovery

Standard IR is narrow and unstable. At $H = 3$, sampled kill improves from 36% to 52% and chop from 43% to 51%, but bury remains near zero. Aggregate sampled success rises only to 35.7%. At $H = 6$ it falls to 29.3%; at $H = 15$ it drops below the corrupted baseline to 21.0%; at $H = 20$ it rebounds to 34.3%. Deterministic

success is damaged at every horizon, ranging from 16.7% to 25.3% versus 41.3% corrupted. Standard IR can discover a useful local direction for one policy/task combination, but its teaching signal is not robust.

Entropy removal does not solve the failure. Entropy-off sampled means remain between 24.7% and 28.7%; deterministic means remain between 15.3% and 20.0%. This agrees with Study IIB’s objective decomposition. Entropy may influence individual decisions, but it is neither necessary nor sufficient for the broad harm.

Removing critic bootstrap changes the outcome decisively. Reward-only IR reaches sampled means of 54.3%, 58.7%, 73.0%, and 68.0% as H increases from 3 to 20. Deterministic means are 74.0%, 74.7%, 76.0%, and 85.3%. The result is not merely recovery to the clean actor: the best reward-only conditions exceed it by 16.7 sampled points and 36.0 deterministic points.

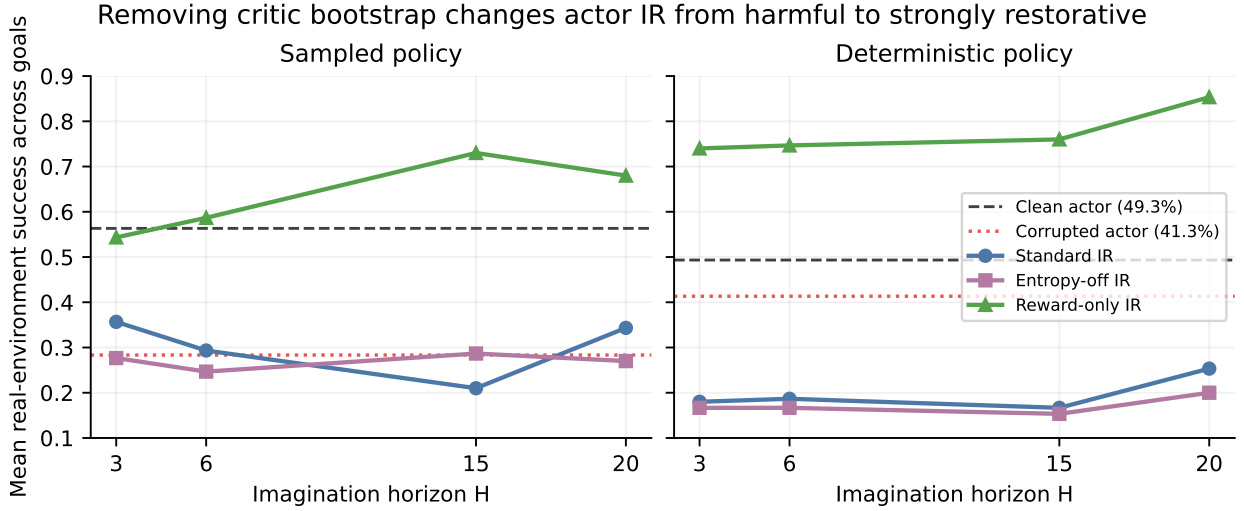


Figure 21: Aggregate horizon response. Standard and entropy-off IR track the corrupted baseline or fall below it. Reward-only IR crosses the clean-actor line and generally benefits from longer imagination. This is the central causal ablation of the chapter.

The per-goal pattern is informative. At sampled $H = 15$, reward-only IR obtains 64% kill, 96% bury, and 59% chop. Relative to paired corrupted episodes, changes are +28 points (95% interval [+15, +40]), +90 [+83, +96], and +16 [+2, +30]. All exclude zero. At $H = 20$, corresponding changes are +29 [+17, +41], +85 [+77, +92], and +5 [−8, +19]. Thus $H = 15$ is the most consistent sampled horizon; $H = 20$ is strongest under deterministic action selection.

Reward-only rehearsal recovers different skills at different horizon scales

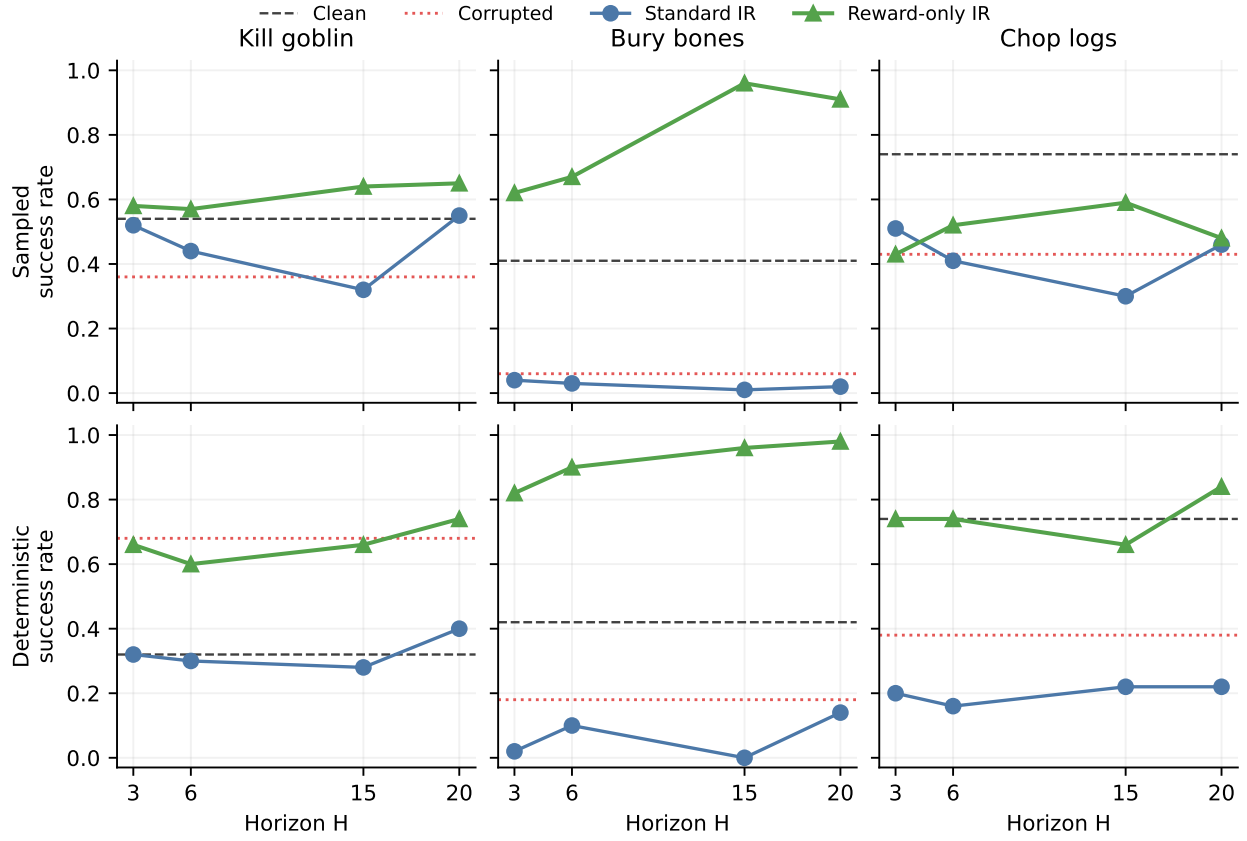


Figure 22: Goal-specific horizon response. Bury bones exhibits dramatic reward-only recovery, kill improves consistently under sampled control, and chop peaks earlier or weakens at sampled $H = 20$. There is no single horizon that dominates every task and policy mode.

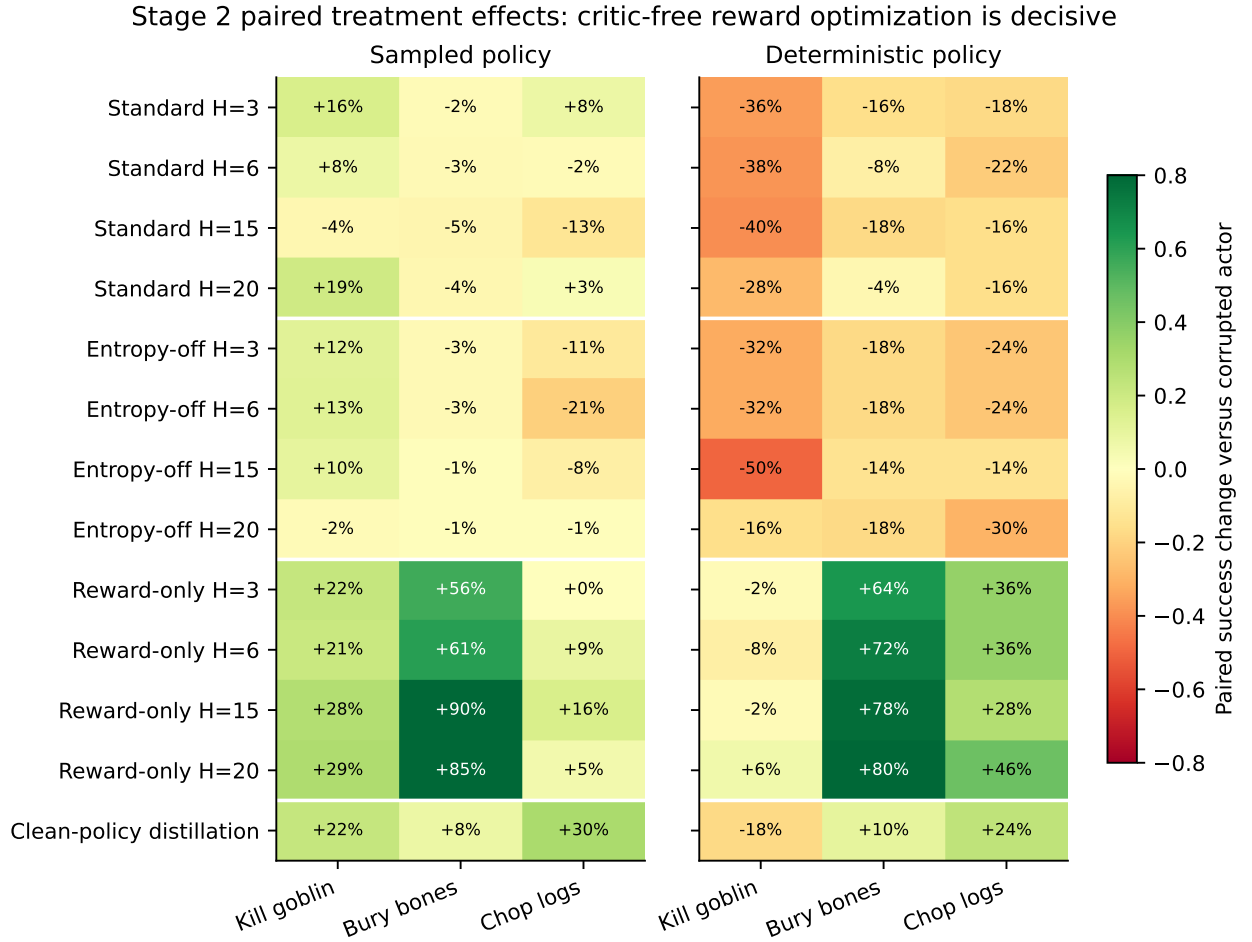


Figure 23: Paired changes relative to the corrupted actor. The green reward-only block contrasts sharply with standard and entropy-off objectives. Deterministic kill requires special caution because actor corruption itself improved its raw deterministic success from 32% clean to 68% corrupted.

Reference distillation provides partial mechanical recovery. Distillation raises sampled aggregate success from 28.3% to 48.3%: kill rises 36%→58%, bury 6%→14%, and chop 43%→73%. It does not exactly recreate the source policy within the limited episode-local updates, especially on bury, but it confirms that the optimizer can move the corrupted actor toward a trusted target. Reward-only IR’s much stronger bury result is therefore not explained by simple cloning of the original actor; it constitutes online policy improvement under the learned reward model.

9.5 Study III interpretation

Study III supports the prediction made before the run. The actor-adaptation machinery is functional; entropy is not the main obstacle; and critic bootstrap is the principal harmful component of the present IR objective. This is a stronger conclusion than a correlation between critic error and behavioral collapse because the assay intervenes on the objective while holding the checkpoint, corruption, posterior samples, optimizer schedule, environment seeds, and frozen parameters fixed.

The reward-only result also changes the interpretation of the learned world model. Study II showed that reward and continuation are not counterfactually goal selective at completion states, yet they still contain enough trajectory-level event structure to guide an actor to genuine real-environment success. In this arena, different goals require different visual/action paths even if the final event detector is shared. A goal-conditioned actor can exploit those reward gradients from a goal-specific starting context. This explains how reward-only IR can work without excusing the semantic defect: in a denser or more ambiguous task landscape, rewarding any completion-like state could become direct cross-goal interference.

Performance above the clean actor should be called *test-time policy improvement*, not memory recovery alone. It is encouraging because success is measured in RLScope rather than by predicted return. It still warrants tra-

jectory inspection: the model may have discovered a narrow but valid strategy, and horizon-specific degradation on chop suggests that compounding model error remains relevant.

10 Study IV: critic-free recovery from natural forgetting

10.1 Question and prospective design

Study III establishes a controlled possibility result: a retained world model can teach a synthetically damaged actor when the critic is removed from the objective. Study IV (canonical Stage 3B) asks the necessary external-validity question: does the same objective repair behavior lost *naturally* during sequential training? No actor corruption is applied. The inputs are the immutable 1.25M checkpoint, at which the three behaviors were relatively strong, and the naturally collapsed 1.5M checkpoint. The experiment was specified before launching any child evaluation and contains five conditions:

1. frozen checkpoint actor;
2. standard critic-bootstrapped IR at the historical $H = 6$ setting;
3. reward-only IR at $H \in \{6, 15, 20\}$.

All adapted conditions start from the unmodified checkpoint actor. Actor updates begin at the first real posterior state, occur once per environment action, accumulate within an episode, and are discarded at reset. Each update uses 128 Monte Carlo posterior samples and learning rate 4×10^{-5} . Standard IR retains the historical entropy coefficient 3×10^{-4} and Dreamer lambda-return bootstrap. Reward-only IR uses Equation 12: entropy and terminal critic bootstrap are zero, while the learned RSSM, reward, and continuation remain active. Encoder, RSSM, decoder, reward, continuation, online value, slow value, replay, normalizers, and persistent checkpoint state are frozen.

Every checkpoint-condition-goal cell is evaluated with 100 sampled and 50 deterministic episodes. The complete design therefore contains

$$2 \text{ checkpoints} \times 5 \text{ conditions} \times 3 \text{ goals} \times (100 + 50) = 4,500 \text{ episodes.} \quad (14)$$

Reset seeds are paired across the five conditions within checkpoint, goal, and policy mode. The primary outcome is real-environment success; secondary outcomes include paired discordance, actor update magnitude, action count, and imagined-compute dose.

Before the new evaluation, cells were labelled *weak* when their historical Stage I sampled success was below 50%. This labels 1.25M bury (historical 49%) and all three 1.5M goals as weak; 1.25M kill (56%) and chop (88%) are competent. The label is an oracle mechanism-analysis stratum, not a proposed deployment gate. The distinction matters because the independent Stage IV frozen schedule measured 1.25M bury at 71% sampled: the precommitted label remains historical, while all treatment effects correctly use the new paired frozen outcome.

10.2 Execution and integrity

The restart-safe workflow completed all 60 evaluation units on their first attempt in 23.60 hours. All 4,500 requested episodes and reset identities were present, all child processes returned successfully, and the pairing audit reported no mismatch. Persistent parameter-integrity digests were unchanged in every unit. The experiment used approximately 623,247 real actions. Adapted arms performed approximately 489,099 episode-local actor updates and 689 million imagined latent transitions; none of these imagined transitions is new environment experience.

10.3 Absolute recovery results

Figure 24 gives the primary sampled-policy matrix. At 1.25M, the frozen mean is 70.3%. Reward-only IR raises it at every horizon: 81.3% at $H = 6$, 83.7% at $H = 15$, and 87.0% at $H = 20$. Standard $H = 6$ IR instead reduces it to 40.3%. At 1.5M, the frozen mean is only 13.3%. Reward-only IR raises it to 21.7%, 39.7%, and 43.3%; standard IR reaches 19.0%.

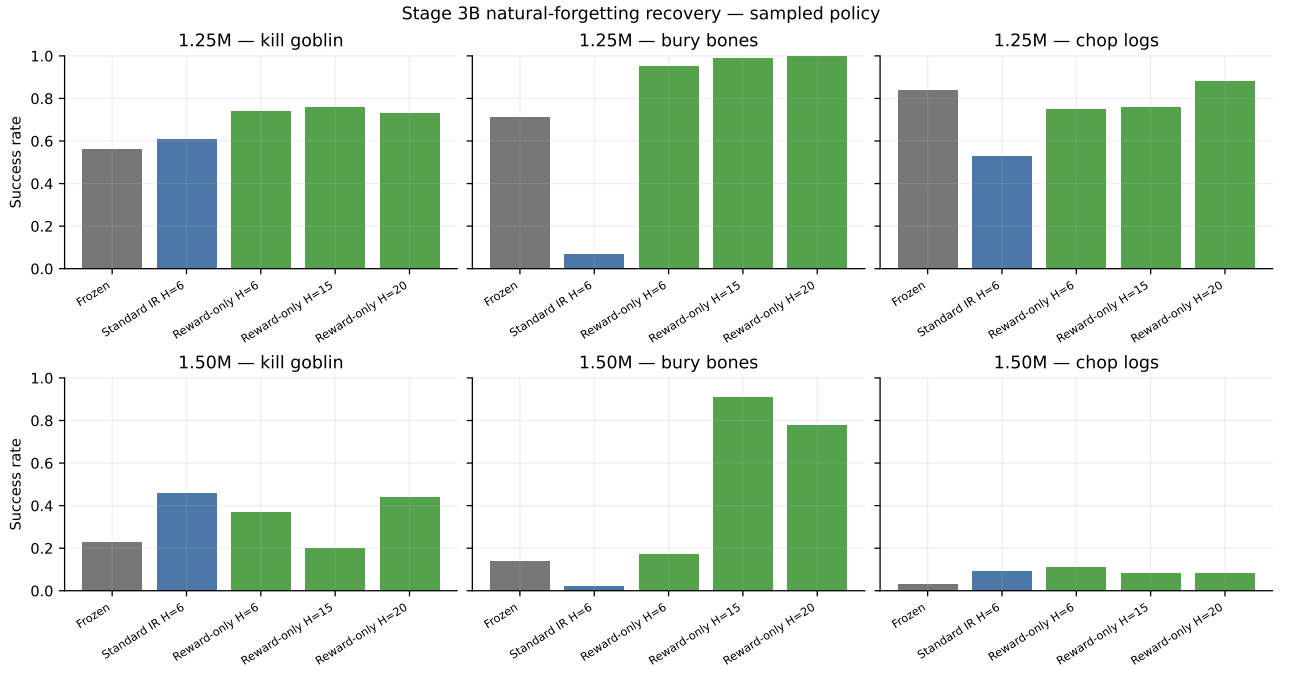


Figure 24: Study IV sampled-policy success (100 real-environment episodes per bar). Reward-only IR improves aggregate behavior at both checkpoints. The major natural recovery is 1.5M bury bones; standard IR catastrophically damages 1.25M bury and chop.

The deterministic evaluation reproduces the main result rather than merely reflecting exploratory action sampling. At 1.25M, reward-only $H = 20$ raises mean success from 59.3% to 84.7%; standard IR lowers it to 35.3%. At 1.5M, reward-only $H = 15$ is strongest, raising the mean from 11.3% to 42.7%. The different aggregate optimum— $H = 20$ for 1.25M but $H = 15$ for 1.5M deterministic control—is an early warning against a universal horizon rule.

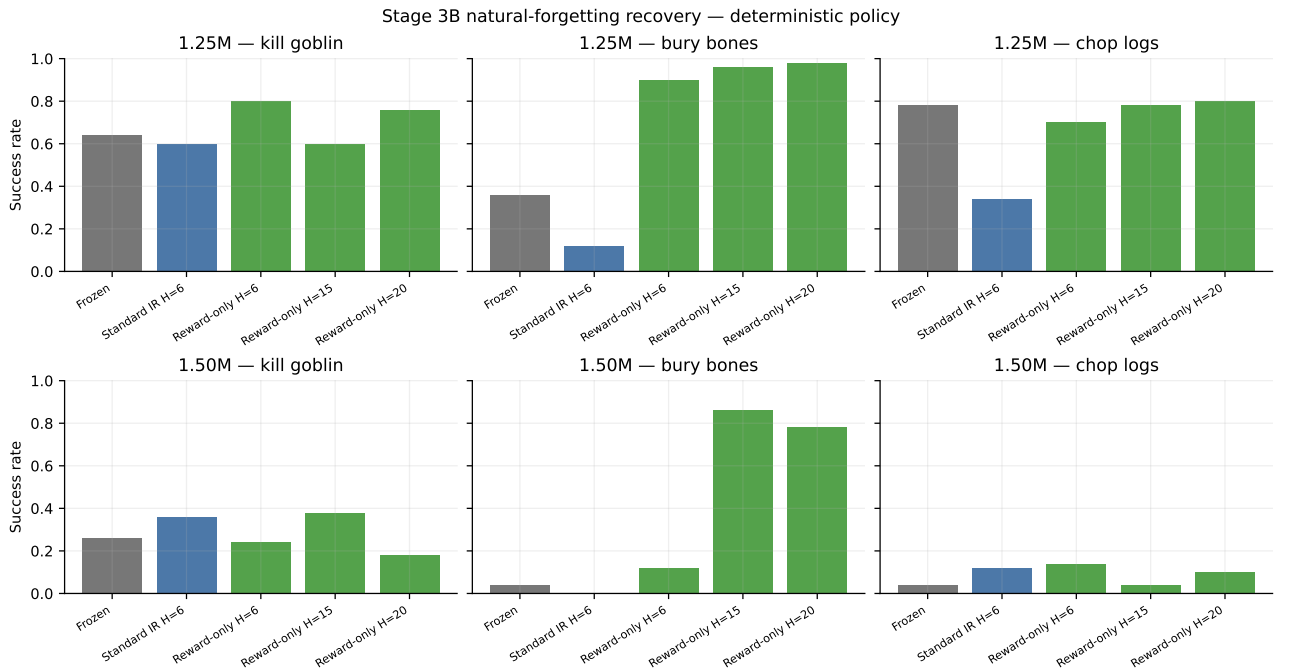


Figure 25: Study IV deterministic-policy success (50 episodes per bar). The bury recovery remains large without action sampling, while goal-specific horizon preferences become more apparent.

Table 4: Mean success across three goals in Study IV. Each sampled entry pools 300 episodes and each deterministic entry 150. Changes in parentheses are arithmetic mean changes from the paired frozen cells.

Checkpoint	Condition	Sampled	Deterministic
1.25M	Frozen	.703	.593
	Standard IR $H = 6$	.403 (−.300)	.353 (−.240)
	Reward-only $H = 6$	.813 (+.110)	.800 (+.207)
	Reward-only $H = 15$	.837 (+.133)	.780 (+.187)
	Reward-only $H = 20$	.870 (+.167)	.847 (+.253)
1.50M	Frozen	.133	.113
	Standard IR $H = 6$	.190 (+.057)	.160 (+.047)
	Reward-only $H = 6$	.217 (+.083)	.167 (+.053)
	Reward-only $H = 15$	.397 (+.263)	.427 (+.313)
	Reward-only $H = 20$	.433 (+.300)	.353 (+.240)

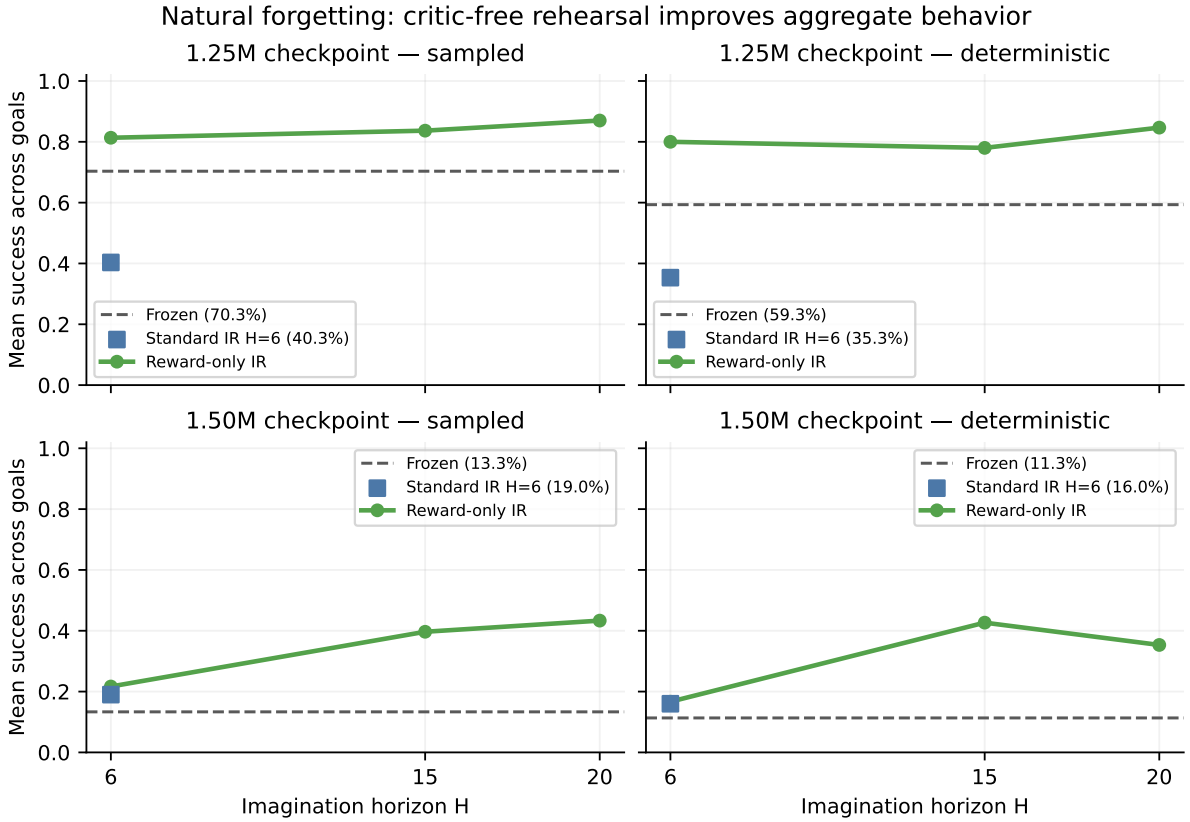


Figure 26: Aggregate horizon response on unmodified checkpoint actors. Reward-only IR exceeds the frozen mean in all twelve checkpoint-mode-horizon aggregates. Standard IR’s aggregate gain at 1.5M masks severe goal-specific asymmetry.

10.4 Paired effects and task-specific exceptions

The most compelling cell is naturally forgotten 1.5M bury bones. Reward-only $H = 15$ changes sampled success from 14% to 91%, a paired change of $+.77$ with 95% episode-bootstrap interval $[+.68, +.85]$. Deterministic success changes from 4% to 86%, $+.82[+.70, +.92]$. At $H = 20$ the corresponding changes remain large but are lower: $+.64[+.53, +.74]$ sampled and $+.74[+.62, +.86]$ deterministic. This is the first direct evidence in the programme that an imagined, critic-free test-time update can recover a severe natural forgetting deficit.

Other cells show both transfer and limits. At 1.25M, reward-only $H = 15$ improves sampled kill by $+.20$, with interval $[+.08, +.32]$. At $H = 20$, bury improves by $+.29$, with interval $[+.20, +.38]$. Strong chop is slightly damaged at shorter horizons: $-.09$ at $H = 6$ and $-.08$ at $H = 15$, before returning to $+.04$ at $H = 20$; all three chop intervals include zero. At 1.5M, $H = 20$ improves sampled kill by $+.21[+.09, +.33]$ but changes

deterministic kill by $-.08[-.26, +.10]$. Across the 12 prespecified weak sampled condition–horizon cells, reward-only IR improves 11 and harms one; across six competent cells it improves four and harms two.

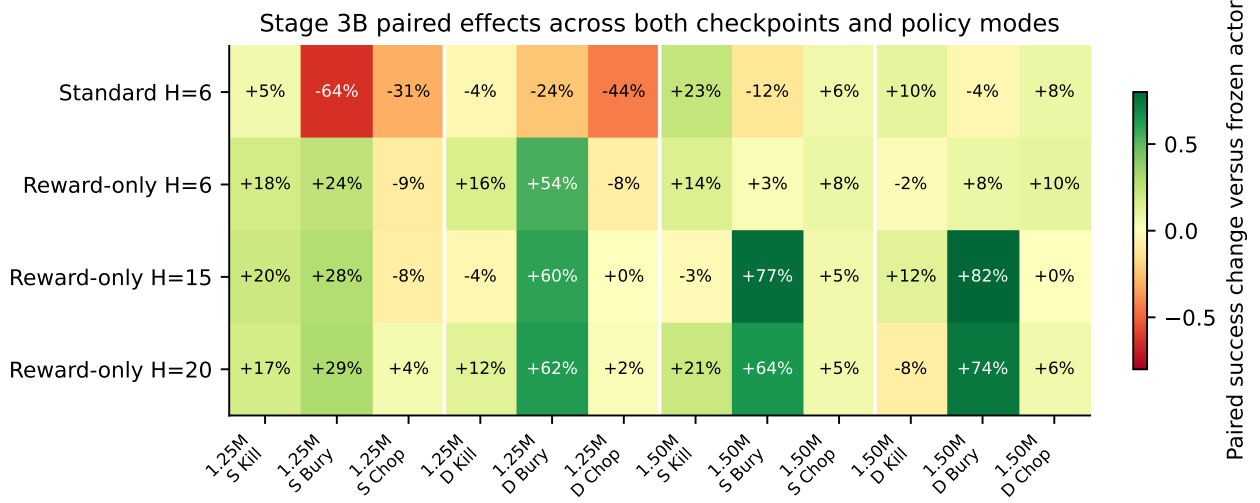


Figure 27: Complete Stage IV paired treatment-effect matrix. S and D denote sampled and deterministic policies. Critic-free recovery is broad but not uniform; the optimum depends on checkpoint, task, and action-selection mode.

Standard IR is conditionally useful but unsafe. It raises 1.5M sampled kill from 23% to 46%, $+.23[+.10, +.36]$, yet lowers 1.25M bury from 71% to 7%, $-.64[-.74, -.54]$, and chop from 84% to 53%, $-.31[-.42, -.20]$. A positive aggregate at the collapsed checkpoint therefore does not rehabilitate the critic objective; it demonstrates that the critic can occasionally point in a useful direction while remaining unsuitable for unconditional adaptation.

10.5 Study IV interpretation and claim boundary

Study IV supports its precommitted hypothesis. The Stage III result was not specific to an independent zero-mask perturbation: the same critic-free learned reward objective repairs at least one severe policy deficit created by real sequential training and improves aggregate behavior at both audited checkpoints. Because success is measured after the update in RLScapes rather than by predicted reward, the bury result is genuine behavioral recovery rather than merely an increase in model score.

The study also strengthens the causal account linking Studies II and III. Standard and reward-only arms share the checkpoint, actor optimizer, posterior starts, learned transition/reward/continuation models, environment protocol, and paired seed schedule. Removing the critic-dominated teaching signal reverses catastrophic 1.25M damage and enables large 1.5M recovery. Entropy is also absent in reward-only IR, but Study III’s entropy-off ablation already showed that entropy removal alone does not explain the reversal.

This remains episode-local test-time policy adaptation, not persistent continual learning. There is one training seed, no learned gate, and no guarantee that the best horizon can be known in advance. The prespecified weak label uses historical evaluation and is diagnostic privilege. Stage IV therefore establishes *where IR can work* and supplies target behavior for a later gate and trust region: retain the large weak-task gains without adapting destructively on already competent states.

11 Study V: critic provenance and calibration

Studies II–IV establish that the critic-dominated IR objective is harmful and that removing its bootstrap enables real behavioral recovery. They do not determine where the critic error originates. Canonical Stage 3A therefore decomposes four candidate explanations: value error already present on real posterior states; error accumulated as imagination depth increases; optimistic action sequences selected by the actor; and drift associated with weak goal conditioning. The audit is read-only and reuses the immutable 1.25M “strong” and 1.50M “weak” checkpoints.

11.1 Fixed state bank and estimands

The source is the same 384-episode Stage 1A state bank: 64 successful and 64 failed episodes for each trained goal, containing 48,299 factual states. From these episodes the audit extracts 1,433 valid posterior anchors and evaluates all five one-hot goal queries. For every anchor-goal pair it samples 128 Monte Carlo posterior trajectories to $H = 20$, then analytically reuses prefixes at $H \in \{1, 3, 6, 15, 20\}$. Actor-selected proposals are compared with recorded-action proposals from the same posterior anchors whenever sufficient future actions exist.

For a factual replay state, critic calibration uses the realized discounted reward remaining in its recorded episode as the target. Counterfactual goals and actor-selected imagined trajectories are not assigned false oracle returns. Their role is comparative: goal queries are contrasted at the same state, and actor proposals are contrasted with recorded actions under the same posterior start. For each prefix the exact IR teaching signal is decomposed into

$$G_H^{\text{IR}} = G_H^{\text{reward}} + G_H^{\text{bootstrap}}, \quad (15)$$

alongside continuation, online value, slow value, entropy, and imagined-state diagnostics. Neither agent parameters nor optimizer, replay, or checkpoint state are updated.

Both audits completed with identical before/after parameter digests. Each checkpoint contains 35,825 actor-prefix and 26,365 recorded-prefix summaries, or 62,190 rows; the combined audit contains 124,380 summaries. The actor proposal alone represents 7,165 anchor-goal rollouts and 18,342,400 imagined latent transitions per checkpoint.

11.2 The critic is already wrong on real posterior states

Figure 28 and Table 5 give the central result. The two checkpoints see exactly the same 48,299 factual states and targets, whose state-weighted mean is 0.176. The online critic instead predicts 1.359 at 1.25M and 1.771 at 1.50M. Because these episodes contain at most one sparse completion reward, values above one are structurally overoptimistic rather than merely conservatively high.

Table 5: Real-posterior critic calibration on the identical factual state bank. Correlations are unweighted means over the three trained goals; target, prediction, bias, and MAE are state-count weighted.

Checkpoint	Target	Prediction	Bias	MAE	Pearson	Spearman
1.25M strong	.176	1.359	+1.184	1.192	.198	.213
1.50M weak	.176	1.771	+1.595	1.595	.058	.081

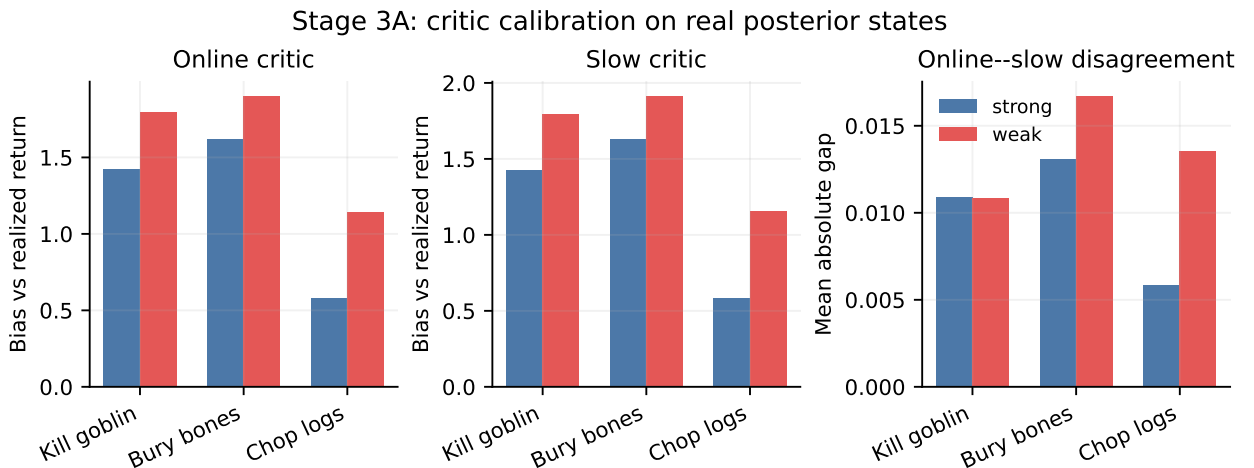


Figure 28: Critic calibration on real posterior states. Bias grows by 0.412 (34.8%) at the collapsed checkpoint while value ranking nearly disappears. The slow critic closely tracks the online critic rather than providing an independent check.

The failure is not only a common intercept error. Goal-level biases at 1.25M are +1.421 for kill, +1.621 for bury, and +0.579 for chop; at 1.50M they rise to +1.793, +1.900, and +1.142. Mean Pearson correlation falls from

.198 to .058 and Spearman from .213 to .081. The critic therefore loses both calibration and useful ordering as behavior collapses. A scalar offset correction would leave an extremely weak policy-improvement signal.

The slow critic does not solve this problem. Its weighted biases are +1.190 and +1.603, and its mean absolute prediction gap from the online critic is only .0098 and .0136. It is a delayed stabilizer for the same learned target, not an independently calibrated evaluator or uncertainty estimate.

11.3 Horizon decomposition: no late runaway

If open-loop model error were progressively inflating the target, longer prefixes should produce continuing growth. The observed pattern is different. Table 6 and Figure 29 show that the factual actor-selected target is already almost entirely bootstrap at $H = 1$ and plateaus by $H = 3$ –6. From $H = 6$ to $H = 20$, the strong full target changes from 1.480 to 1.482 and the weak target from 1.891 to 1.891.

Table 6: Actor-selected factual IR target decomposition. Values are unweighted means over the three trained goals, matching the plotted estimand.

Checkpoint	H	Full target	Reward-only	Bootstrap	Bootstrap fraction
1.25M	1	1.443	.052	1.391	95.9%
1.25M	6	1.480	.072	1.407	94.4%
1.25M	20	1.482	.106	1.376	91.7%
1.50M	1	1.848	.036	1.812	97.5%
1.50M	6	1.891	.039	1.852	97.4%
1.50M	20	1.891	.050	1.841	96.9%

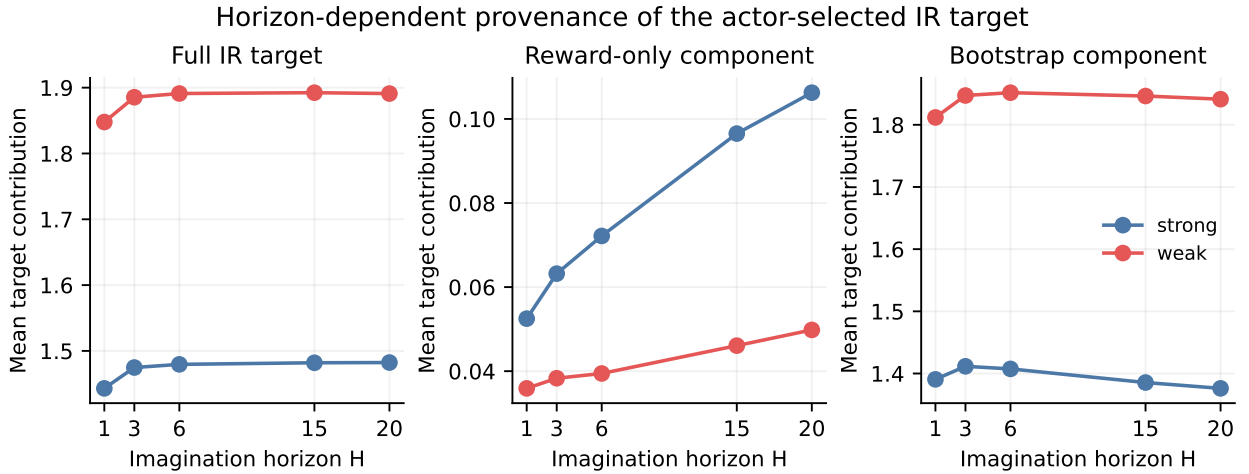


Figure 29: IR target provenance across horizon. Longer imagination does not create a $H = 6$ to $H = 20$ explosion. The collapsed checkpoint imagines less goal reward but assigns substantially more bootstrap value.

At 1.25M, added depth increases predicted reward while bootstrap slightly falls. At 1.50M, both components are nearly flat after $H = 3$. Crucially, the weak checkpoint predicts *less* reward than the strong checkpoint while assigning much larger total return. This locates the checkpoint difference in value bootstrap rather than accumulating reward-model hallucinations. Recorded-action proposals are likewise optimistic: their mean return bias spans +1.17 to +1.37 at 1.25M and +1.50 to +1.73 at 1.50M across the tested horizons. Actor exploitation is therefore unnecessary for the error to exist.

11.4 Actor selection amplifies the critic error

Although the actor does not create the critic’s bad scale, it preferentially selects trajectories with still higher bootstrap values. Relative to paired recorded actions, the actor raises the full target at every checkpoint and horizon (Table 7). At the collapsed checkpoint the apparent advantage comes entirely from bootstrap while predicted reward is lower.

Table 7: Actor-minus-recorded factual proposal differences from the same posterior anchors. Positive bootstrap and negative reward identify critic-seeking rather than predicted task progress.

Checkpoint	H	Full target	Bootstrap	Reward-only
1.25M	1	+.139	+.211	-.072
1.25M	6	+.126	+.138	-.012
1.25M	20	+.142	+.129	+.013
1.50M	1	+.212	+.298	-.086
1.50M	6	+.175	+.194	-.019
1.50M	20	+.182	+.196	-.015

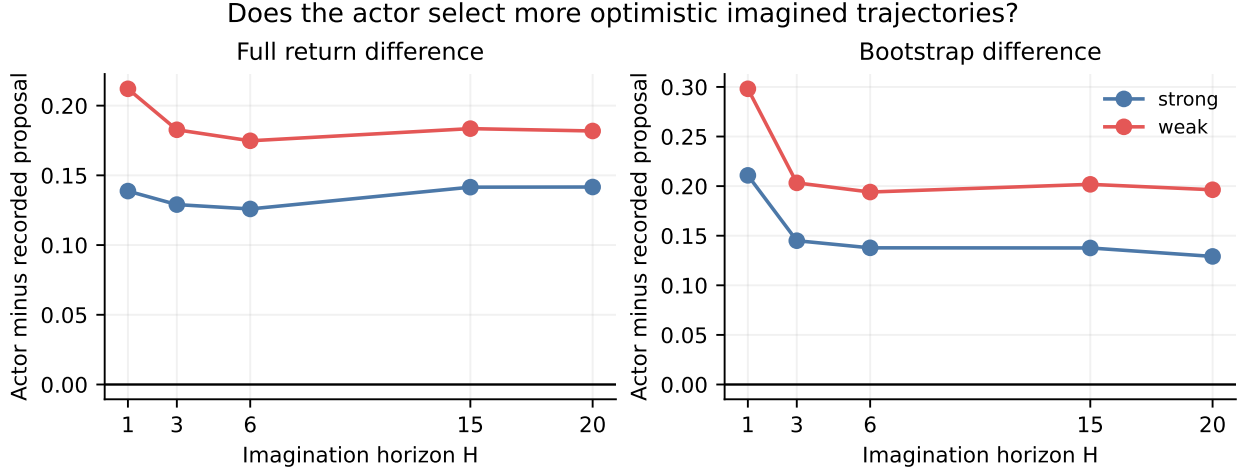


Figure 30: Actor-selected minus recorded-action imagined targets. The actor consistently selects higher-value prefixes; at 1.50M this occurs despite lower predicted reward. Model/critic exploitation is real, but it adds roughly 0.18 target units atop real-state bias of roughly 1.60.

The signature is clearest on bury bones. At $H = 6$, actor-minus-recorded bootstrap is +.266 at 1.25M and +.289 at 1.50M, while reward differences are $-.036$ and $-.043$. Thus the actor has learned to navigate the critic’s optimistic landscape rather than the reward model’s completion landscape. The effect is a secondary amplifier: it adds approximately .13–.14 full-return units at the strong checkpoint and .17–.21 at the weak checkpoint, compared with real-state biases already exceeding one.

Finally, actor-selected factual and pooled counterfactual targets remain nearly indistinguishable. Their mean full-target difference is below .001 at $H = 6$ and .008 at $H = 20$ at both checkpoints; reward-only differences remain below .005 at the headline horizons. This agrees with the direct Study IIa confusion matrix: checkpoint identity and value scale dominate the requested one-hot goal. Because Stage V pools four alternative goal queries, this observation supports but does not replace Study IIa’s stronger same-state counterfactual test.

11.5 Study V interpretation and claim boundary

Stage V refines the earlier “model exploitation” hypothesis. Exploitation is present, but it is not the origin of the optimism. The primary failure is a structurally overoptimistic, poorly ranked critic on real posterior states; the continual checkpoint change worsens that landscape; and the actor then selects trajectories that exploit its bootstrap. Increasing horizon from six to twenty does not cause runaway target growth, and the slow critic supplies no independent protection.

This is a provenance audit, not a unique causal diagnosis of why critic training drifted. Candidate upstream causes include recursive bootstrapping on biased targets, insufficiently goal-discriminative rewards and values, distribution shift across sequential phases, and the absence of task-semantic bounds on a success-return value. The audit nevertheless rules out three tempting remedies: simply shorten the existing critic-backed horizon, replace the online critic with its slow copy, or subtract one global calibration offset. Any future critic reuse should first demonstrate bounded calibration and useful ranking on held-out real states.

12 Study VI: counterfactual goal semantics and bounded-value repair

Studies II and V identified two different defects in the retained teaching signals. Reward and continuation detect factual completion events but barely change when the same latent state is queried under another goal; the critic is both weakly ranked and far outside the attainable sparse-return scale. Studies III and IV showed that removing the critic permits useful actor adaptation, but did not test whether either defect can be repaired. Canonical Stage 4A is the corresponding intervention study. It asks three ordered questions:

1. Can explicit counterfactual event labels make reward and continuation depend on the requested goal while factual-only matched training does not?
2. Can a separate Bernoulli success-value head recover a calibrated, bounded teaching signal without modifying the historical Dreamer critic?
3. Do these head-level repairs improve paired real-environment IR beyond the already successful original reward-only objective?

The ordering is important. A head can pass a held-out semantic audit without providing useful gradients along actor-imagined trajectories; conversely, the simple RLScape goals may be behaviorally separable even when their terminal event detector is aliased. Study VI therefore treats real-environment success as the primary behavioral endpoint and the held-out audit as a mechanistic endpoint, rather than assuming one implies the other.

12.1 Disjoint repair and held-out state banks

The source agent is the immutable reservoir checkpoint at 1.25M environment steps. A fresh episode-identity split is drawn from its preserved goal-stratified replay. The repair bank contains 96 successful and 96 failed episodes for each of kill goblin, bury bones, and chop logs (576 episodes); the held-out bank contains 64 successful and 64 failed episodes per goal (384 episodes). No episode occurs in both banks. The held-out audit contains 47,916 posterior states and 192 genuine completion events. FIFO-original, reservoir-original, and every repaired clone are evaluated on this identical reservoir-originating state bank. FIFO results consequently measure cross-bank generalization, not performance on a native FIFO sample.

Posterior latents are cached under the frozen reservoir encoder and RSSM. Repair samples 24 states per episode, balances event and ordinary states with event fraction 0.5, and runs 2,000 reward/continuation optimizer updates with batch size 256 and learning rate 4×10^{-5} . The factual-only control repeats the factual goal query in all five query slots, matching the counterfactual arm’s tensor shape, examples per update, and optimizer dose. The counterfactual arm instead queries all five one-hot goals. For recorded goal g_t , queried goal g , sparse factual reward r_t , factual goal event e_t , and physical terminal event d_t , the targets are

$$r_t^{(g)} = r_t \mathbf{1}[g = g_t], \quad (16)$$

$$c_t^{(g)} = 1 - \mathbf{1}[d_t \vee (e_t \wedge g = g_t)]. \quad (17)$$

Completion of goal g_t is therefore rewarding and stopping only when that goal is queried. A physical death stops every goal. The labels are valid for the event represented by the transition; they deliberately do not assign counterfactual values or claim what an alternative-goal policy would later have achieved.

The third clone receives the identical counterfactual reward/continuation updates and then 2,000 additional updates to a separate bounded head

$$b_\omega(s_t, g_t) = \Pr_\omega(\text{discounted factual completion} \mid s_t, g_t) \in [0, 1]. \quad (18)$$

Its targets are factual discounted Monte Carlo completion returns from the recorded episode. The encoder, RSSM, decoder, actor, historical online and slow critics, and all normalizers remain frozen. Integrity manifests verify that source and exported actors are identical and that the counterfactual and counterfactual-plus-bounded reward/continuation parameter digests are bit-identical. Thus the bounded intervention adds a new head rather than silently clipping or rewriting the old critic.

12.2 Paired actor-adaptation assay

The real-environment assay contains eight prespecified conditions:

1. original frozen actor;
2. original standard Dreamer IR at $H = 6$;
3. original reward-only IR at $H = 15$;
4. factual-repaired reward-only IR at $H = 15$;
5. counterfactual-repaired reward-only IR at $H = 15$;
6. counterfactual-plus-bounded reward-only IR at $H = 15$;
7. counterfactual-plus-bounded bounded-bootstrap IR at $H = 6$;
8. the same bounded-bootstrap objective at $H = 15$.

Reward-only IR uses Equation 12. The bounded objective replaces the historical critic in the lambda return with b_ω and uses the bounded prediction as the actor baseline:

$$G_t^B = \lambda\text{-return}(\hat{r}_{t:t+H}, \hat{c}_{t:t+H}, b_\omega), \quad (19)$$

$$A_t^B = G_t^B - b_\omega(\hat{s}_t, g). \quad (20)$$

All adapted actors receive one temporary update per real action, beginning at the first action, with 128 Monte Carlo posterior samples and learning rate 4×10^{-5} . Repaired, reward-only, and bounded objectives use zero actor entropy; the historical standard control retains 3×10^{-4} . The world model and every head are frozen during adaptation, and actor changes are discarded at episode reset.

Each condition-goal cell uses 100 sampled and 50 deterministic episodes. The assay therefore contains

$$8 \text{ conditions} \times 3 \text{ goals} \times (100 + 50) = 3,600 \text{ episodes}. \quad (21)$$

All 48 cells, five audits, and three repair arms completed without a failed child attempt. Reset identities match the frozen control in every goal-policy-mode stratum, and all requested episode counts are exact. The 56 scientific units ran from 16 August 11:50 to 17 August 03:03 (15.21 hours). After writing the result CSV/JSON, the supervisor encountered a plotting-only type error and did not write its terminal marker. The maintained plotting code is corrected; no numerical or experimental unit was affected.

12.3 Counterfactual labels repair held-out semantics

Figure 31 and Table 8 show a clean mechanistic intervention. At completion states, reservoir-original reward top-1 goal accuracy is 33.3%, only modestly above five-goal chance (20%); the matching-minus-nonmatching reward margin is 0.004. Matched factual-only updates change these to 34.4% and 0.020. Counterfactual updates instead reach 97.4% and 0.567. Continuation changes in the same direction: at a completed event its predicted continuation is 0.108 for the completed goal but 0.698 for other goals. The reservoir original predicts 0.043 and 0.057, effectively stopping every queried goal at the event.

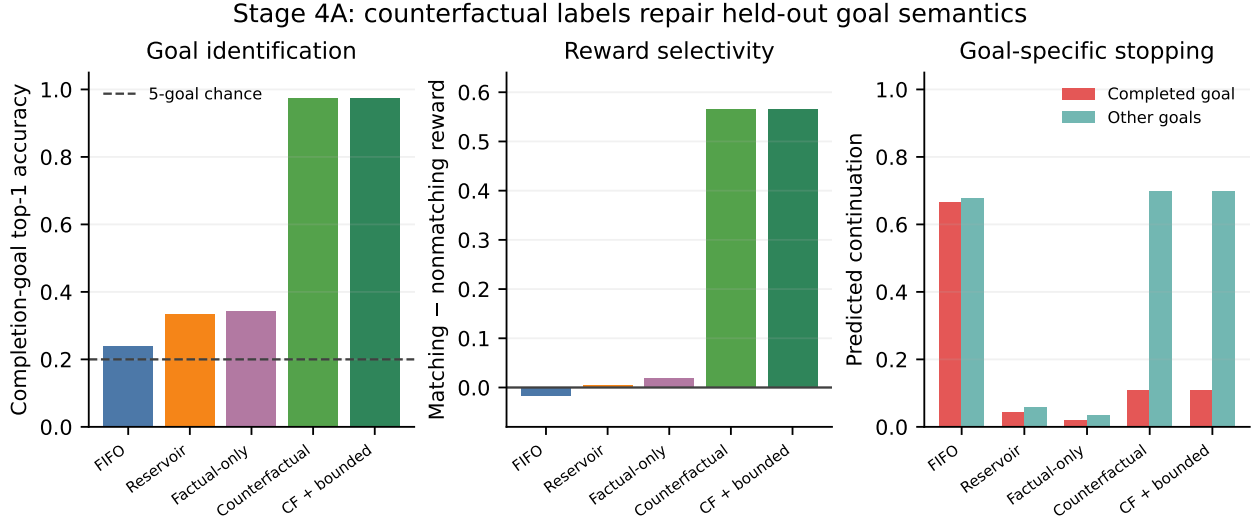


Figure 31: Study VI held-out semantic audit. Factual-only matched training preserves the original goal aliasing. Counterfactual event labels make reward and continuation discriminate the completed goal. For continuation, the desired direction is low for the completed goal and high for other goals.

Table 8: Held-out completion-event selectivity in Study VI (192 events). Reward margin is matching minus mean nonmatching prediction. Continuation columns are probabilities of continuing after the event.

Checkpoint/repair	Reward top-1	Reward margin	Continue match	Continue other
FIFO original	.240	−.017	.666	.677
Reservoir original	.333	.004	.043	.057
Factual-only	.344	.020	.020	.034
Counterfactual	.974	.567	.108	.698
Counterfactual + bounded	.974	.567	.108	.698

This semantic gain does not come from losing factual event recognition. Counterfactual reward AUROC remains 0.9996 and continuation AUROC 0.9993 over 47,916 factual states. Positive-event reward confidence becomes less sharp—mean predicted reward falls from 0.959 in the reservoir original to 0.785—but ordinary-state false positives remain small. The intended trade is therefore visible: slightly softer factual confidence in exchange for a large, held-out counterfactual margin.

12.4 The bounded head repairs scale and ranking

The historical critic remains severely optimistic on the new held-out bank. For kill, bury, and chop it predicts means 1.623, 1.694, and 0.819 against realized discounted targets 0.194, 0.078, and 0.224. The bounded head predicts 0.180, 0.069, and 0.226. Its signed biases are consequently -0.014 , -0.009 , $+0.002$, compared with $+1.429$, $+1.616$, $+0.595$ for the original critic. Held-out MAE is 0.179, 0.084, and 0.173; Brier score is 0.129, 0.064, and 0.123.

The intervention also improves ranking rather than only output scale. Initial-state success AUROC changes from 0.400 to 0.568 for kill, 0.109 to 0.848 for bury, and 0.570 to 0.632 for chop. Bury is the clearest repair: the historical critic strongly ranks failures above successes, whereas the bounded head becomes a useful discriminator. Kill ranking remains only slightly above chance, so bounding is not a complete value solution.



Figure 32: Study VI value repair. Left: the separately trained bounded head closely matches held-out mean return scale while the unchanged critic exceeds the attainable sparse-return range. Right: initial-state success ranking improves for all goals, most strongly for bury bones.

12.5 Behavioral result: reward-only remains strongest

Figure 33 gives every real-environment cell. The frozen actor scores 69.7% sampled and 59.3% deterministic on average. The original standard $H = 6$ objective reproduces the earlier failure, falling to 34.0% and 38.0%. Sampled bury collapses from 69% to 7%, a paired change of $-.62[-.72, -.52]$, and chop from 85% to 49%, $-.36[-.48, -.24]$.

All reward-only variants are much stronger. Original reward-only reaches 80.3% sampled and 78.0% deterministic. Factual-only reward repair is the best sampled condition: 88.0% mean and 82% worst-goal success, with kill/bury/chop at 86%/96%/82%. Its paired sampled gains are $+.31[+.20, +.42]$ for kill and $+.27[+.17, +.37]$ for bury; chop changes by $-.03[-.12, +.06]$. Counterfactual reward-only remains effective at 81.7%/77.3% sampled/deterministic, but does not beat the matched factual control. Better completion-state goal semantics therefore do not automatically produce the best actor gradient in this compact arena.

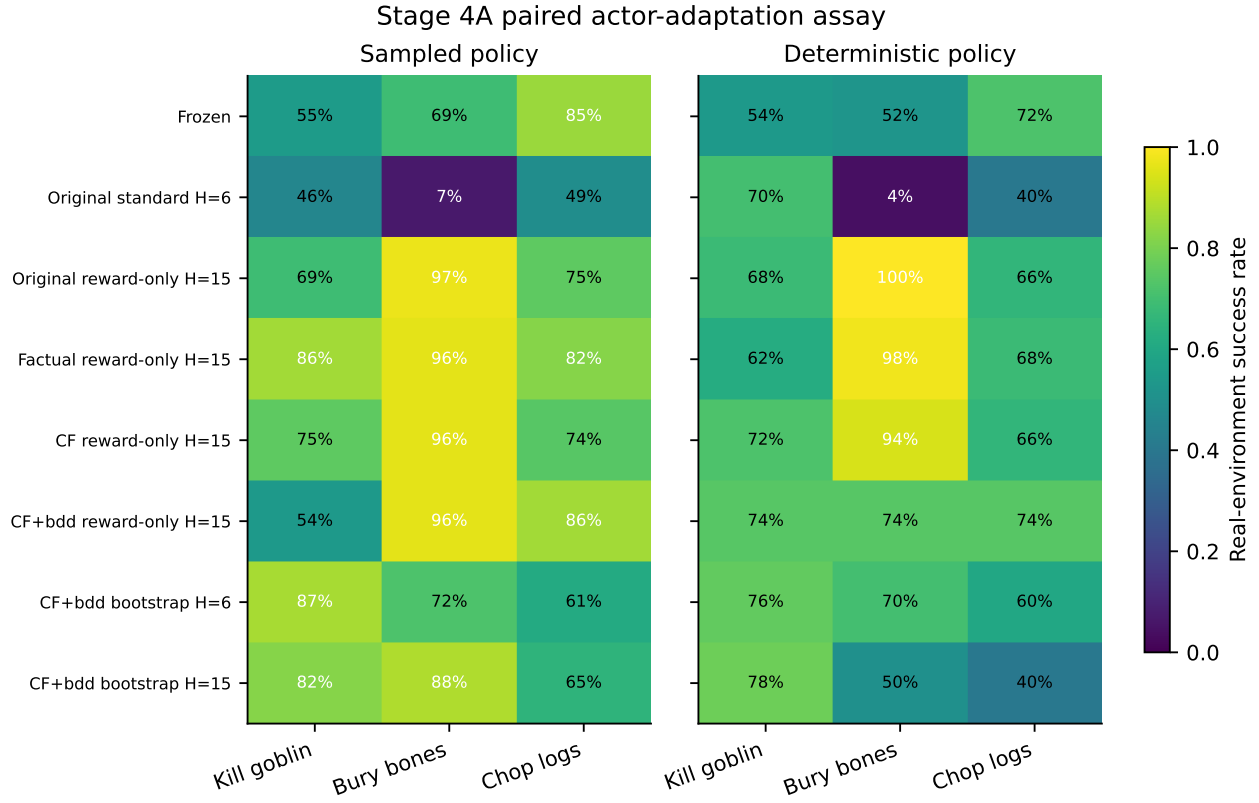


Figure 33: Complete Study VI real-environment success matrix. Each sampled cell contains 100 episodes and each deterministic cell 50. Factual-only reward IR gives the strongest sampled and worst-task profile; the repaired bounded bootstrap is markedly safer than the original critic but remains horizon- and goal-dependent.

Table 9: Mean and worst-goal real-environment success in Study VI. Changes are arithmetic mean changes from the paired frozen cells.

Condition	Sampled mean	Sampled worst	Det. mean	Det. worst
Frozen original	.697	.550	.593	.520
Original standard $H = 6$	.340 (−.357)	.070	.380 (−.213)	.040
Original reward-only $H = 15$	.803 (+.107)	.690	.780 (+.187)	.660
Factual reward-only $H = 15$	.880 (+.183)	.820	.760 (+.167)	.620
CF reward-only $H = 15$	.817 (+.120)	.740	.773 (+.180)	.660
CF+B reward-only $H = 15$	.787 (+.090)	.540	.740 (+.147)	.740
CF+B bounded $H = 6$	.733 (+.037)	.610	.687 (+.093)	.600
CF+B bounded $H = 15$	.783 (+.087)	.650	.560 (−.033)	.400

Aggregate performance and weakest-task safety

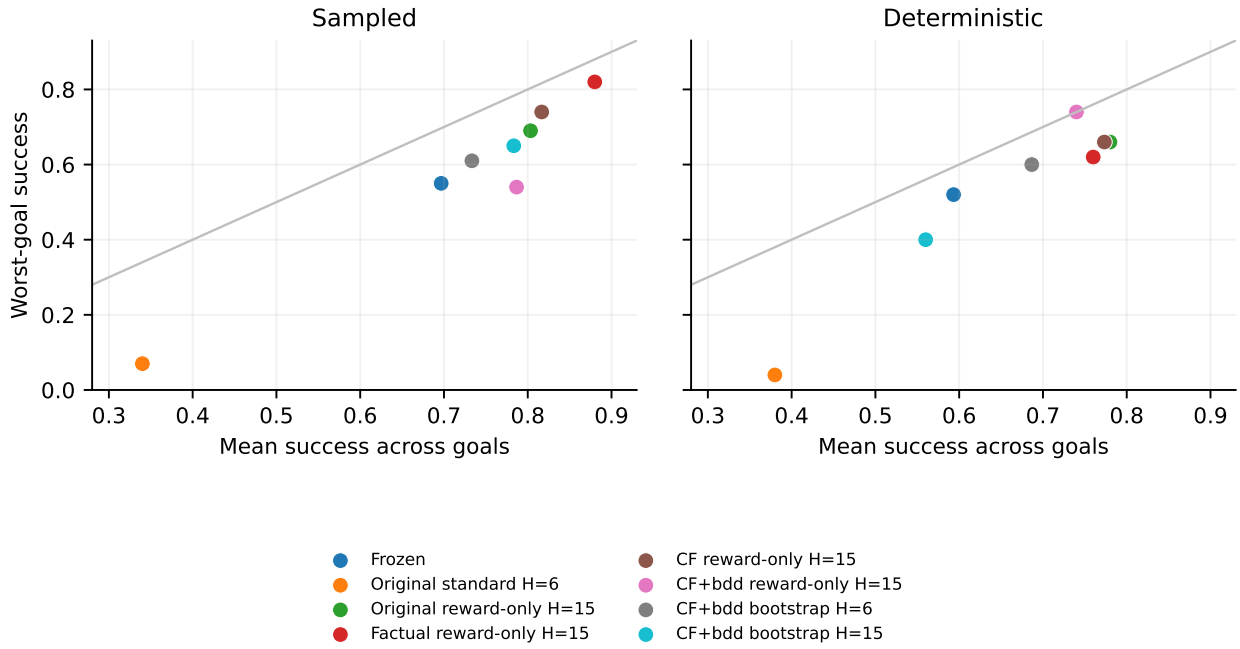


Figure 34: Study VI mean success versus weakest-goal success. A useful actor teacher should move toward the upper right rather than conceal one damaged goal behind aggregate gains. Factual reward-only IR gives the strongest sampled Pareto point; the bounded reward-only clone gives the most even deterministic profile.

Bounded-bootstrap IR is substantially safer than the historical critic in aggregate, but not uniformly safe. At $H = 6$ it improves sampled kill by $+32[+.21, +.43]$ and deterministic kill by $+.22[+.02, +.42]$, while harming sampled chop by $-.24[-.36, -.12]$. At $H = 15$, sampled kill and bury improve by $+.27$ and $+.19$, but sampled chop falls by $-.20[-.31, -.09]$ and deterministic chop by $-.32[-.50, -.14]$. A calibrated range is therefore necessary but not sufficient: ranking error, imagined-state distribution shift, baseline subtraction, and repeated update dose can still produce a bad actor direction.



Figure 35: Study VI paired changes from the contemporaneous frozen actor. S and D denote sampled and deterministic control. Reward-only improvement is broad but not universal; the bounded objective trades strong kill gains against chop damage.

12.6 Study VI interpretation and claim boundary

Study VI cleanly accepts the head-repair hypotheses and rejects their simplest behavioral implication. Explicit counterfactual labels can teach one latent completion event to mean different things under five goal queries, and a separate Bernoulli return head can remove gross critic optimism. Yet factual reward-only IR is behaviorally strongest under sampled control. In these three tasks, the actor may already separate goals through their starting context, action paths, and visual affordances; sharper goal semantics at the final event need not add useful gradients along the imagined path. Counterfactual training also softens positive factual reward confidence, which may trade signal strength for semantic purity.

One negative-control comparison requires special caution. The counterfactual and counterfactual-plus-bounded checkpoints have identical actors and bit-identical reward/continuation heads, so reward-only behavior should be identical under fully paired adaptation randomness. It is not: sampled means are 81.7% and 78.7%, with larger goal-specific differences. Environment reset IDs were paired, but Monte Carlo posterior and action RNG streams inside separate adaptation processes were not asserted equal. This contrast must be read as a stochastic reproducibility diagnostic, not a causal effect of an unused bounded head. Future assays should pair agent PRNG streams and/or repeat adaptation seeds.

The study remains single-checkpoint, single-training-seed, and episode-local. It establishes that both diagnosed head defects are repairable and that a bounded critic is less hazardous than the original one. It does not establish that the current bounded objective is ready for persistent continual updates. The immediate target is a dose-controlled conservative value ablation against the reward-only teacher, followed by a gate and trust region only after the source of remaining chop harm is understood.

13 Study VII: conservative bounded-value dose and stability

Study VI produced an apparently contradictory result. The separately bounded success head was far better calibrated than the historical critic and improved held-out success ranking, yet using it as the full actor teacher remained task-dependent. Reward-only IR was safer behaviorally, but by construction discarded any longer-horizon information the repaired value head might contain. Canonical Stage 4B therefore asks a narrower intervention question: can a small amount of bounded value advice improve reward-only IR without sacrificing the weakest goal?

This study also corrects the stochastic-assay weakness exposed in Study VI. Environment resets alone are not sufficient pairing when each episode contains Monte Carlo posterior sampling, sampled policy actions, and a changing temporary actor. Stage 4B explicitly indexes both posterior and policy random keys by replicate, episode, and environment step. The comparison is consequently a paired dose assay rather than a collection of merely seed-matched processes.

13.1 Causal interpolation and fixed controls

All conditions load the same Stage 4A `counterfactual_bounded` checkpoint. The encoder, RSSM, decoder, counterfactually repaired reward and continuation heads, bounded value head, historical online/slow critics, and persistent actor are fixed at load. Only an episode-local actor clone changes. Let the normalized critic-free reward advantage and bounded-value advantage be

$$A_t^R = R_{t,R}^\lambda / \sigma_R, \quad (22)$$

$$A_t^B = (R_{t,B}^\lambda - V_{t,B}) / \sigma_R. \quad (23)$$

The mixed teacher is

$$A_t^\beta = A_t^R + \beta \text{clip}(A_t^B - A_t^R, -c, c). \quad (24)$$

Thus $\beta = 0$ is exactly reward-only IR and unclipped $\beta = 1$ is exactly the Study VI bounded objective. The correction is the incremental difference $A^B - A^R$, so imagined reward is not counted twice. All adapted conditions use one actor update at every nonterminal real step, 128 Monte Carlo posterior samples, learning rate 4×10^{-5} , and zero entropy bonus. Every learned model and prediction head remains frozen.

Nine conditions were fixed before evaluation: frozen; reward-only $H = 15$; unclipped $H = 15$ doses $\beta \in \{.05, .10, .25, 1\}$; $\beta = .25$ with a .50 correction clip; $\beta = .25$ with a post-Adam actor-update RMS cap of 1.5×10^{-5} ; and $\beta = .25, H = 6$ as a horizon control. The update cap was chosen from Study VI update magnitudes before Stage 4B outcomes were observed and preserves the Adam update direction.

13.2 Pairing, budget, and decision rule

Three prespecified replicates use distinct environment and agent seeds. Within each replicate, all conditions share exact goal/reset sequences and explicit episode-step posterior and action keys. Each condition runs all three goals for 50 sampled and 25 deterministic episodes per replicate:

$$9 \text{ conditions} \times 3 \text{ goals} \times 3 \text{ replicates} \times (50 + 25) = 6,075 \text{ episodes.} \quad (25)$$

All 162 evaluation units completed with zero failed attempts. The pairing audit passed, all requested episode counts are exact, the source checkpoint digest matches the immutable specification, and the restart-safe queue records every child command. A stale unit created by the initial non-tmux launcher was reconciled on restart without duplicating or altering its completed episodes.

The primary endpoints are macro mean success, weakest-goal success, paired macro change from reward-only, and weakest-goal paired change from reward-only. Advancement required positive sampled macro change, positive sampled macro change in at least two of three replicates, and sampled and deterministic weakest-goal changes no lower than $-.05$. This rule was fixed before results were read; deterministic harm cannot be hidden by a better sampled mean.

13.3 Small value doses strongly improve sampled behavior

Figure 36 shows a non-monotonic dose response. Reward-only reaches 73.8% sampled macro success and 51.3% weakest-goal success, already improving the frozen actor’s 62.0%/50.0%. A five-percent bounded correction raises these to 87.6%/78.7%. Its paired macro gain is $+.138$, positive in every replicate ($+.067, +.133, +.213$), and its per-goal changes are $+.273$ kill goblin, $+.040$ bury bones, and $+.100$ chop logs. The gain is therefore neither compensation by one easy task nor one lucky adaptation seed.

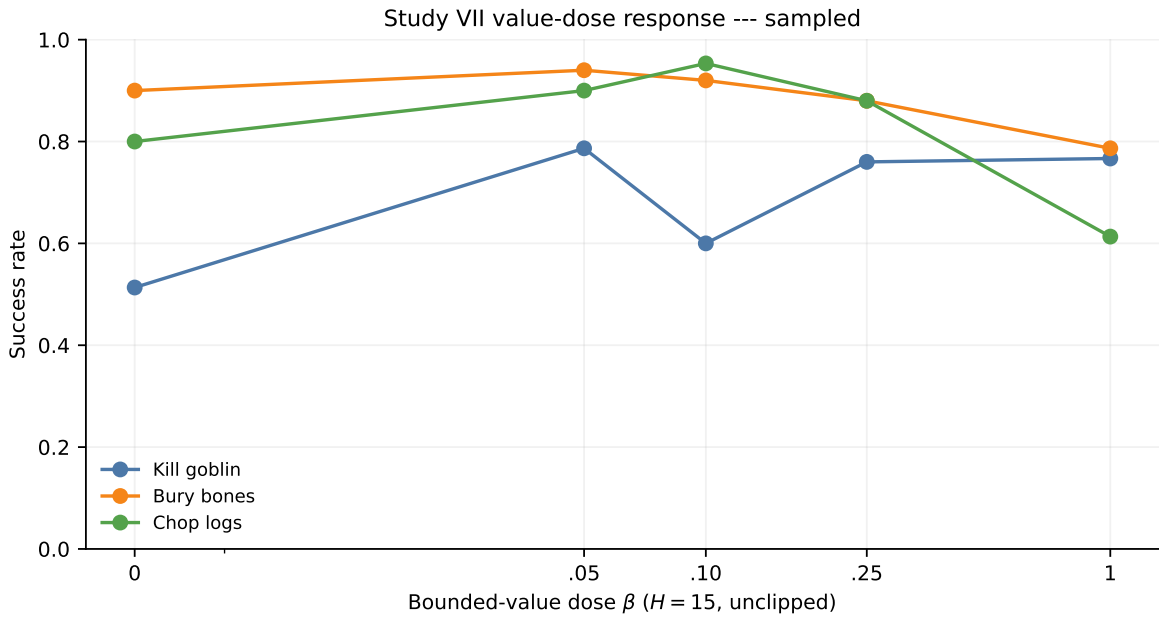


Figure 36: Study VII sampled-policy dose response for the unclipped $H = 15$ family. Small value corrections improve every goal, with the best aggregate and weakest-goal profile at $\beta = .05$. Pure bounded-value teaching ($\beta = 1$) loses the benefit.

Table 10: Study VII aggregate sampled-policy result. Deltas and weakest deltas are relative to reward-only $H = 15$.

Condition	Mean	Weakest	Δ mean	Δ weakest
Frozen	.620	.500	-.118	-.300
Reward-only $H = 15$	.738	.513	.000	.000
$\beta = .05, H = 15$	.876	.787	+.138	+.040
$\beta = .10, H = 15$	.824	.600	+.087	+.020
$\beta = .25, H = 15$	.840	.760	+.102	-.020
$\beta = .25$, correction clip .50	.862	.747	+.124	.000
$\beta = .25$, update RMS cap	.836	.767	+.098	-.080
$\beta = .25, H = 6$	.811	.773	+.073	-.027
$\beta = 1, H = 15$	.722	.613	-.016	-.187

The remaining controls constrain the mechanism. Moderate dose $\beta = .25$ is useful but less good than $\beta = .05$; clipping its incremental correction raises the sampled mean to 86.2% but does not beat the five-percent weakest-goal result. The post-Adam update cap changes the profile without removing worst-goal harm. Shortening H from 15 to 6 lowers the moderate-dose mean from 84.0% to 81.1%. Pure bounded teaching reaches only 72.2%, below reward-only. The bounded head therefore contributes useful information as a correction, not as a replacement teacher.

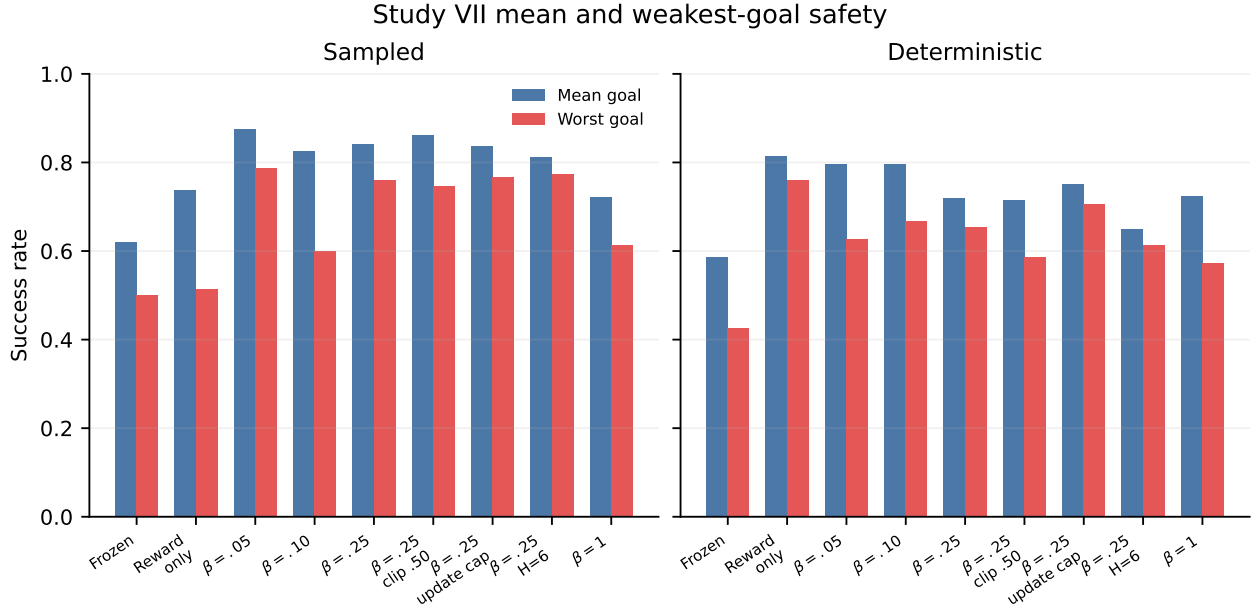


Figure 37: Study VII macro and weakest-goal success. The five-percent correction is the strongest sampled treatment. Reward-only remains the strongest deterministic treatment, exposing a policy-mode trade-off hidden by sampled success alone.

13.4 Deterministic control rejects automatic advancement

The sampled gain does not transfer uniformly to the policy mode. Reward-only is the best deterministic condition at 81.3% macro and 76.0% weakest-goal success. Both $\beta = .05$ and $.10$ reach 79.6% macro, but their weakest goals fall to 62.7% and 66.7%. For $\beta = .05$, the loss is concentrated in kill goblin ($-.133$ relative to reward-only), while chop improves by $+.093$ and bury changes by $-.013$. The actor update has placed more useful probability mass into the sampled distribution without reliably improving its modal action sequence.

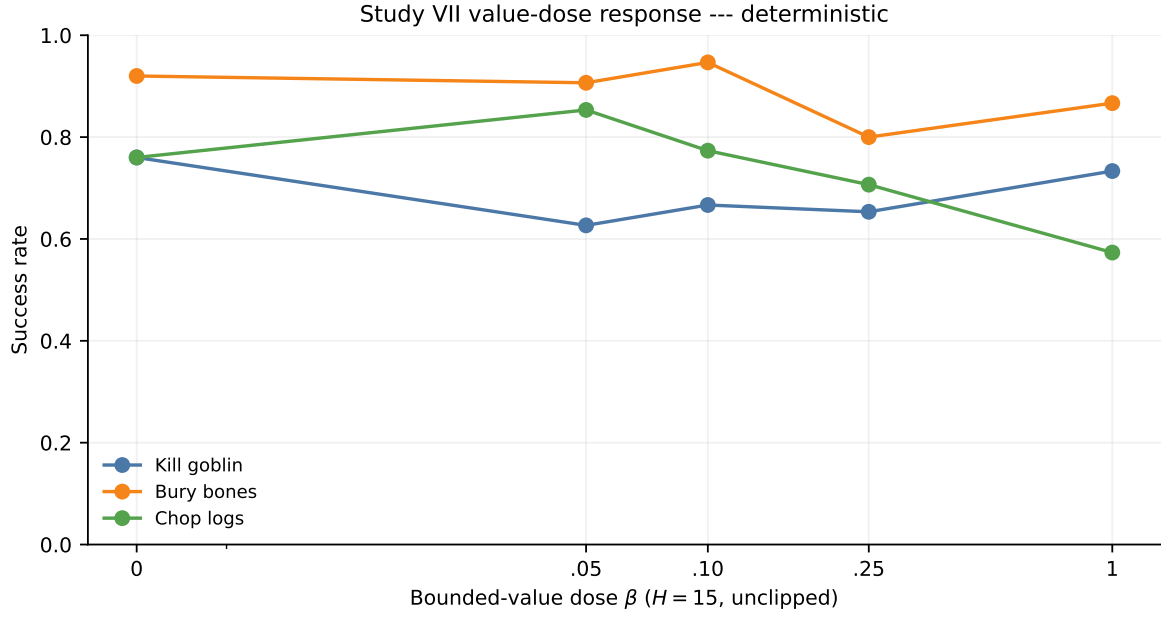


Figure 38: Study VII deterministic-policy dose response for the unclipped $H = 15$ family. Goal-specific effects do not share the sampled optimum: the five-percent dose improves chop but damages kill, while larger doses trade among all three goals.

Table 11: Study VII aggregate deterministic-policy result. No mixed condition respects the prespecified $-.05$ weakest-goal safety floor.

Condition	Mean	Weakest	Δ mean	Δ weakest
Frozen	.587	.427	-.227	-.333
Reward-only $H = 15$	.813	.760	.000	.000
$\beta = .05, H = 15$	.796	.627	-.018	-.133
$\beta = .10, H = 15$	.796	.667	-.018	-.093
$\beta = .25, H = 15$	.720	.653	-.093	-.120
$\beta = .25$, correction clip .50	.716	.587	-.098	-.173
$\beta = .25$, update RMS cap	.751	.707	-.062	-.213
$\beta = .25, H = 6$	.649	.613	-.164	-.280
$\beta = 1, H = 15$	.724	.573	-.089	-.187

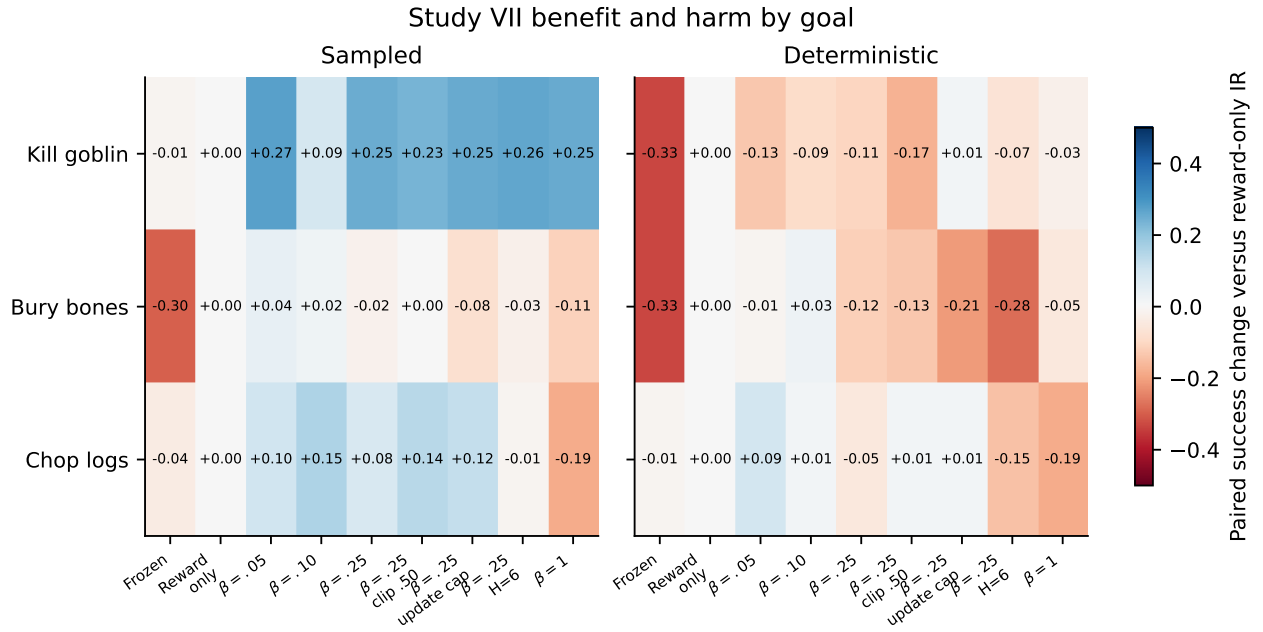


Figure 39: Study VII paired goal-level change from reward-only IR. Blue indicates benefit and red harm. Small value doses broadly improve sampled control, especially kill goblin, while deterministic effects are weaker and often negative.

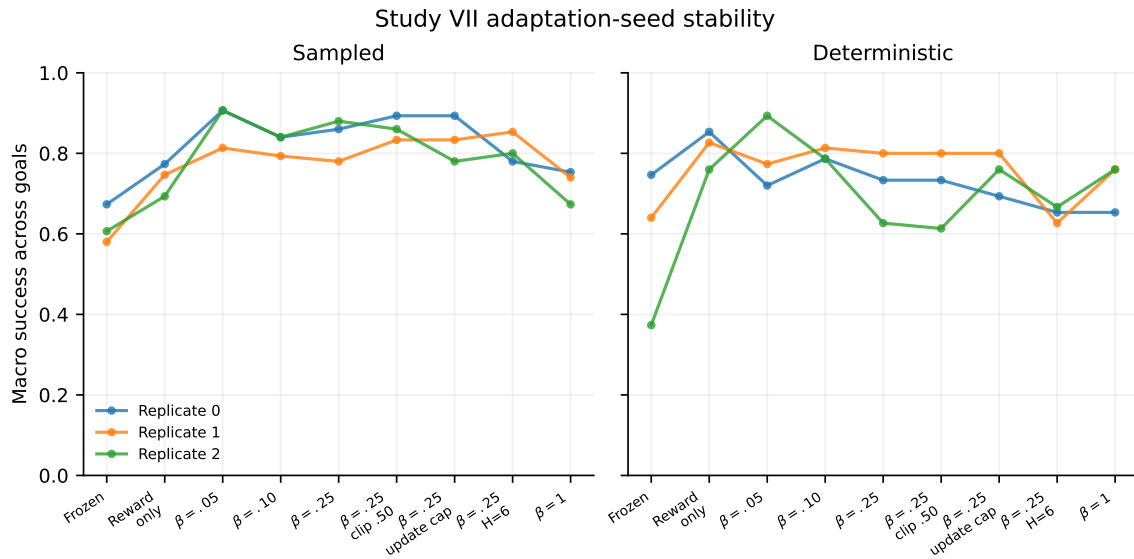


Figure 40: Study VII adaptation-seed stability. The five-percent sampled macro gain is positive in all three fully paired replicates. Deterministic ordering is less stable and remains below reward-only on average.

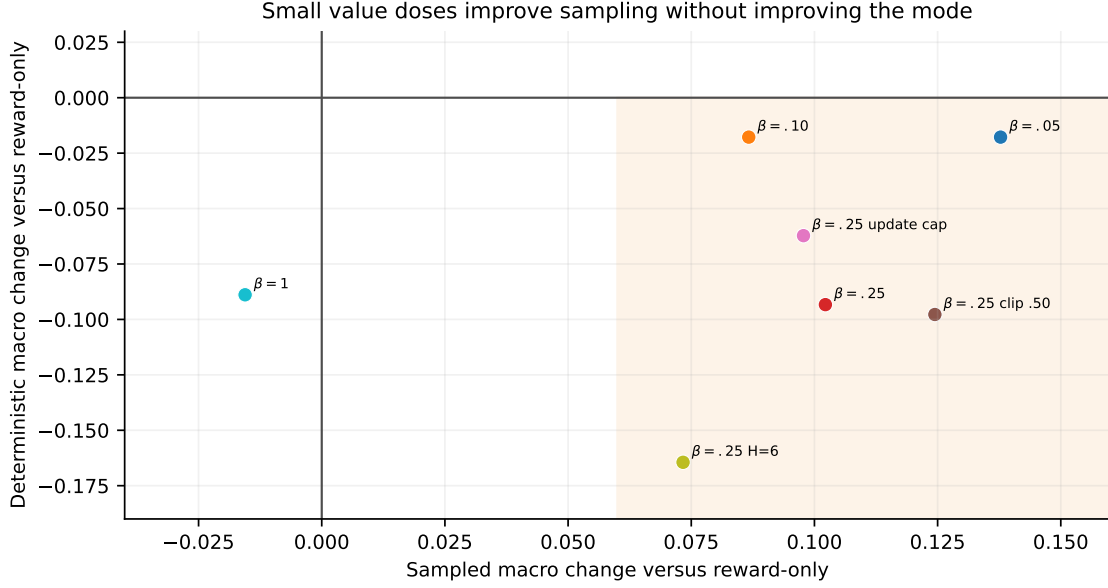


Figure 41: Macro changes relative to reward-only in the two policy modes. Every intermediate-dose treatment lies in the sampled-benefit/deterministic-harm quadrant; the pure bounded anchor is worse in both. This is the central safety result of Study VII.

13.5 Study VII decision and claim boundary

No mixed condition passes the complete advancement rule. The two smallest doses satisfy the sampled clauses, but their deterministic weakest-goal changes are $-.133$ and $-.093$, below the $-.05$ floor. Reward-only $H = 15$ therefore remains the approved teacher for Stage 5. The five-percent correction is retained as the leading gated or regularised candidate, not silently promoted to the default algorithm.

Scientifically, the result is positive but bounded. It establishes that the repaired value head contains behaviorally useful information: a small dose produces a large, replicated, all-goal sampled improvement. It also establishes that calibration on held-out replay states does not guarantee a safe modal actor update on imagined states. This is one repaired checkpoint from one continual-training seed, with actor changes reset every episode. Persistent continual updates, independent world-model seeds, longer task sequences, and training-time critic regularisation remain untested.

14 Study VIII: interpretable gating and compute accounting

14.1 Question, signals, and frozen selection

Stage 5A begins the gating programme. The checkpoint is the Stage VI counterfactual-plus-bounded clone, but the approved actor objective is still reward-only $H = 15$. All model components, heads, normalizers, and persistent checkpoint parameters remain frozen; only an episode-local actor and its optimizer may change. The question is whether a prospective state signal can avoid unnecessary updates without losing the recovery supplied by always-on IR.

Four signal families are measured at eligible factual posterior states. Let $\pi_m(a | s, g)$ be the action distribution induced by posterior sample m and $\bar{\pi} = M^{-1} \sum_m \pi_m$ for $M = 128$. The main uncertainty statistic is the normalized MCPB disagreement

$$D_{JS}(s, g) = H(\bar{\pi}) - \frac{1}{M} \sum_{m=1}^M H(\pi_m), \quad (26)$$

reported separately from factual actor entropy $H(\pi)$ and mean posterior entropy. High entropy can mean that several actions are interchangeable; high JS instead means that posterior beliefs imply different policies. A more expensive $H = 15$ consequence audit also measures imagined goal-success frequency and divergence between the actor mixture and the success-conditioned first-action distribution.

Three fixed gate families follow. An entropy gate and a JS gate accept when their respective statistic exceeds

a frozen quantile threshold. A two-stage gate first applies an entropy-or-JS screen, then pays for consequence imagination and accepts only when at least 4/128 rollouts succeed and their successful-action divergence exceeds a second threshold. If accepted, the same consequence batch is reused for the update. Thresholds are selected in audit and qualification phases before a three-replicate confirmation phase; the selection rule maximizes sampled macro success subject to a deterministic weakest-goal floor of $-.05$ relative to the corresponding always-on objective. No gate candidate actually satisfies that rule, so the least-bad candidate from each family is retained for diagnostic confirmation rather than falsely described as qualified.

Confirmation compares no IR and always-on, entropy-gated, JS-gated, and two-stage-gated variants of standard, reward-only, and $\beta = .05$ mixed IR. Each goal-condition-replicate receives 50 sampled and 25 deterministic episodes: 234 units and 8,775 recorded confirmation episodes. Reset schedules, posterior keys, and action keys are paired. Timers are synchronized and record gate time, update time, posterior samples, imagined transitions, wall time, throughput, and actor updates.

14.2 Fixed gates save work but lose the recovery signal

Table 12 summarizes the central anchors. Always-on reward-only is decisively strongest: sampled macro success is .822, versus .611 for both entropy- and JS-gated reward-only and .549 for the two-stage version. The result is not simply too much updating. Standard IR is a bad teacher, and gating it can improve its poor baseline; entropy-gated standard rises from .418 to .469 sampled and from .289 to .449 deterministic. That does not make standard IR useful relative to reward-only. The scientific failure is that the gates cannot preserve the beneficial objective while withholding only harmful or redundant applications.

Table 12: Study VIII confirmation macro success and update density. Success is averaged over three goals and three paired replicates. Update density is actor updates per real environment step.

Condition	Sampled	Deterministic	Updates/step S	Updates/step D
No IR	.476	.378	.000	.000
Always standard	.418	.289	.993	.994
Entropy standard	.469	.449	.201	.197
JS standard	.371	.338	.503	.496
Two-stage standard	.480	.436	.042	.040
Always reward-only	.822	.653	.990	.991
Entropy reward-only	.611	.516	.245	.273
JS reward-only	.611	.596	.527	.477
Two-stage reward-only	.549	.338	.053	.041

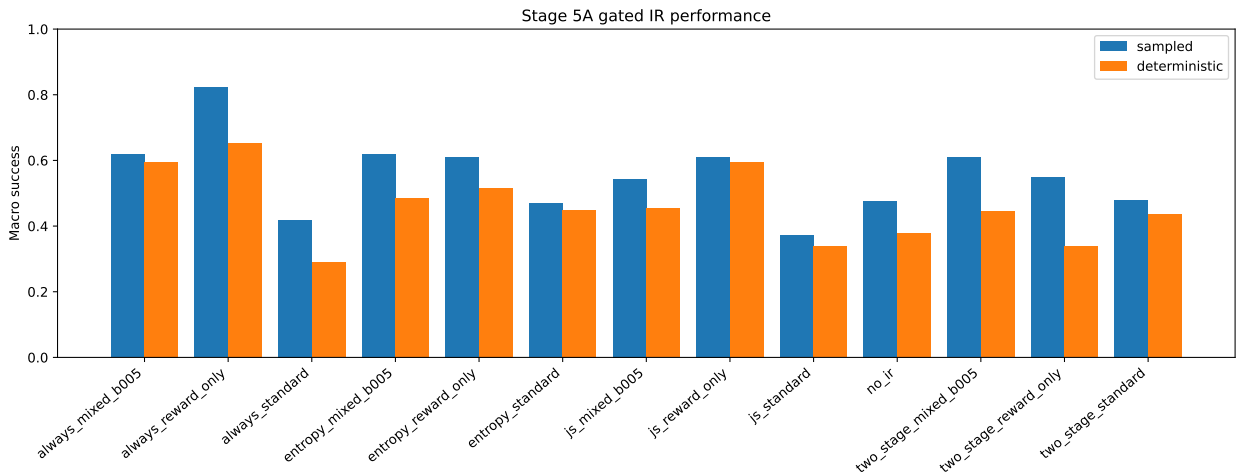


Figure 42: Study VIII real-environment confirmation performance. Fixed gates frequently reduce success relative to their always-on reward-only or mixed anchors; improvements over standard IR do not close the gap to the approved reward-only teacher.

The compute result is real but must be described precisely. Entropy and JS gates remove roughly half to four-fifths of direct update rollouts, while the two-stage gate applies only about four to five updates per hundred

real steps. However, its consequence screen is itself an imagination workload: despite a 95% reduction in actor updates, it saves only about 30% of imagined transitions relative to always-on reward-only. On this machine, gating still reduces the *incremental* wall-time overhead of always-on adaptation, but behavioral damage dominates the trade-off.

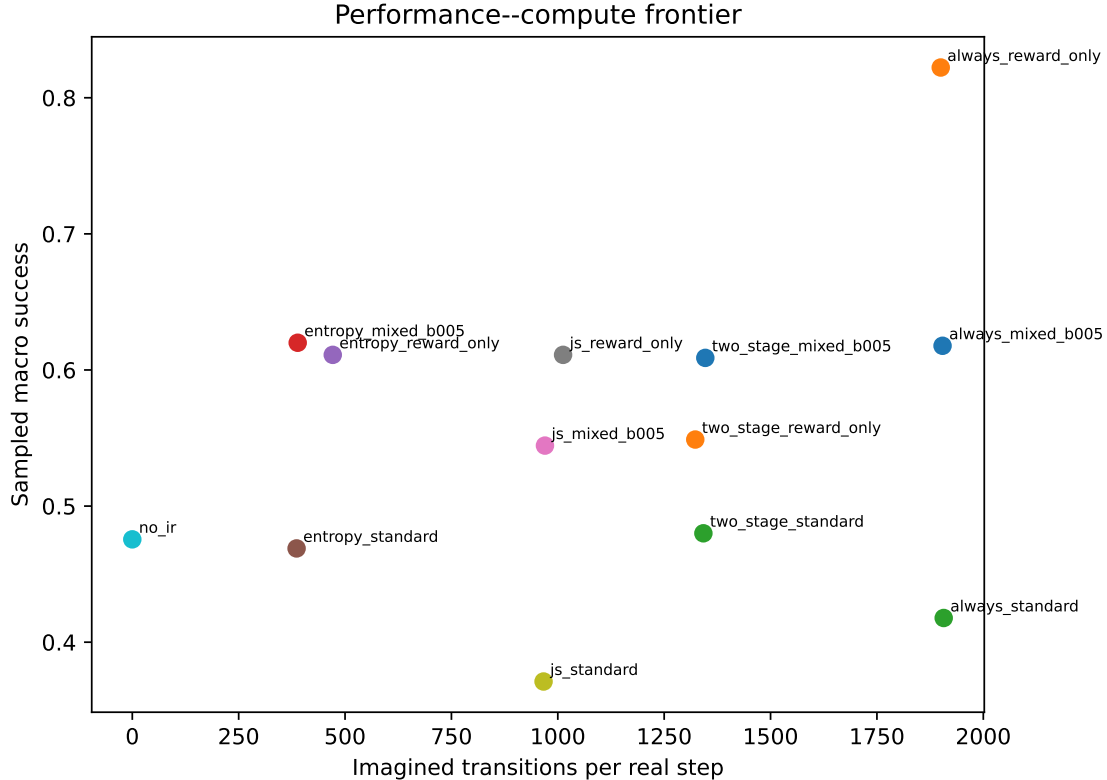


Figure 43: Study VIII performance-compute frontier. The upper-right reward-only anchor is expensive but behaviorally dominant. Moving left by thresholding reduces imagined work, yet no fixed gate lies on a frontier that preserves its success.

14.3 Study VIII decision

No gate advances. The result rejects entropy, a global JS threshold, and the tested consequence threshold as stand-alone necessity rules. It does not reject gating as a research direction. Stage VIII starts every episode from a checkpoint deliberately selected for actor repair and therefore omits the critical negative case: a policy which is already competent and should be left alone. Stage IX addresses that case directly.

15 Study IX: weak-versus-competent IR necessity discrimination

15.1 Calibration and confirmation design

Stage 5B first evaluates all three goals with no IR, always-on standard IR, and always-on reward-only IR. A weak task must gain at least .05 from its best always-on treatment; a distinct strong task must achieve at least .60 without adaptation. The frozen calibration selects *bury bones* as weak (no-IR competence 0, calibration reward-only gain +.50) and *chop logs* as strong (no-IR competence .80). This selection is hashed before confirmation.

The confirmation matrix contains no IR, always-on standard and reward-only, and JS- and two-stage-gated versions of each objective. It uses three paired replicates with 50 sampled and 25 deterministic episodes per cell: 84 units and 3,150 episodes after 270 calibration episodes. Actor changes remain episode-local. The intended signature is separation: retain the weak-task IR benefit while reducing strong-task updates and preventing damage.

15.2 The JS direction is plausible, but behavioral discrimination fails

Table 13: Study IX success on the calibrated competent chop task and weak bury task. S/D denote sampled/deterministic policy evaluation.

Condition	Chop S	Chop D	Bury S	Bury D
No IR	.820	.747	.427	.000
Always standard	.407	.373	.167	.040
JS standard	.713	.373	.080	.133
Two-stage standard	.693	.493	.180	.373
Always reward-only	.860	.760	.653	.653
JS reward-only	.867	.840	.320	.480
Two-stage reward-only	.900	.720	.280	.240

JS does move in the expected qualitative direction: reward-only activation is .582/.496 updates per step on weak sampled/deterministic bury versus .384/.398 on strong chop. Yet it still withholds too much useful repair, losing 33.3 sampled and 17.3 deterministic percentage points relative to always-on reward-only on bury. The two-stage rule is worse as a necessity detector. It activates only .013 updates per weak bury step but about .155-.167 per strong chop step—the opposite of the required ordering.

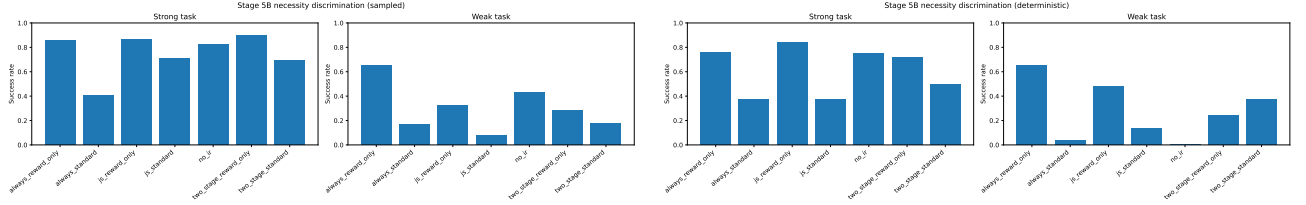


Figure 44: Study IX sampled (left) and deterministic (right) success. The gated conditions protect competent chop reasonably well but do not retain always-on recovery on weak bury.

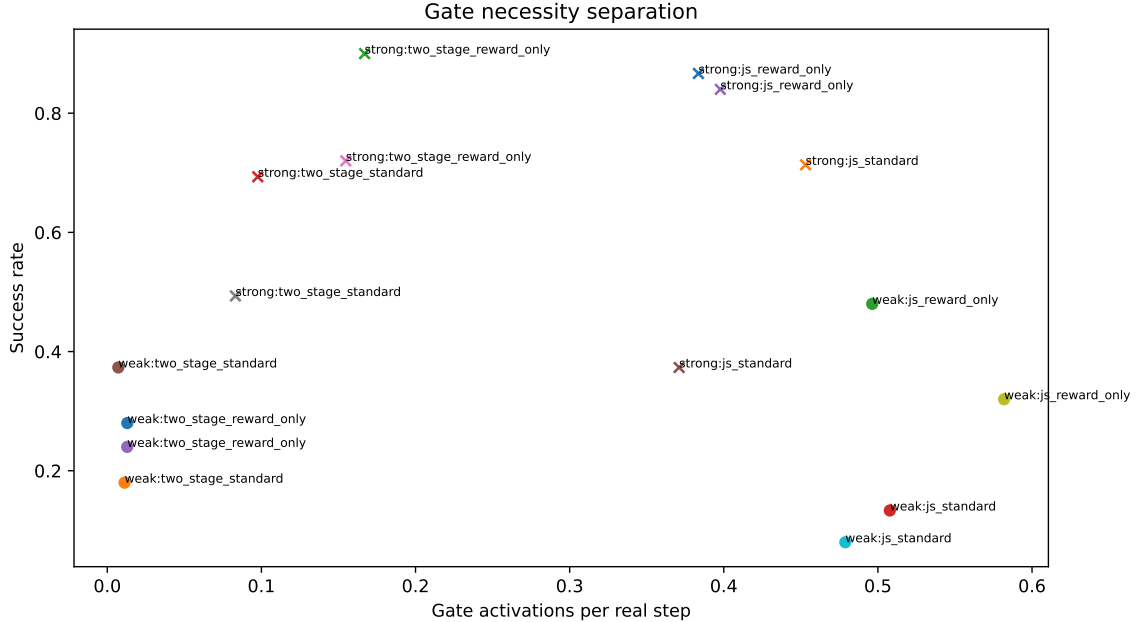


Figure 45: Study IX gate activation against real success. JS activates more on weak bury, but this does not translate into sufficient recovery. The two-stage gate is nearly dormant on the task that most needs adaptation.

Episode lengths are retained in every unit in addition to binary success, so successful agents can be distinguished after reaching 100%. In this experiment success remains the binding endpoint: no gated weak-task condition approaches the always-on rate, so shorter completion among the surviving successes cannot reverse the decision.

15.3 Study IX decision

Stage 5C persistent recovery is paused. Persisting a gate already shown to misclassify the weak and competent cases would turn a diagnostic failure into multi-episode policy damage. The next experiment therefore returns to the recorded prospective signals and asks which of two logically separate facts they contain: whether the current actor will underperform, and whether IR will actually repair it.

16 Study X: offline performance-gate audit

16.1 Prospective features and retrospective labels

Stage 5B.1 is a non-interventional audit of 90 reset-matched episode pairs from the Stage VIII traces. Predictors use only the first 1, 5, 10, or 25 factual states and therefore precede the outcome. Each no-IR episode is paired with its always-on reward-only counterpart and assigned one retrospective class:

`ir_needed` : no IR fails, IR succeeds,
`ir_unnecessary` : both succeed,
`ir_harmful` : no IR succeeds, IR fails,
`not_repaired` : both fail.

The resulting counts are 21, 27, 18, and 24, respectively. Labels are used only after execution to measure separability; a deployable gate never observes them.

The predictors include mean/max JS, factual and posterior actor entropy, imagined $H = 15$ success fraction, successful-action concentration/divergence, and combinations. “Predicted failure” is one minus the imagined success fraction. AUROC is interpreted as ranking information, not a safe threshold. Task-specific optimal thresholds are then applied across every other task to measure transfer directly.

16.2 Imagined failure predicts performance; JS does not calibrate globally

Table 14: Study X pooled feature AUROC. Underperformance asks whether no IR will fail; IR needed asks whether the paired reward-only intervention converts failure to success.

Predictor	First 10 states		First 25 states	
	Underperform	IR needed	Underperform	IR needed
Mean JS	.501	.618	.527	.564
Factual actor entropy	.349	.476	.376	.518
Posterior actor entropy	.340	.467	.366	.506
Predicted failure	.669	.584	.772	.640
JS $\times$ predicted failure	.517	.629	.548	.575

The simplest result is decisive: pooled ten-state JS is chance for later underperformance. Entropy is worse than chance because confident/uncertain behavior is task- and policy-mode-dependent, not a monotonic competence score. Imagined failure is the best current performance estimator, improving from AUROC .669 at ten states to .772 at 25. It is less effective at answering the second question—whether IR repairs the failure—where JS and the JS–failure interaction retain some modest information.

The pooled numbers hide severe cell variation. At ten states, predicted failure AUROC ranges from .484 for sampled bury to .875 for deterministic kill. JS underperformance AUROC ranges from .238 for deterministic bury to .684 for sampled chop. Consequently, a threshold tuned on one task often has poor true-positive/false-positive behavior off diagonal. A fixed global threshold is therefore the wrong abstraction even when a feature has useful within-task ranking.

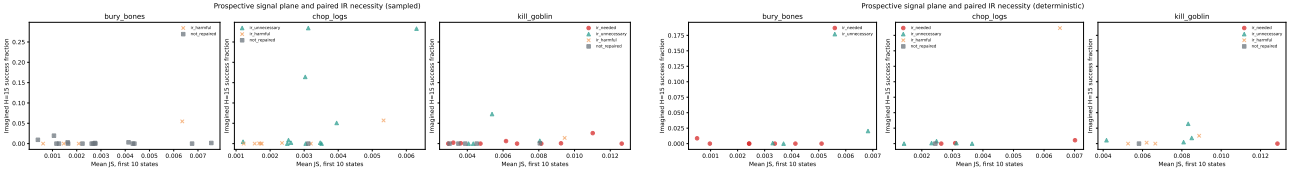


Figure 46: Study X prospective signal planes. Mean JS and imagined success overlap substantially across later necessity classes; the geometry also changes with task and policy mode.

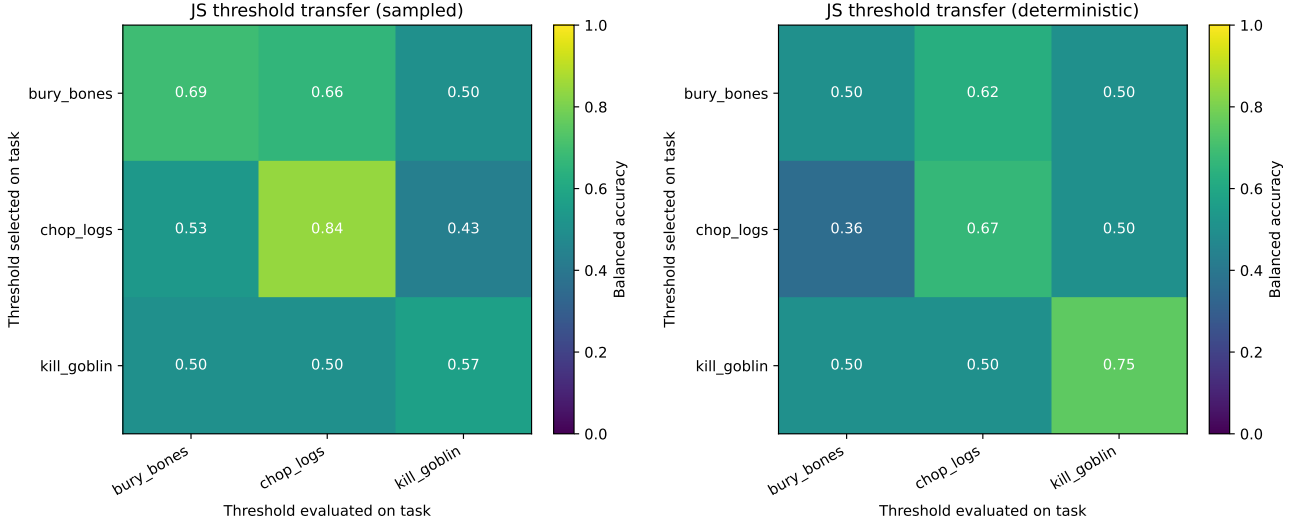


Figure 47: Study X JS-threshold transfer. Rows tune the threshold and columns evaluate it; weak off-diagonal behavior shows that the numerical boundary is task-specific.

16.3 Study X decision

Gating is separated into two estimands. First estimate “will this actor underperform?”; imagined failure is the best available signal. Then estimate “would this particular update improve it?”; JS alone is insufficient. The historical traces do not contain a held-out pre/post score for an update, so Study XI adds exactly that missing prospective comparison.

17 Study XI: paired virtual-update acceptance

17.1 Causal gate and lifecycle

At a probe state, Stage 5B.2 evaluates the current actor on held-out imagination batch B , performs one tentative reward-only update on independent batch A , and reevaluates the tentative actor on B using identical posterior and action-sampling random numbers. Formally,

$$\theta' = U_A(\theta), \quad \Delta_{\text{virt}} = \hat{J}_B(\theta') - \hat{J}_B(\theta), \quad \theta \leftarrow \begin{cases} \theta' & \Delta_{\text{virt}} > 0, \\ \theta & \text{otherwise.} \end{cases} \quad (27)$$

The score is the mean reward-only imagined return. Rejection restores both actor and adaptation-optimizer state exactly. Batch separation prevents training-batch approval, while common random numbers reduce comparison variance. The world model, reward/continuation heads, all value heads, representation, dynamics, normalizers, and source checkpoint stay frozen.

Every episode starts dormant and probes immediately. If rejected, it probes again after 25 or 100 real steps. An accepted probe activates stepwise IR; three consecutive rejections return the gate to dormancy. The absolute baseline-score screen is intentionally disabled because Study X rejected a portable threshold, but its score is still logged for diagnosis.

The four conditions are no IR, always-on reward-only, and virtual gates with 25- and 100-step dormant cadence. Each uses $H = 15$, 128 posterior samples, one actor update at learning rate 4×10^{-5} , and zero entropy bonus.

Three goals, sampled/deterministic modes, three paired replicates, and 50/25 episodes per cell produce 2,700 recorded episodes. The gate additionally logs every proposal, predicted change, acceptance, active/dormant transition, rollback, posterior sample, imagined transition, and synchronized runtime.

17.2 Mixed task behavior rejects the virtual score as a safety test

Table 15: Study XI sampled real-environment success (150 episodes per goal and condition).

Condition	Kill	Bury	Chop	Mean
No IR	.480	.373	.833	.562
Always reward-only	.653	.660	.807	.707
Virtual, dormant 25	.653	.000	.867	.507
Virtual, dormant 100	.487	.160	.853	.500

Table 16: Study XI deterministic real-environment success (75 episodes per goal and condition).

Condition	Kill	Bury	Chop	Mean
No IR	.347	.373	.813	.511
Always reward-only	.533	.653	.747	.644
Virtual, dormant 25	.453	.200	.907	.520
Virtual, dormant 100	.440	.187	.787	.471

The gate appears promising if only chop is inspected. The 25-step condition raises sampled chop from .833 no-IR and .807 always-on to .867, and deterministic chop to .907. It also matches always-on sampled kill at .653. Bury overturns that conclusion: the same gate reduces sampled success to zero and deterministic success to .200. The 100-step cadence is less destructive but also recovers less kill and remains below no IR on bury. This is precisely why all three tasks and both policy modes are required for a safety claim.

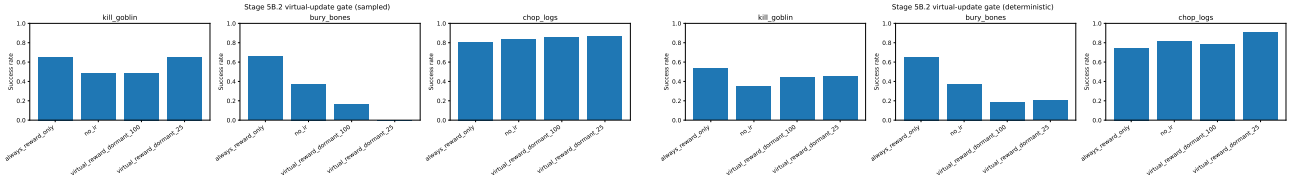


Figure 48: Study XI real-environment success. Virtual acceptance is highly task dependent: strong chop gains coexist with catastrophic bury damage.

Retrospective paired labels expose why. With 100-step cadence, 534 episodes receive at least one IR update, 27 known repair opportunities receive none, and 272 already competent episodes are nevertheless updated. With 25-step cadence these counts are 645, 6, and 346. The first virtual improvement ranks true IR need at AUROC .528 and .470, respectively—chance. Even the maximum observed virtual improvement reaches only .558–.561. Initial predicted failure is more informative about real underperformance (AUROC .602 and .659), reproducing Study X’s separation between performance need and intervention utility.

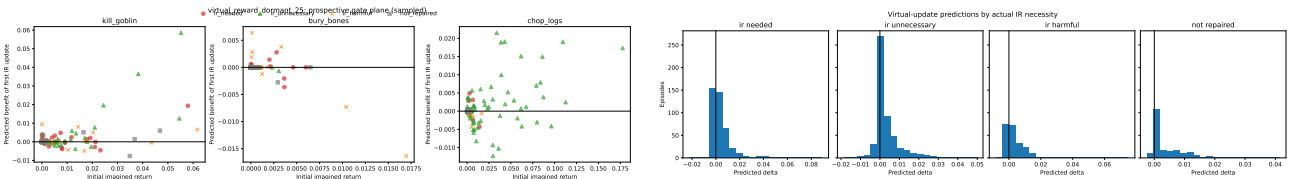


Figure 49: Study XI decision diagnostics. Predicted improvement overlaps across real necessity classes; a positive imagined pre/post score is not evidence of positive real-environment effect.

17.3 Compute and completion speed

No IR takes .1197 seconds per real step on average and always-on reward-only takes .1296. The 100-step virtual gate averages .1275: it saves part of the *adaptation overhead*, but only about .0022 seconds per total step. The 25-step gate averages .1371 and is therefore 5.7% slower than always-on IR. Its accepted update density is only .239 per step, but each tentative decision requires a before rollout, a training rollout/update, and an after rollout. Avoided updates do not imply avoided compute.

Successful-episode length is also reported rather than allowing a 100% success rate to hide efficiency. For example, sampled kill successes average 72.0 actions under the 25-step gate versus 81.2 always-on; deterministic chop successes average 69.3 versus 53.4, despite the higher gated success rate. These secondary differences are useful but do not compensate for the bury failure.

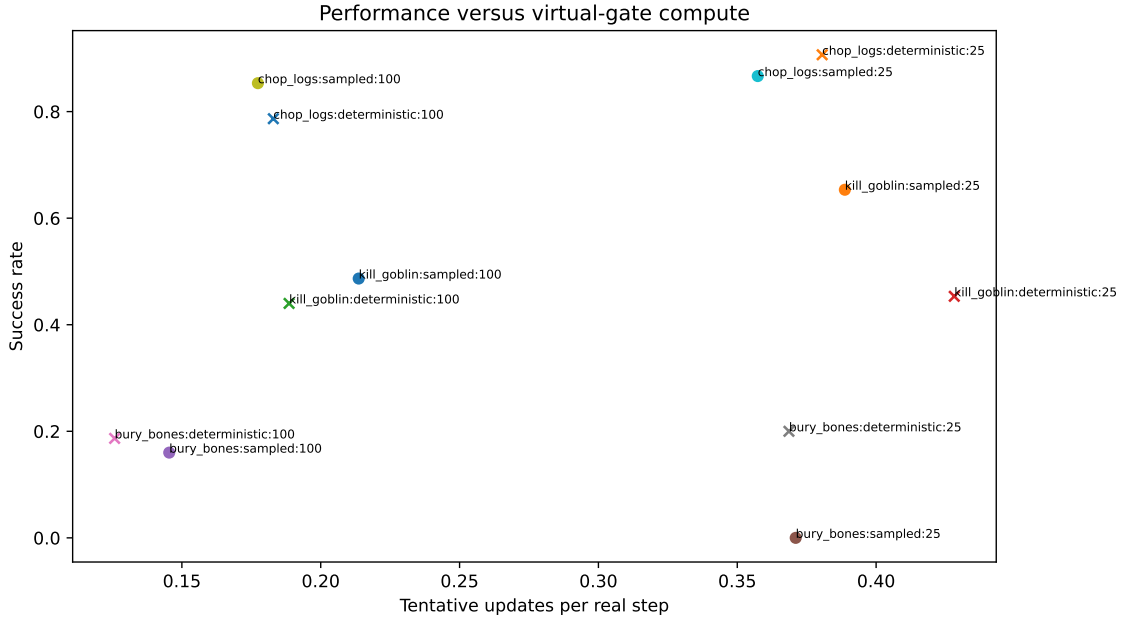


Figure 50: Study XI virtual-gate performance versus tentative-update density. Sparse probing lowers proposals, but performance is task dependent and the 25-step condition is not a wall-time saving.

17.4 Study XI decision and claim boundary

The simple virtual-improvement gate is rejected as a deployable safety mechanism. Its paired design is technically sound and its negative result is scientifically strong: even an independent held-out imagination batch, exact common-random-number comparison, and optimizer rollback do not make the model a reliable judge of its own policy update. The failure is not mere threshold calibration; the ordering statistic itself is close to chance for true IR benefit.

This does not contradict the earlier result that reward-only imagination can train an actor. A model can provide a useful aggregate learning gradient while remaining too miscalibrated to rank the real consequence of individual updates. Stage 5C remains deferred. The next gate should combine an explicit underperformance estimate with uncertainty or conservative evidence about update utility, while hard KL/update rollback is developed as a separate damage bound rather than mistaken for a necessity detector.

18 Study XII: bury-bones virtual-gate autopsy

18.1 Question and controlled design

Study XI's sampled bury result is too severe to treat as ordinary gate noise: the 25-step virtual gate falls from .660 under always-on reward-only IR to exactly zero. Study XII reproduces this cell and asks three narrower questions. First, is the actor passing through a harmful partially repaired state before enough IR updates accumulate? Second, does the virtual score select updates that are worse than an equal opportunity to update at random real steps? Third, does a small, nominally independent value signal regularize a reward model that

may otherwise be exploited?

The experiment starts every episode from the same repaired Stage VI counterfactual-plus-bounded checkpoint. Only the actor and its adaptation optimizer can change, and both reset at the next episode. Dynamics, representation, decoder, reward, continuation, original critic, slow critic, bounded value head, and all normalizers remain frozen. The task is bury bones, the policy is sampled, and episodes are capped at 200 real actions. Every IR proposal uses horizon $H = 15$, 128 posterior starts, one actor update at learning rate 4×10^{-5} , and zero entropy bonus. The environment and agent seed bases, 121000 and 131000, exactly reproduce the Study XI sampled schedule.

Let $u_t \in \{0, 1\}$ indicate whether a candidate actor update is committed at real step t . Prefix-dose conditions use

$$u_t = \mathbb{I}[t \leq B], \quad B \in \{16, 46, 80, 127\}, \quad (28)$$

whereas the random control samples a fixed set S of at most 46 step indices per episode and uses $u_t = \mathbb{I}[t \in S]$. Because successful episodes can terminate before all selected steps, requested and realized update counts are reported separately. Virtual-positive commits when the paired held-out score from Study XI satisfies $\Delta_{\text{virt}} > 0$; virtual-negative reverses the decision. The latter is an intentionally anti-causal diagnostic, not a candidate algorithm.

The reward-only actor loss is compared with an exact mixed loss

$$\mathcal{L}_{.005} = .995 \mathcal{L}_{\text{reward}} + .005 \mathcal{L}_{\text{value}}. \quad (29)$$

This coefficient is exactly .005; it must not be confused with the earlier historical label `mixed_b005`, which denoted $\beta = .05$. Four mixed conditions repeat always-on, 46-prefix, random-46, and virtual-positive schedules. No IR, a diagnostic no-IR world-audit condition, and always-on reward-only complete the 14-condition matrix. Three paired replicates use 50 episodes per main condition and 20 per world-audit condition, giving 2,010 recorded episodes. Success, episode length, realized updates, full real-action histograms, virtual before/after scores, predicted reward events and peaks, and actions immediately preceding predicted peak reward are retained.

18.2 Behavioral result: update placement dominates nominal dose

Table 17: Study XII sampled bury-bones result over three paired replicates. Mean updates are realized actor commits per episode; successful episodes can terminate before a cap or random schedule is exhausted.

Condition	Success	Mean length	Mean updates
No IR	.340	158.5	0.0
No IR + world audit	.233	175.0	0.0
Always reward-only	.607	124.2	124.2
Reward prefix cap 16	.127	186.2	15.8
Reward prefix cap 46	.113	187.6	44.9
Reward prefix cap 80	.207	179.7	77.3
Reward prefix cap 127	.613	119.8	91.4
Reward random 46	.540	134.0	30.9
Virtual-positive reward, dormant 25	.000	200.0	48.6
Virtual-negative reward, dormant 25	.173	179.9	16.1
Always mixed $\beta = .005$	.000	200.0	200.0
Mixed $\beta = .005$ prefix cap 46	.127	184.0	44.7
Mixed $\beta = .005$ random 46	.540	135.7	31.2
Virtual-positive mixed $\beta = .005$, dormant 25	.260	172.6	39.2

Always-on reward-only IR improves success by 26.7 percentage points over no IR, and the 127-update prefix reaches an essentially identical .613. The shorter caps do not interpolate smoothly between those endpoints: 16, 46, and 80 updates all fall below no IR, reaching only .127, .113, and .207. Consecutive early updates can therefore move the actor into a partial-repair region that is worse than leaving it untouched. Sufficient continued updates eventually pass through that region.

The random control is decisive. Randomly distributing at most 46 update opportunities reaches .540 and a mean length of 134.0, whereas committing the first 46 reaches .113 and length 187.6. Random successes terminate after a mean of 30.9 realized commits, so the result is not explained by receiving more optimization. Spreading adaptation across new posterior states is much safer than applying the same nominal budget consecutively near episode start.

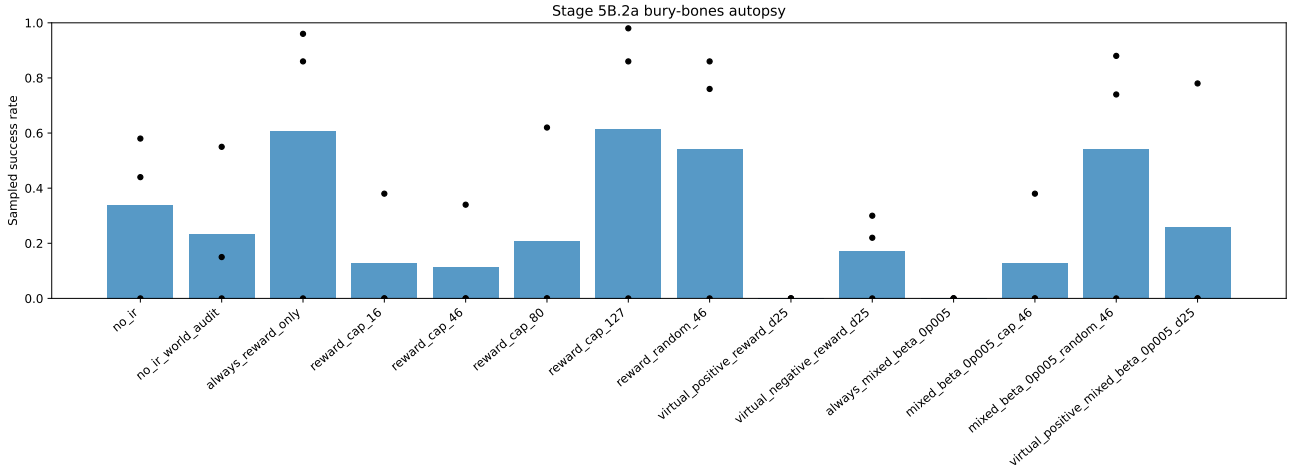


Figure 51: Study XII aggregate and replicate-level sampled success. Bars pool 150 episodes for main conditions and 60 for the audit; black points are replicate rates and reveal substantial seed interaction.

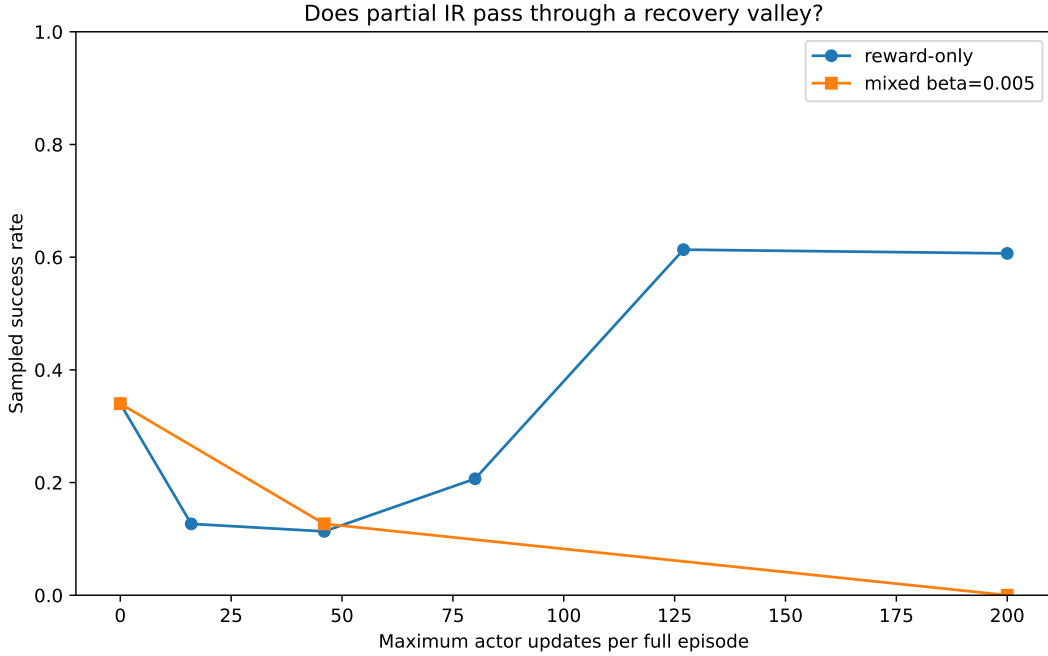


Figure 52: Prefix-dose response. Reward-only IR exhibits a pronounced partial-repair region before recovering near the 127-update cap. The exact $\beta = .005$ mixture is not a benign interpolation: always-on mixed adaptation collapses.

18.3 Virtual selection is worse than random selection

Virtual-positive reward IR commits 48.6 updates per episode—close to the 46-update prefix and more than the successful random schedule—yet produces zero successes in all three replicates. Reversing the decision reaches .173. This inversion remains below the .340 frozen actor and is not useful as a policy, but its strict improvement over the intended direction supports the specific conclusion that the learned score orders at least some proposals in the wrong real-world direction. The result strengthens Study XI: failure is not only an ill-chosen scalar threshold. Positive imagined improvement selects an especially harmful subset of candidate updates.

The scatter in Figure 53 shows the missing link directly. Virtual-positive reward proposals receive positive imagined before/after changes over a broad range, but every corresponding real episode fails. Successful virtual-negative and mixed episodes occupy overlapping imagined-score ranges. Mean predicted change is therefore neither calibrated to real success nor a safe ordinal utility statistic.

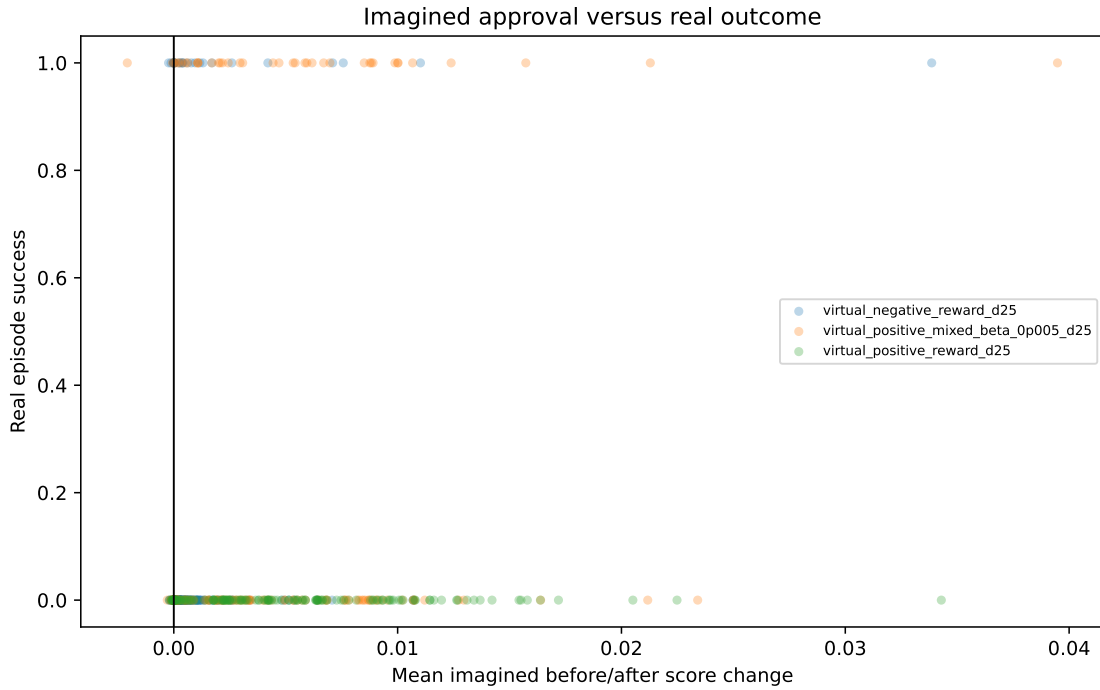


Figure 53: Imagined approval versus real outcome for the three virtual conditions. Positive model-predicted change does not separate real successes, and the intended reward-only selector occupies only the failure row.

18.4 A tiny value term is a qualitative intervention

The exact $\beta = .005$ control does not provide general protection. Always-on mixed IR fails all 150 episodes, compared with .607 reward-only, even though the value term carries only one two-hundredth of the weight used in Study VII’s best sampled condition. Under random scheduling, reward-only and mixed IR both produce exactly 81 successes (.540). Under the virtual selector, the mixed variant reaches .260 while reward-only remains at zero, but this aggregate is driven entirely by replicate 1 (.780 versus zero in replicates 0 and 2). The value gradient can rotate the adaptation trajectory dramatically, but the direction is neither stable nor predictably conservative.

Table 18: Selected Study XII success rates by paired replicate. Variation is across environment and adaptation random streams around one frozen learned model, not across independently trained models.

Condition	Replicate 0	Replicate 1	Replicate 2
No IR	.00	.44	.58
Always reward-only	.00	.86	.96
Reward prefix cap 46	.00	.00	.34
Reward prefix cap 127	.00	.98	.86
Reward random 46	.76	.00	.86
Virtual-positive reward	.00	.00	.00
Virtual-negative reward	.00	.30	.22
Always mixed $\beta = .005$	.00	.00	.00
Mixed random 46	.74	.00	.88
Virtual-positive mixed $\beta = .005$	.00	.78	.00

These interactions are scientifically important. Always-on reward-only is beneficial in two seeds and ineffective in one; random scheduling reverses which seed benefits; and virtual mixed repair appears in only one seed. Three episode-level replicates establish that the aggregate failure is not a single bad rollout, but they do not establish population stability across independently trained world models.

18.5 Action attractors and the failed no-intervention audit

The failed policies are not distinguished by one scalar action-concentration measure. The real late-episode top-click fraction is .350 for virtual-positive reward, compared with .332 for no IR and .409 for the successful random schedule. Figure 54 nevertheless shows that conditions redistribute probability among a small set of interface regions; virtual-positive reward places the strongest mass near grid column 23, row 6. The actions preceding the model’s peak predicted reward instead concentrate near column 21, row 8 (Figure 55). This visual misalignment is consistent with a distorted imagined strategy, but does not by itself establish a causal hallucinated-reward loop.

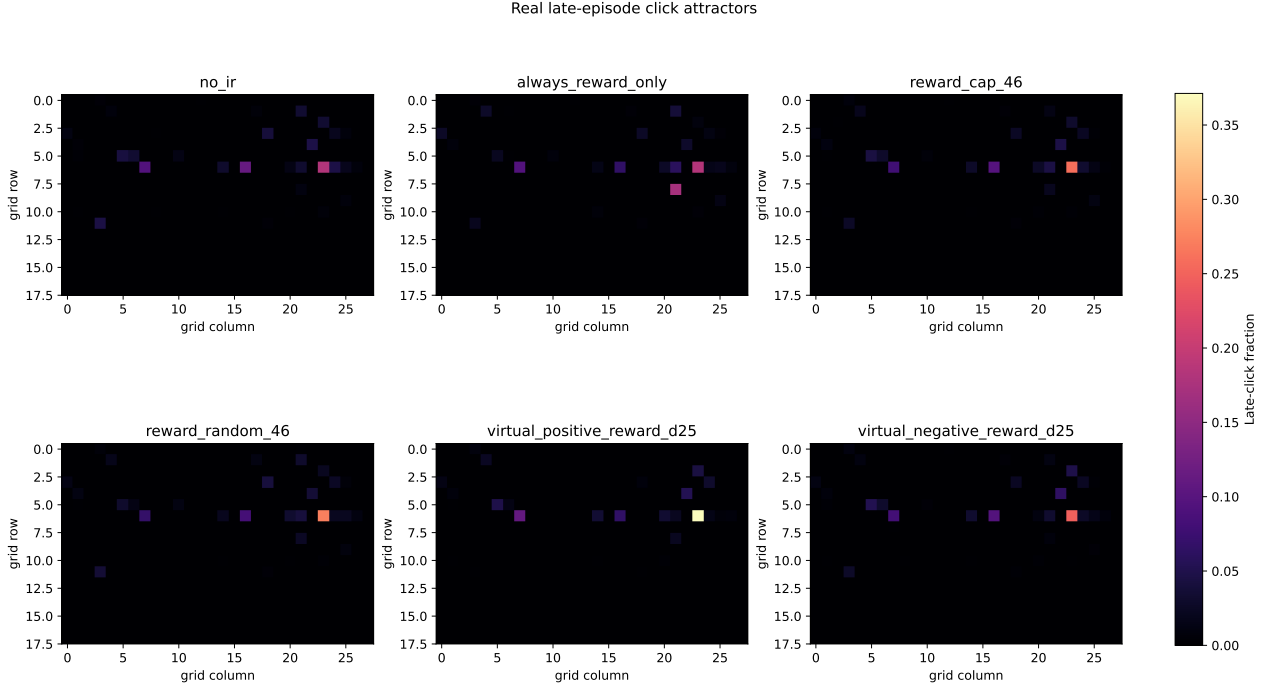


Figure 54: Late-episode real click distributions for selected conditions. Virtual-positive reward forms a strong spatial attractor, but action concentration alone does not distinguish recovery from collapse.

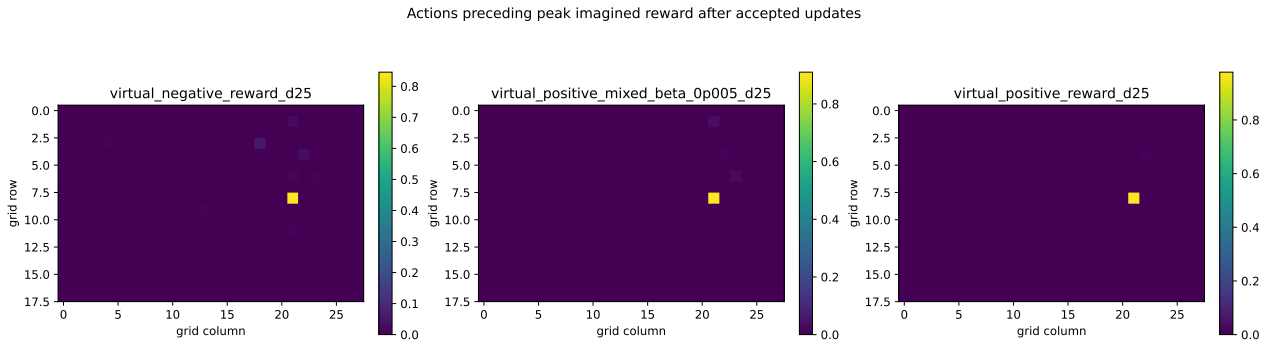


Figure 55: Actions immediately preceding peak reward in frozen-model imagined rollouts after accepted virtual updates. All virtual variants concentrate on a narrow predicted-reward action region.

The intended no-intervention check fails. Reset seeds are identical, but the no-IR world-audit condition matches only 1,276 of 8,916 paired real actions (.143). Its success rate, .233 over 60 episodes, also differs from the .340 150-episode no-IR reference, most visibly in replicate 2 (.150 versus .580). The likely implementation-level explanation is that computing live model probes perturbs or consumes a random stream used by the sampled actor; an error in action alignment is another possibility. Consequently, the 29.5 MB imagined-reward diagnostic table is retained for provenance, but it is not treated as a paired no-op counterfactual. The behavioral dose/scheduling/virtual comparisons remain valid paired-seed experiments, but this study cannot cleanly prove

that a particular learned-reward hallucination caused them.

18.6 Study XII decision and claim boundary

The original virtual-positive gate remains rejected. Study XII localizes two mechanisms behind the bury catastrophe: consecutive early adaptation has a wide partial-repair region, and the positive imagined selector is substantially worse than random placement at a comparable nominal budget. Exact $\beta = .005$ mixed IR is also rejected as a generic safety regularizer. The best current operational reference remains externally invoked, episode-local reward-only IR; when it is known that repair is required, continuing through roughly 127 update opportunities is safer than stopping after a small prefix. Random scheduling is promising as an adaptation-cadence control, not yet as a necessity gate.

The model-exploitation hypothesis remains open. Before another live model autopsy, probe computation must use an isolated RNG namespace and demonstrate exact no-IR action equality under sampled control. Only then should imagined reward peaks be correlated with later real actions and task progress. Stage 5C remains deferred: persistence would compound both the partial-repair valley and the demonstrated selection error.

19 Cross-study synthesis

The experiments support the following chain of evidence:

1. Reservoir replay retains old observations and can temporarily recover all three behaviors, but balanced availability does not stabilize continual optimization.
2. Factual completion detection survives collapse, so failure is not wholesale erasure of reward examples.
3. Factual-only replay does not teach the same latent completion state to mean different things under different goals; reward and continuation largely ignore the requested goal in counterfactual queries.
4. The critic is badly overoptimistic, loses ranking ability, and grows more optimistic after real behavioral collapse.
5. The standard six-step IR return is 94%–97% critic bootstrap, so actor updates mostly inherit critic error rather than predicted task reward.
6. Entropy is small in the audited loss and removing it does not rescue controlled recovery.
7. Removing critic bootstrap produces large, paired, real-environment recovery. IR is therefore viable, but the current Dreamer value target is not a safe test-time teacher.
8. Critic-free recovery transfers from artificial actor masking to unmodified continual checkpoints: it improves all twelve checkpoint–mode–horizon aggregate comparisons and nearly restores naturally forgotten 1.5M bury behavior.
9. The transfer is not uniform. Strong chop can be damaged, and the best horizon changes by checkpoint and policy mode; a deployable method therefore requires a gate and conservative update constraint rather than unconditional optimization.
10. Critic error is already severe on real posterior states and worsens on an identical state bank after continual collapse; imagination depth is not required to produce it.
11. Actor-selected imagination is a secondary amplifier: it chooses higher-bootstrap trajectories that often have lower predicted reward than paired recorded actions.
12. The slow critic closely reproduces the online critic’s scale and error, so online–slow agreement cannot serve as a safety gate in its current form.
13. Counterfactual event labels repair the measured goal-semantic defect: held-out completion-goal accuracy rises from 33.3% to 97.4%, while matched factual-only updates leave it near chance.
14. A separately bounded factual success head repairs critic scale and improves initial-state ranking, most strongly for bury, without changing the historical critic or actor.

15. Head-level repair is not equivalent to behavioral superiority. Factual-only reward IR gives the strongest sampled mean and worst-task result; bounded-bootstrap IR remains harmful on chop at both tested horizons.
16. Environment-seed pairing is insufficient for a low-variance IR assay when posterior and policy randomness inside adaptation are not also paired. The unused-bounded-head reward-only contrast exposes this requirement directly.
17. Under full environment and agent-PRNG pairing, small bounded-value corrections contain reproducible behavioral information: $\beta = .05$ improves sampled macro success by 13.8 percentage points over reward-only in all three replicates and improves every sampled goal.
18. Value dose is non-monotonic. Pure bounded teaching is worse than reward-only, and neither correction clipping, an actor-update RMS cap, nor shortening H makes the mixed objective uniformly safer.
19. Sampled improvement is not modal-policy safety. Every mixed condition improves the sampled macro but harms the deterministic macro relative to reward-only; no condition passes the prespecified deterministic weakest-goal floor.
20. Fixed entropy, JS, and consequence gates reduce actor updates and imagined work, but none preserves always-on reward-only recovery. Compute reduction and behavioral selectivity are therefore separate properties.
21. JS has the right coarse activation direction in the weak-versus-competent assay, yet its global threshold removes too much weak-task repair. The two-stage consequence gate activates in the wrong task direction.
22. Prospective imagined failure predicts later real underperformance better than entropy or JS, reaching pooled AUROC .772 over the first 25 states. Its numerical calibration and the JS boundary nevertheless change substantially across tasks and policy modes.
23. Predicting underperformance is not the same as predicting IR utility. The paired necessity classes contain both harmful IR and unrepaired failures, so “the actor looks weak” cannot by itself justify an update.
24. A paired virtual pre/post update test does not solve that second problem. Its first predicted improvement ranks true repair need near chance, approves many competent episodes, and catastrophically damages bury despite exact rollback of rejected proposals.
25. Gating compute must include the decision itself. A 25-step virtual cadence is slower than always-on IR because each tentative decision evaluates before and after on held-out imagination in addition to the training rollout.
26. The bury catastrophe is not explained by update count alone. A 46-update prefix reaches 11.3%, a matched random schedule reaches 54.0%, and virtual-positive selection reaches zero while committing 48.6 updates per episode.
27. Actor repair has a harmful partial-dose region. Prefix caps of 16–80 all underperform the frozen actor, whereas a 127-update cap reaches 61.3% and matches always-on reward-only IR.
28. Reversing the virtual selector improves reward-only success from zero to 17.3% but remains harmful, supporting wrong-direction ranking without making the inverse a viable gate.
29. A value coefficient as small as $\beta = .005$ is not automatically conservative. Always-on mixed IR fails all 150 bury episodes, while random placement makes reward-only and mixed conditions tie at 54.0%.
30. Live model auditing is itself an intervention unless its random streams are isolated. The Study XII no-IR audit matches only 14.3% of paired sampled actions, so its imagined-reward traces cannot yet prove model exploitation.

The initial hypothesis—that model exploitation slowly inflates critic targets during continual learning—is therefore only partly correct. Actor exploitation is measured directly, but the larger error precedes it: the critic is already badly calibrated at real posterior states. Nor does value grow progressively from $H = 6$ to $H = 20$. Study VI further separates critic scale from actor-teacher quality. Bounding removes the impossible value scale but does not eliminate bad task-specific gradients; counterfactual labels remove goal aliasing but do not improve over factual reward-only adaptation. Study VII then shows that this bounded head is not useless: at low weight it carries reproducible policy information that reward-only IR lacks. The remaining defect is dose- and policy-mode-dependent rather than a binary good/bad critic distinction. Recursive target bias, imagined-state

generalization, objective construction, continual distribution shift, and modal-policy concentration therefore remain distinct candidate failure mechanisms rather than one undifferentiated “weak conditioning” explanation.

The gating programme adds a second important distinction. The world model can supply a useful reward-only gradient over many states without supplying a calibrated one-step estimate of whether a particular policy update will help in the real environment. Aggregate actor learnability and update-level self-evaluation are not equivalent capabilities. Study X suggests a practical decomposition: first estimate the probability of underperformance, then estimate intervention benefit conditional on uncertainty about that estimate. Study XI shows that raw imagined mean improvement is not yet a sufficient second estimator. Study XII goes further: the same nominal update budget is useful when distributed randomly, harmful as an early prefix, and catastrophic when chosen by positive virtual improvement. Any next acceptance test must therefore account for both model error and adaptation cadence, and it must demonstrate that its own diagnostic computation does not perturb the sampled policy.

20 Revised experimental ladder

The chronology changes the priorities proposed after Study I.

1. **Stage 4A (completed): counterfactual and bounded head repair.** Counterfactual labels repaired held-out reward/continuation semantics and the bounded head repaired value scale and ranking. The behavioral hypothesis was mixed: factual reward-only IR remained strongest, while bounded IR retained goal-specific harm.
2. **Stage 4B (completed): conservative value-dose and stability assay.** Fully paired replication identified a large sampled benefit at $\beta = .05$, but every mixed condition failed the deterministic weakest-goal safety floor. Reward-only remains the approved base teacher; the small correction remains a candidate to activate only under a gate or stronger constraint.
3. **Stage 5A (completed): interpretable gate and compute audit.** Entropy, global MCPB JS, and consequence gates reduce updates but do not preserve reward-only recovery. None passes its frozen advancement rule.
4. **Stage 5B (completed): necessity discrimination.** JS activates more often on weak bury than competent chop but still loses most of the weak-task gain; the two-stage gate orders the tasks incorrectly.
5. **Stage 5B.1 (completed): prospective signal audit.** Imagined failure is the strongest current under-performance predictor; JS contributes weak intervention information but has task-specific thresholds.
6. **Stage 5B.2 (completed): paired virtual-update gate.** Held-out imagined improvement fails as an update-utility estimator and the 25-step cadence costs more than always-on IR. This mechanism does not advance.
7. **Stage 5B.2a (completed): bury-bones virtual-gate autopsy.** Matched random scheduling preserves most reward-only recovery, early-prefix adaptation passes through a harmful partial-repair region, and virtual-positive selection is worse than either. Exact $\beta = .005$ is not a safe regularizer. The live world-model audit must be repaired before making a hallucinated-reward claim.
8. **Stage 5C (deferred): persistent recovery.** Do not carry actor changes across episodes until a gate can both recover a weak task and abstain on a competent one under episode-local controls. Persistence would otherwise amplify a known decision error.
9. **Stage 5D: hard update safety and conservative utility.** Bound accepted action-distribution drift with hard KL or update rollback independently of the necessity gate. In parallel, test whether uncertainty-aware or repeated virtual evidence, rather than mean imagined improvement, predicts real benefit. Performance estimation and update safety should remain distinct ablations.
10. **Training-time critic regularisation.** Keep this as a separate intervention rather than changing the Stage 5 teacher post hoc. Candidate controls include bounded/return-range targets, held-out calibration and ranking audits, counterfactual goal supervision, and regularisation against optimistic drift across continual phases. Compare any retrained critic against the existing pinned baseline before using it in persistent IR.

11. **Replay-mixture continual experiment.** Preserve Algorithm R retention but train primarily on the current task plus a controlled old-goal reservoir mixture, with update dose matched where possible. Compare gated reward-only/conservative IR against a no-IR baseline at pinned checkpoints.
12. **Stage 6: full validation.** Replicate across training seeds and task orders, then extend from three to five RLScope goals. Add return phases only after the mechanism is stable, to quantify relearning speed, backward transfer, and real-sample savings.

21 Global limitations

The combined evidence is unusually deep for one training trajectory but remains exploratory.

- **One continual-training seed.** Episode-level paired intervals do not quantify variation across independently trained world models and critics.
- **Adaptive research sequence.** Studies II and III were designed after inspecting Study I, Study IV after Study III, Study V after the critic-free recovery result, Study VI after the provenance audit, and Study VII after the repair result. Each workflow was specified before its child evaluations, but the overall sequence remains mechanism discovery rather than an independent confirmatory programme.
- **Corruption calibration and deterministic anomaly.** The mask was selected on sampled episodes. Under the independent main schedule, corruption improves deterministic kill success from 32% to 68%; aggregate deterministic “recovery fractions” are therefore not meaningful for that goal. Raw success and paired changes are reported instead.
- **Reward-only is model based, not oracle based.** It removes the critic but still trusts learned reward, continuation, and transition predictions. Success in the real environment validates outcomes, not calibration everywhere in latent space.
- **No persistent learning in Studies III, IV, VI, or VII.** Adaptation resets each episode. The studies establish test-time recovery and mechanism repair, not yet a continual-learning update rule that safely carries knowledge forward.
- **Historical weak labels are not gates.** Study IV fixed weak/competent status from the earlier sampled evaluation. The independent frozen schedule shifted some absolute rates, most visibly 1.25M bury from historical 49% to 71%. Treatment effects use paired contemporary baselines, but the label itself cannot be deployed online.
- **Stage V is diagnostic rather than interventional.** Its same-state design localizes critic error and actor amplification, but does not uniquely identify whether recursive bootstrapping, weak goal semantics, sequential distribution shift, or another optimizer interaction caused critic drift.
- **Stage VI uses one repaired checkpoint.** The counterfactual and bounded heads are fitted and tested from the reservoir 1.25M representation. Their calibration and behavioral effects may change at the collapsed checkpoint, under FIFO representations, or across independent training seeds.
- **Stage VII repeats adaptation seeds, not training seeds.** Its three replicates vary environment and agent random streams around one fixed repaired actor/world model. They demonstrate stability of the episode-local treatment effect, not population variation across independently trained critics.
- **Sampled-deterministic divergence remains mechanistically unresolved.** Small value doses improve sampled behavior but not deterministic control. The current assay does not distinguish beneficial distribution broadening from unstable logit shifts, state-specific mode changes, or later-step compounding.
- **Agent randomness was not fully paired.** Stage VI pairs reset identities but not every posterior/policy random draw inside separate adaptation processes. The differing reward-only results from functionally identical counterfactual and counterfactual-plus-bounded teachers quantify this otherwise hidden source of assay variance.
- **Event semantics are narrower than policy semantics.** Counterfactual labels are valid for the observed completion transition, but do not identify alternative-goal values or guarantee correct reward gradients on actor-imagined off-distribution states.

- **No oracle return for imagined counterfactuals.** Realized returns calibrate factual recorded states. Actor-selected and alternative-goal paths are assessed through paired contrasts, not falsely labelled with unobserved ground truth.
- **Shared simple action interface.** All tasks use a 504-way left-click grid and the same compact arena. Results may change with harder RLScape tasks, longer temporal dependencies, or physical robotic control.
- **Objective comparisons are not compute neutral.** Longer horizons require more imagined transitions and reward-only conditions often finish episodes sooner. Scientific comparison should report both behavioral gain and test-time compute.
- **Gate labels are retrospective.** The four necessity classes in Studies X and XI are scientifically valid audit labels but unavailable online. AUROC quantifies prospective signal information; it does not make the labels deployable inputs.
- **One repaired checkpoint and three familiar goals.** Studies VIII–XI vary episode and adaptation randomness around the same Stage VI actor/world model. Gate behavior may differ across independent learned models or unseen tasks.
- **Model-based self-validation shares model error.** The independent A/B batches in Study XI prevent direct overfitting and common random numbers reduce variance, but both are generated by the same frozen world model. They are not independent evidence about model bias.
- **Study XII is a single-task sampled-policy autopsy.** It deliberately concentrates 2,010 episodes on the anomalous bury cell and does not estimate whether the same dose valley or scheduling effect transfers to kill, chop, or deterministic control.
- **World-audit non-interference failed.** The diagnostic no-IR condition shares reset seeds with no IR but only 14.3% of aligned sampled actions. Its imagined reward/action traces are exploratory visualizations, not valid paired causal evidence about model exploitation.
- **Update budgets are opportunities, not equal realized compute.** Random and capped conditions terminate successful episodes before exhausting their nominal budget. Both requested limits and realized actor commits are required to interpret the result.
- **Compute measurements are host specific.** Relative timings are internally synchronized on the same 4090 host and are appropriate for within-study comparison, but absolute seconds per step will change with accelerator and compiler cache state.

22 Chapter conclusion

The programme began with a replay hypothesis: preserve older goals in a reservoir so that goal-conditioned Dreamer heads remain useful for IR. Reservoir replay did preserve information and yielded striking intermediate multi-goal competence, but it did not prevent synchronized collapse. Direct audits then revealed why ordinary IR was unreliable. The reward and continuation heads recognize completion-like events but lack counterfactual goal semantics; the critic is severely optimistic; and almost the entire standard IR target is critic bootstrap.

Controlled actor corruption turns that diagnosis into a causal result. Standard IR and entropy-off IR do not reliably recover the actor. Reward-only IR does, reaching 73.0% sampled and 85.3% deterministic mean success and exceeding the original clean actor under the same evaluation schedules. Study IV then removes the artificial corruption: on naturally damaged actors, reward-only IR improves aggregate success at every tested horizon and recovers 1.5M bury from 14% to 91% sampled and 4% to 86% deterministic. Ordinary IR still catastrophically damages competent 1.25M behaviors.

Study V explains the failed teacher more precisely. On identical real states, critic bias grows from +1.184 to +1.595 as ranking collapses; the slow critic mirrors the error; target scale plateaus by short horizons; and the actor selects additional high-bootstrap, often lower-reward trajectories. Study VI then demonstrates that the diagnosed defects can be changed independently. Counterfactual supervision produces near-perfect held-out completion-goal identification, and a separate bounded head returns predictions to the attainable success scale. These are genuine mechanism repairs, not merely post-hoc correlations. Study VII converts the bounded repair into a dose-response result: a five-percent correction raises sampled success from 73.8% to 87.6% relative to reward-only and does so in all three fully paired replicates, whereas pure bounded teaching is inferior.

The behavioral result prevents an overly simple conclusion. Factual-only reward IR, not the semantically strongest counterfactual head, gives the best Stage VI sampled profile; Stage VII then shows that a small

bounded correction can improve that general strategy, but only in sampled control. Every mixed treatment reduces deterministic macro success relative to reward-only, so no candidate passes the prespecified safety rule. The research direction therefore remains viable, but its center of gravity changes again: the immediate opportunity is not simply “fix the critic and turn standard IR back on.” It is **reward-only adaptation as the approved reference teacher, conservative value information activated only under evidence-based gating and policy constraints, fully paired stochastic assays, and replay that preserves old experience without overwhelming current-task learning.**

Studies VIII–XII show why the phrase “evidence-based gating” must be treated as a research problem rather than an implementation detail. Global entropy and JS thresholds save compute but remove useful repair. Imagined failure is a meaningful early estimate of future underperformance, but cannot say whether IR will fix that failure. Even a tentative reward-only update validated on an independent, paired imagination batch does not answer that second question: the virtual score is near chance for true repair need and approves catastrophic bury damage. The safest current algorithm is consequently still episode-local always-on reward-only IR when repair is externally known to be required; it is not yet autonomous continual adaptation.

The bury autopsy replaces one vague catastrophe with two concrete design constraints. First, adaptation is path dependent: stopping after a short prefix can be worse than no adaptation, while continuing to roughly 127 opportunities recovers the actor. Second, an imagined utility score can select updates that are much worse than randomly timed updates at a similar nominal budget. The exact $\beta = .005$ experiment also shows that “small” objective weight does not imply small behavioral effect. Finally, the failed no-op audit demonstrates a methodological constraint for future online gating: diagnostic imagination must have a private random stream, otherwise measuring the policy changes the sampled policy being measured.

This remains a positive result in the intended scientific sense. It identifies conditions under which IR works strongly on synthetic and natural policy deficits, identifies and localizes the learned component that makes the default method fail, directly repairs goal semantics and value scale, measures how much repaired value information can be useful, and rejects increasingly plausible but unsafe gating schemes under real-environment evaluation. It also identifies adaptation cadence as an independent control variable rather than an implementation detail. The next constructive work should first repair RNG-isolated model auditing, then combine an explicit underperformance estimator with conservative uncertainty about update utility, a cadence that avoids partial repair, and a hard, separately tested update constraint. A training-time branch should continue to investigate why the critic becomes optimistic and weakly goal conditioned. Another costly continual-training run is not yet required.

A Complete frozen-policy evaluation table

Table 19: Frozen success by checkpoint. S=sampled (100 episodes); D=deterministic (50 episodes). Dash means the goal had not yet been introduced.

Replay	Goal/mode	250k	500k	750k	1.0M	1.25M	1.5M
FIFO	Kill S	.91	.96	.75	.06	.98	.73
	Kill D	.80	.94	.70	.04	.92	.76
	Bury S	–	–	.21	1.00	.08	.12
	Bury D	–	–	.30	1.00	.28	.04
	Chop S	–	–	–	–	1.00	1.00
	Chop D	–	–	–	–	1.00	1.00
Reservoir	Kill S	.49	.78	.70	.02	.56	.16
	Kill D	.26	.74	.76	.02	.46	.24
	Bury S	–	–	.51	.04	.49	.08
	Bury D	–	–	.48	.00	.34	.04
	Chop S	–	–	–	–	.88	.05
	Chop D	–	–	–	–	.60	.06

B Reservoir IR effects and paired intervals

Table 20: Actor IR minus frozen success. Intervals are episode-order paired bootstrap 95% intervals; S uses 100 episodes and D uses 50. These intervals do not capture training-seed uncertainty.

Checkpoint	Goal	Mode	Change	95% interval
250k	Kill	S	−.13	[−.25, .00]
250k	Kill	D	+.02	[−.10, .14]
500k	Kill	S	−.05	[−.16, .06]
500k	Kill	D	+.04	[−.12, .20]
750k	Kill	S	−.02	[−.12, .08]
750k	Kill	D	−.10	[−.22, .02]
750k	Bury	S	−.22	[−.33, −.10]
750k	Bury	D	−.12	[−.32, .08]
1.0M	Kill	S	+.01	[−.02, .05]
1.0M	Kill	D	+.02	[.00, .06]
1.0M	Bury	S	−.03	[−.08, .01]
1.0M	Bury	D	.00	[.00, .00]
1.25M	Kill	S	+.03	[−.10, .16]
1.25M	Kill	D	+.10	[−.08, .28]
1.25M	Bury	S	−.18	[−.30, −.06]
1.25M	Bury	D	−.18	[−.34, −.02]
1.25M	Chop	S	−.36	[−.47, −.25]
1.25M	Chop	D	−.18	[−.36, .00]
1.5M	Kill	S	+.12	[.02, .23]
1.5M	Kill	D	+.16	[−.04, .36]
1.5M	Bury	S	−.02	[−.09, .05]
1.5M	Bury	D	−.02	[−.08, .04]
1.5M	Chop	S	+.07	[.00, .14]
1.5M	Chop	D	+.02	[−.08, .12]

C Reproducibility manifest

Table 21: Key configuration recorded in the reservoir experiment manifest.

Item	Value
Git commit	ae2e1d25ed5eddc1289dddf36a946e24242d0c22
RLScape	0.1.6; 28×18 click grid; 200-action limit; sparse success reward
Curriculum	kill goblin $\rightarrow$ bury bones $\rightarrow$ chop logs; 500k steps each; seed 0
Dreamer	“50m” preset; batch 8; sequence 32; train ratio 32; imagination horizon 15; bfloat16
Goal conditioning	Five-way one-hot; reward, continuation, actor, value, slow value conditioned; RSSM state goal-agnostic
Reservoir	1,000 complete episodes per goal; Algorithm R; uniform goal $\rightarrow$ episode $\rightarrow$ sequence sampling; persistent across phases
Checkpoints	250k, 500k, 750k, 1.0M, 1.25M, 1.5M; all pinned
IR	Actor only; every action; one update; $H = 6$; 128 posterior samples; $\alpha = 4 \times 10^{-5}$; entropy 3×10^{-4} ; critic/world model frozen
Evaluation	100 sampled and 50 deterministic episodes per condition; frozen/IR seed schedules paired

The report directory includes `generate_figures.py` and compact CSV extracts for the evaluation matrix, 50k training bins, and resource milestones. Running

```
MPLCONFIGDIR=/tmp/rlscape-report-mpl python generate_figures.py
latexmk -pdf report.tex
```

regenerates the figures and paper from the compact audited summaries.

D Study II–XI reproducibility manifest

Table 22: Key configuration for the diagnostic, recovery, repair, and gating assays.

Item	Value
Study II checkpoints	Strong 1.25M and weak 1.5M milestones from the reservoir run
Study II state bank	384 episodes: 64 successful and 64 failed per trained goal; 48,299 states; source bank pinned at 1.25M
Head sweep	Reward, continuation, online value, slow value under all five one-hot goal IDs
Imagination audit	1,433 valid anchors; five-goal sweep; 128 paired posterior samples; $H = 6$; 7,165 rollouts and 5,502,720 imagined transitions per checkpoint
Study III source	1.25M checkpoint; actor-only independent zero mask; seed 0; selected strength 0.4
Study III objectives	Standard lambda-return IR; entropy-off standard IR; reward-only IR; clean-reference policy distillation
Study III horizons	$H \in \{3, 6, 15, 20\}$; one actor update per real action; 128 posterior samples; $\alpha = 4 \times 10^{-5}$
Frozen components	Encoder, RSSM, decoder, reward, continuation, online critic, slow critic, and persistent checkpoint parameters
Study III evaluation	20 sampled calibration episodes per goal/candidate; main 100 sampled + 50 deterministic episodes per goal/condition; 6,750 main episodes
Pairing and integrity	Same goal/mode/reset seeds across conditions; 21/21 calibration and 90/90 main units complete; zero failed attempts; persistent parameter digests unchanged
Stage IV sources	Unmodified 1.25M and 1.50M natural reservoir checkpoints; no actor corruption and no continual retraining
Stage IV conditions	Frozen; standard IR $H = 6$; reward-only IR $H \in \{6, 15, 20\}$; one update per real action; 128 posterior samples; $\alpha = 4 \times 10^{-5}$

Continued on next page

Table 22: Key configuration for the diagnostic, recovery, repair, and gating assays (continued).

Item	Value
Stage IV evaluation	Three goals; 100 sampled + 50 deterministic episodes per cell; 60/60 units and 4,500/4,500 episodes complete on first attempt
Stage IV integrity	Same checkpoint/goal/mode/reset seeds across conditions; no pairing mismatch; persistent parameter digests unchanged; 23.60-hour workflow
Stage V sources	Immutable 1.25M and 1.50M checkpoints; same 384-episode Stage 1A bank; 48,299 factual states and 1,433 valid anchors
Stage V proposals	All five goals; actor-selected and paired recorded actions; 128 Monte Carlo posterior samples; prefix reuse at $H \in \{1, 3, 6, 15, 20\}$
Stage V measurements	Real-state value calibration; online/slow value; full, reward-only, and bootstrap returns; actor-minus-recorded contrasts; continuation and entropy diagnostics
Stage V integrity	124,380 prefix summaries; 18,342,400 actor-imagined transitions per checkpoint; both parameter digests unchanged; canonical specification digest verified
Stage VI source/split	Reservoir 1.25M checkpoint; disjoint 576-episode repair bank and 384-episode held-out bank; 47,916 held-out states and 192 completion events; FIFO original audited on the same reservoir-originating bank
Stage VI repair	Factual-only and all-five-goal counterfactual reward/continuation arms; 2,000 updates; batch 256; $\alpha = 4 \times 10^{-5}$; event fraction .5; separate bounded factual-success head with 2,000 updates
Stage VI evaluation	Frozen; original standard $H = 6$; four reward-only $H = 15$ arms; bounded-bootstrap $H \in \{6, 15\}$; three goals; 100 sampled + 50 deterministic episodes per cell
Stage VI integrity	5 audits + 3 repairs + 48 evaluation cells; 3,600/3,600 episodes; zero failed child attempts; reset identities paired; actor digests equal; counterfactual reward/continuation digests bit-identical
Stage VII source/objective	Stage VI counterfactual-plus-bounded checkpoint; $A^\beta = A^R + \beta \text{clip}(A^B - A^R, -c, c)$; all model/heads frozen; episode-local actor only
Stage VII conditions	Frozen; reward-only $H = 15$; $\beta \in \{.05, .10, .25, 1\}$ at $H = 15$; $\beta = .25$ correction clip .50; update RMS cap 1.5×10^{-5} ; $H = 6$ control
Stage VII adaptation	One actor update per nonterminal real step; 128 posterior samples; $\alpha = 4 \times 10^{-5}$; zero entropy; three environment/agent-seed replicates
Stage VII evaluation/integrity	Three goals; 50 sampled + 25 deterministic episodes per cell; 162/162 units and 6,075/6,075 episodes; zero failed attempts; reset, posterior, and action keys fully paired
Stage VIII source/signals	Stage VI counterfactual-plus-bounded checkpoint; factual and posterior actor entropy; 128-sample MCPB JS; $H = 15$ predicted success and successful-action divergence
Stage VIII gates	Entropy, global JS, and two-stage consequence gates; audit $\rightarrow$ qualification $\rightarrow$ three-replicate confirmation; thresholds frozen before confirmation
Stage VIII evaluation	No IR plus 12 always/gated objective conditions; three goals; 50 sampled + 25 deterministic episodes per replicate/cell; 234 confirmation units and 8,775 episodes; synchronized compute accounting
Stage IX calibration	Weak task requires $\geq .05$ best IR gain; strong task requires $\geq .60$ no-IR competence; frozen selection chooses bury bones and chop logs
Stage IX evaluation	Seven conditions; three paired replicates; 50 sampled + 25 deterministic episodes per cell; 270 calibration + 3,150 confirmation episodes; episode-local actor
Stage X data/audit	90 reset-matched no-IR/reward-only pairs; first 1/5/10/25 factual-state prefixes; four retrospective necessity classes; feature AUROC, calibration, and cross-task threshold transfer
Stage XI virtual gate	Independent training batch A and held-out batch B ; common random numbers before/after; commit iff mean reward-only return improves; exact actor/optimizer rollback otherwise
Stage XI lifecycle	Probe on reset; dormant cadence 25 or 100; accepted proposals activate stepwise checks; three consecutive rejects return to dormancy; $H = 15$, batch 128, $\alpha = 4 \times 10^{-5}$, zero entropy
Stage XI evaluation/integrity	Four conditions; three goals and two modes; three paired replicates; 2,700/2,700 recorded episodes; complete pairing and probe-level diagnostics

E Complete Study III success table

Table 23: Sampled-policy real-environment success in Study III (100 episodes per cell).

Condition	Kill	Bury	Chop	Mean
Clean actor	.54	.41	.74	.563
Corrupted actor	.36	.06	.43	.283
Standard IR $H = 3$	.52	.04	.51	.357
Standard IR $H = 6$	.44	.03	.41	.293
Standard IR $H = 15$	.32	.01	.30	.210
Standard IR $H = 20$	.55	.02	.46	.343
Entropy-off IR $H = 3$	.48	.03	.32	.277
Entropy-off IR $H = 6$	.49	.03	.22	.247
Entropy-off IR $H = 15$	.46	.05	.35	.287
Entropy-off IR $H = 20$	.34	.05	.42	.270
Reward-only IR $H = 3$	.58	.62	.43	.543
Reward-only IR $H = 6$	.57	.67	.52	.587
Reward-only IR $H = 15$	.64	.96	.59	.730
Reward-only IR $H = 20$	.65	.91	.48	.680
Clean-policy distillation	.58	.14	.73	.483

Table 24: Deterministic-policy real-environment success in Study III (50 episodes per cell). Corruption was calibrated under sampled control and accidentally improved deterministic kill.

Condition	Kill	Bury	Chop	Mean
Clean actor	.32	.42	.74	.493
Corrupted actor	.68	.18	.38	.413
Standard IR $H = 3$	.32	.02	.20	.180
Standard IR $H = 6$	.30	.10	.16	.187
Standard IR $H = 15$	.28	.00	.22	.167
Standard IR $H = 20$	.40	.14	.22	.253
Entropy-off IR $H = 3$	.36	.00	.14	.167
Entropy-off IR $H = 6$	.36	.00	.14	.167
Entropy-off IR $H = 15$	.18	.04	.24	.153
Entropy-off IR $H = 20$	.52	.00	.08	.200
Reward-only IR $H = 3$	.66	.82	.74	.740
Reward-only IR $H = 6$	.60	.90	.74	.747
Reward-only IR $H = 15$	.66	.96	.66	.760
Reward-only IR $H = 20$	.74	.98	.84	.853
Clean-policy distillation	.50	.28	.62	.467

F Complete Study IV success tables

Table 25: Sampled-policy real-environment success in Study IV (100 episodes per cell).

Checkpoint	Condition	Kill	Bury	Chop	Mean
1.25M	Frozen	.56	.71	.84	.703
	Standard IR $H = 6$	.61	.07	.53	.403
	Reward-only IR $H = 6$	.74	.95	.75	.813
	Reward-only IR $H = 15$	.76	.99	.76	.837
	Reward-only IR $H = 20$	.73	1.00	.88	.870
1.50M	Frozen	.23	.14	.03	.133
	Standard IR $H = 6$	.46	.02	.09	.190
	Reward-only IR $H = 6$	.37	.17	.11	.217
	Reward-only IR $H = 15$	.20	.91	.08	.397
	Reward-only IR $H = 20$	.44	.78	.08	.433

Table 26: Deterministic-policy real-environment success in Study IV (50 episodes per cell).

Checkpoint	Condition	Kill	Bury	Chop	Mean
1.25M	Frozen	.64	.36	.78	.593
	Standard IR $H = 6$	.60	.12	.34	.353
	Reward-only IR $H = 6$	.80	.90	.70	.800
	Reward-only IR $H = 15$	.60	.96	.78	.780
	Reward-only IR $H = 20$	.76	.98	.80	.847
1.50M	Frozen	.26	.04	.04	.113
	Standard IR $H = 6$	.36	.00	.12	.160
	Reward-only IR $H = 6$	.24	.12	.14	.167
	Reward-only IR $H = 15$	.38	.86	.04	.427
	Reward-only IR $H = 20$	.18	.78	.10	.353

G Complete Study V diagnostic tables

Table 27: Goal-level real-state online-critic calibration in Study V. The same factual state and target bank is used for both checkpoints.

Checkpoint	Goal	Target	Prediction	Bias	Pearson	Spearman
1.25M	Kill goblin	.198	1.618	+1.421	.182	.121
1.25M	Bury bones	.104	1.725	+1.621	.107	.121
1.25M	Chop logs	.215	.794	+.579	.307	.396
1.50M	Kill goblin	.198	1.991	+1.793	.048	.057
1.50M	Bury bones	.104	2.005	+1.900	.064	.103
1.50M	Chop logs	.215	1.357	+1.142	.061	.082

Table 28: Complete actor-selected factual horizon decomposition in Study V. Means are unweighted across the three trained goals.

Checkpoint	H	Full target	Reward-only	Bootstrap	Bootstrap fraction
1.25M	1	1.443	.052	1.391	.959
	3	1.475	.063	1.412	.952
	6	1.480	.072	1.407	.944
	15	1.482	.097	1.385	.924
	20	1.482	.106	1.376	.917
1.50M	1	1.848	.036	1.812	.975
	3	1.885	.038	1.847	.974
	6	1.891	.039	1.852	.974
	15	1.892	.046	1.846	.971
	20	1.891	.050	1.841	.969

H Complete Study VI success tables

Table 29: Sampled-policy real-environment success in Study VI (100 episodes per cell). CF+B denotes the counterfactual-repaired checkpoint with the additional bounded success head.

Condition	Kill	Bury	Chop	Mean
Frozen original	.55	.69	.85	.697
Original standard $H = 6$	.46	.07	.49	.340
Original reward-only $H = 15$	.69	.97	.75	.803
Factual reward-only $H = 15$	.86	.96	.82	.880
Counterfactual reward-only $H = 15$	.75	.96	.74	.817
CF+B reward-only $H = 15$	.54	.96	.86	.787
CF+B bounded-bootstrap $H = 6$	.87	.72	.61	.733
CF+B bounded-bootstrap $H = 15$	.82	.88	.65	.783

Table 30: Deterministic-policy real-environment success in Study VI (50 episodes per cell).

Condition	Kill	Bury	Chop	Mean
Frozen original	.54	.52	.72	.593
Original standard $H = 6$	.70	.04	.40	.380
Original reward-only $H = 15$	.68	1.00	.66	.780
Factual reward-only $H = 15$	.62	.98	.68	.760
Counterfactual reward-only $H = 15$	.72	.94	.66	.773
CF+B reward-only $H = 15$	.74	.74	.74	.740
CF+B bounded-bootstrap $H = 6$	.76	.70	.60	.687
CF+B bounded-bootstrap $H = 15$	.78	.50	.40	.560

Table 31: Study VI bounded-value held-out diagnostics. Original is the unchanged reservoir critic inside the counterfactual-plus-bounded clone; bounded is the separately trained Bernoulli success head.

Goal	Target	Original pred.	Bounded pred.	Original bias	Bounded MAE	Bounded AUROC
Kill goblin	.194	1.623	.180	+1.429	.179	.568
Bury bones	.078	1.694	.069	+1.616	.084	.848
Chop logs	.224	.819	.226	+.595	.173	.632

I Complete Study VII success tables

Table 32: Sampled-policy real-environment success in Study VII, aggregated over three fully paired replicates (150 episodes per goal and condition).

Condition	Kill	Bury	Chop	Mean
Frozen	.500	.600	.760	.620
Reward-only $H = 15$	.513	.900	.800	.738
$\beta = .05, H = 15$	.787	.940	.900	.876
$\beta = .10, H = 15$	.600	.920	.953	.824
$\beta = .25, H = 15$	.760	.880	.880	.840
$\beta = .25$, correction clip .50	.747	.900	.940	.862
$\beta = .25$, update RMS cap	.767	.820	.920	.836
$\beta = .25, H = 6$	.773	.873	.787	.811
$\beta = 1, H = 15$	.767	.787	.613	.722

Table 33: Deterministic-policy real-environment success in Study VII, aggregated over three fully paired replicates (75 episodes per goal and condition).

Condition	Kill	Bury	Chop	Mean
Frozen	.427	.587	.747	.587
Reward-only $H = 15$	.760	.920	.760	.813
$\beta = .05, H = 15$	.627	.907	.853	.796
$\beta = .10, H = 15$	.667	.947	.773	.796
$\beta = .25, H = 15$	.653	.800	.707	.720
$\beta = .25$, correction clip .50	.587	.787	.773	.716
$\beta = .25$, update RMS cap	.773	.707	.773	.751
$\beta = .25, H = 6$	.693	.640	.613	.649
$\beta = 1, H = 15$	.733	.867	.573	.724

J Complete Study VIII–XII result tables

Table 34: Study VIII confirmation macro success across all prespecified conditions. Each cell aggregates three goals and three paired replicates.

Condition	Sampled	Deterministic
No IR	.476	.378
Always standard	.418	.289
Always reward-only	.822	.653
Always $\beta = .05$	.618	.596
Entropy standard	.469	.449
Entropy reward-only	.611	.516
Entropy $\beta = .05$	.620	.484
JS standard	.371	.338
JS reward-only	.611	.596
JS $\beta = .05$	.544	.453
Two-stage standard	.480	.436
Two-stage reward-only	.549	.338
Two-stage $\beta = .05$	.609	.444

Table 35: Study IX actor-update activation per real step. The desired gate would activate more on weak bury than competent chop. JS has this coarse direction for reward-only; the two-stage gate reverses it.

Condition	Chop (strong)		Bury (weak)	
	S	D	S	D
JS standard	.453	.371	.479	.508
JS reward-only	.384	.398	.582	.496
Two-stage standard	.098	.083	.011	.007
Two-stage reward-only	.167	.155	.013	.013

Table 36: Study X ten-state cell-level AUROC. Dash indicates that the cell lacks both positive and negative IR-need labels.

Goal	Mode	JS: underperform	Predicted failure	JS: IR needed
Bury bones	Sampled	.609	.484	–
Bury bones	Deterministic	.238	.762	.238
Chop logs	Sampled	.684	.789	–
Chop logs	Deterministic	.583	.583	.714
Kill goblin	Sampled	.427	.693	.610
Kill goblin	Deterministic	.625	.875	1.000

Table 37: Study XI gate diagnostic and compute summary, averaged across goals and policy modes. Seconds per step include environment interaction, gating, and adaptation.

Condition	Tentative/step	Commit/step	Seconds/step	Macro S/D
No IR	.000	.000	.1197	.562/.511
Always reward-only	.000	.991	.1296	.707/.644
Virtual dormant 25	.382	.239	.1371	.507/.520
Virtual dormant 100	.172	.110	.1275	.500/.471

Table 38: Study XI mean action count among successful episodes. Dash denotes zero successes in the cell. Length is secondary to success but distinguishes efficiency after competence saturates.

Mode	Condition	Kill	Bury	Chop
Sampled	No IR	83.4	75.1	79.6
	Always reward-only	81.2	90.0	64.3
	Virtual dormant 25	72.0	–	74.2
	Virtual dormant 100	71.4	83.1	73.6
Deterministic	No IR	81.1	82.0	72.5
	Always reward-only	71.3	64.9	53.4
	Virtual dormant 25	90.2	90.8	69.3
	Virtual dormant 100	70.9	81.8	57.9

Table 39: Study XII real-action concentration and update summary. Entropy is computed from the complete real-action histogram; top fraction is the mass on the most frequent click.

Condition	Success	Updates/episode	Action entropy	Top fraction
No IR	.340	0.0	2.205	.332
Always reward-only	.607	124.2	2.083	.358
Reward cap 46	.113	44.9	2.379	.304
Reward cap 127	.613	91.4	2.110	.355
Reward random 46	.540	30.9	1.988	.409
Virtual-positive reward	.000	48.6	2.287	.350
Virtual-negative reward	.173	16.1	2.437	.303
Always mixed $\beta = .005$	.000	200.0	2.155	.334
Mixed random 46	.540	31.2	1.993	.387
Virtual-positive mixed $\beta = .005$	.260	39.2	2.339	.330

The expanded package contains the original compact Study I tables, Stage II CSV/JSON summaries and vector figures, the complete Study III result, comparison, corruption-selection, and specification files, all 60 Study IV result rows with paired effects, the complete compact Study V calibration and target-provenance artifacts, the Stage VI immutable specification, repair manifests, five held-out audit summaries and all 48 evaluation rows, the complete Stage VII dose assay, and all compact Stage VIII–XII specifications, pairing audits, confirmation rows, episode-level gate features, probe diagnostics, completion markers, and vector/raster figures. Study XII additionally retains the complete 2,010-episode diagnostics, real click histograms, action-pairing audit, and frozen-model predicted-reward/action table. The included `generate_figures.py` regenerates the original report-native plots from audited compact inputs; the Stage VIII–XII launchers generated and froze their own figures beside the authoritative summary tables. Raw transition logs remain on the experiment machines and are not duplicated into the report package.

References

- [1] Marcin Andrychowicz, Filip Wolski, Alex Ray, Jonas Schneider, Rachel Fong, Peter Welinder, Bob McGrew, Josh Tobin, Pieter Abbeel, and Wojciech Zaremba. Hindsight experience replay. In *Advances in Neural Information Processing Systems*, volume 30, 2017. URL <https://proceedings.neurips.cc/paper/2017/hash/453fadb8a1a3af50a9df4df899537b5-Abstract.html>.
- [2] Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman and Hall/CRC, 1993. doi: 10.1201/9780429246593.
- [3] Danijar Hafner, Jurgis Pasukonis, Jimmy Ba, and Timothy Lillicrap. Mastering diverse domains through world models. *arXiv preprint arXiv:2301.04104*, 2023. URL <https://arxiv.org/abs/2301.04104>.
- [4] Jeffrey S. Vitter. Random sampling with a reservoir. *ACM Transactions on Mathematical Software*, 11(1): 37–57, 1985. doi: 10.1145/3147.3165.